

PRISSMM

Memorial Sloan Kettering Cancer Center

Data Guide

Curation Guidelines and Directives

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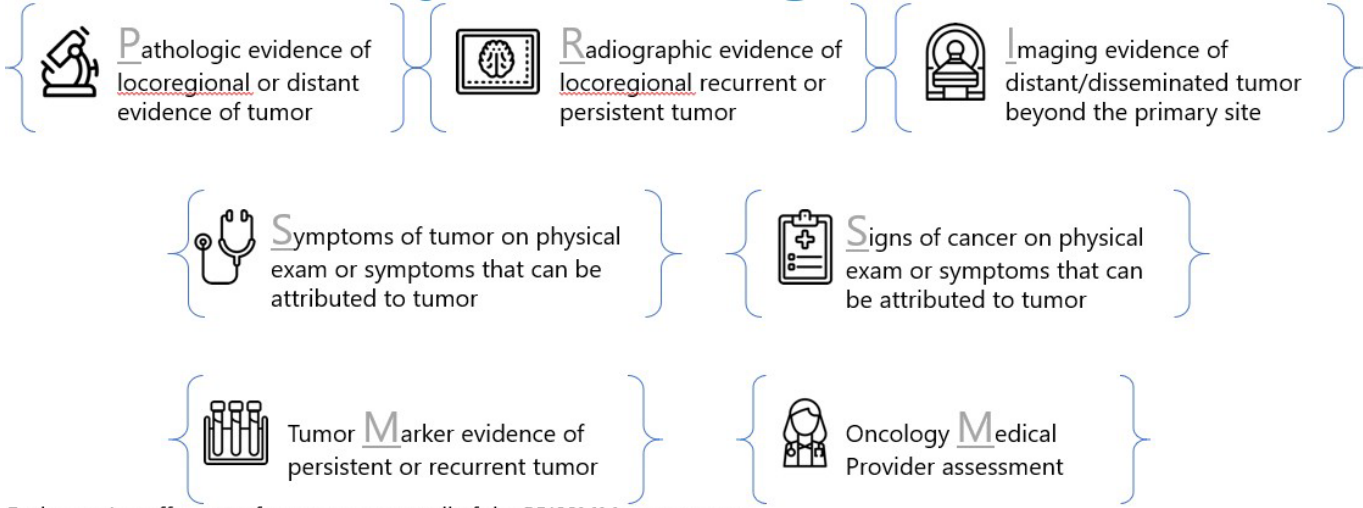
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PRISSMM: A Taxonomy for Defining Cancer Outcomes



Each curation effort may focus on some or all of the PRISSMM components
Signs may be relevant for melanoma outcomes
Markers may not be relevant for lung outcomes

Overview

The Clinical and PRISMM Database is designed for both comprehensive and modular curation; it is created to leverage structured clinical data, existing in institutional data warehouses, and employs widely accepted data standards.

This data guide is to inform the manual curation of patient data from the site's medical record and the data piping of structured data from the sites. The primary audiences for this data guide are the curators reviewing the medical records, the quality assurance (QA) managers monitoring data quality, and the informatic staff setting up the REDCap project and piping in structured clinical data. In addition to the primary audience, analysts and statisticians using the data from REDCap will find this data guide useful.

General Curation Guidelines

- **Do** curate each section starting with the oldest appropriate record, moving to the most recent.
- **Do** use information from past records to inform subsequent records' curations for a given section. For example, while curating the imaging reports, information about a liver metastasis in one imaging report can be used during curation of another imaging report.

The REDCap database is divided into data collection forms based on content as listed below. Several of the sections utilize the repeating instance function of REDCap.

- Curation Initiation and Patient Eligibility
- Patient Characteristics
- Cancer Diagnosis and Staging
- Cancer Directed Drug Therapy
- PRISMM Outcomes
 - Pathology
 - Pathology Special Features
 - Radiology & Imaging
 - Medical Oncology Provider Assessments
 - Tumor Markers
- Cancer Panel Testing

Timed Curation and Timed Sections

One component of monitoring the data quality of curations is recording and analyzing the time that it takes to complete the full patient and the PRISMM pathology and imaging sections, in order to monitor within and between institutional timing metrics for the overall and section curation.

The screenshot displays two sections from the REDCap interface. The top section, 'Start of Cancer Diagnosis Curation', is highlighted in green and shows a timestamp of 10-02-2019 16:39:42 with a 'Now' button. Below it, the 'Stop of Curation Timing - Cancer Diagnosis' section is highlighted in purple and shows a timestamp of 10-02-2019 16:43:34 with a 'Now' button. Both sections include a note: '* must provide value' and instructions to click 'Now' to record the time.

Curation Completion

Each section should be completed for each patient, which is indicated by selecting 'Complete' in the Form Status section at the bottom of each page.

Form Status	
Complete?	Complete

However, if there are questions concerning how to abstract the section for the patient, ‘Unverified’ should be selected; this will indicate that the abstractor has a question for the Project Manager or PI. In addition, if the Quality Assurance (QA) manager finds data quality issues, the QA manager will change the status to unverified to indicate that the curator should review and modify the form.

Data Quality Assurance

Upon the completion of curation, each case is eligible for QA review by source data verification (SDV) by a QA Manager. It is recommended that 10% of the cases be reviewed in this manner. As this occurs the QA manager will complete the QA section at the bottom of the following forms:

- Patient Characteristics
- Cancer Diagnosis and Staging
- Cancer Directed Drug Therapy
- PRISMM Pathology
- PRISMM Radiology & Imaging
- PRISMM Medical Oncology Provider Assessments
- Cancer Panel Test

On each of the data entry forms at the bottom, there is a QA section with major and minor violations listed similar to the figure below.

When the curator marks the form “Complete” the form may be reviewed by the QA manager [A]. The QA Manager will code “Yes” [B] to the question **Cancer Diagnosis QA performed**.

Stop of Curation Timing - Cancer Diagnosis

10-04-2019 15:34:43 **Now** M-D-Y H:M:S
 Click NOW to indicate the time that you complete curating this section by clicking NOW.

* must provide value

The following field should be completed by the QA Manager.

Cancer Diagnosis QA performed?

☐ Yes
☐ No

For QA manager only.

Form Status

Complete?

Complete

If “Yes” is answered, following question will appear: **Were any data quality issues identified?** [C]

The following field should be completed by the QA Manager.

Cancer Diagnosis QA performed?

☒ Yes
☐ No

For QA manager only.

Were any data quality issues identified?

☐ Yes
☐ No

If data quality issues are identified, the following panel will appear [D] along with a data field for the curator to indicate that

the data issues were resolved. [E]

Stop of Curation Timing - Cancer Diagnosis

10-04-2019 15:34:43

Now

Save & ...

-- Cancel --

* must provide value

Click NOW to indicate the time that you complete section by clicking NOW.

For Curator: Are the QA issues resolved?

Yes

No

E

For curator.

reset

The following field should be completed by the QA Manager.

Cancer Diagnosis QA performed?

Yes

No

For QA manager only.

reset

Were any data quality issues identified?

Yes

No

reset

Types of data quality issues identified

Major violation(s)

Minor violation(s)

Other issue(s)

D

Types of major data quality issues identified

Incorrect/missed cancer type

Incorrect/missed dx date [> 1 month off]

Incorrect/missed stage [early v 4]

Missing cancer diagnosis

For QA manager only.

Types of minor data quality issues identified

Incorrect/missed sites of distant metastases at diagnosis

Incorrect/missed cancer-specific questions

Types of other data quality issues identified

Incorrect/missed source of diagnosis date

Other minor mistake/omission

QA issues resolved?

QA Notes

Expand

For QA manager and curator. May contain PHI, data will not be shared.

Form Status

Complete?

Unverified

REDCap Form

curation_initiation_eligibility

This form collects details about the curation like curator, activity, and cohort. In addition, the form can be customized by adding data elements, modifying existing choices, or adding choices to existing data elements. For example, data elements could be added that collect eligibility criteria.

The curator should complete this form prior to beginning the manual curation of the patient's data.

Curation Initiation Curated Fields

Record ID

[record_id] PN0000

Values

Alpha-numeric unique identifier for the case

Description

- Institution-generated or REDCap generated identification number specific to each patient.
- This identifier can be defined and assigned prior to having the data piped into the project's REDCap.
 - For the BPC project, the Record ID should be uploaded to SAGE as the GENIE ID.
 - For the BPC project, study-specific identifiers, like the randomized order number from SAGE, can be used for the Record ID.
- It is recommended that this should not be the medical record number for the patient.

Directives

- No action is required for this data field as it is piped into the project.

PRISMM Curator

[cur_curator] PN0001

Values

List should include the names or identifiers for Institution Study Staff, including:

- Curators
- Project Manager
- QA Manager

Description

- The curator responsible for the manual curation of the case identified by the unique **Record ID**.
- The choice list for this data field should be customized to the site's study staff prior to moving the REDCap project into production in either the REDCap Data Dictionary or the form; See Appendix 12: REDCAP Set-up and Data Piping.
- If your name is not listed in the dropdown list, contact the Project Manager so that it can be added.

Directives

- Select your name from the list; if you do not find your name, please contact your curation manager/project manager for guidance.

- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Curation Cohort

[cur_cohort] PN0002

Values (examples shown are for the BPC GENIE project; however, this field should be customized to meet your project's goals)

- 1, BPC Bladder Cancer
- 3, BPC Breast Cancer
- 5, BPC Colon/Rectum Cancer
- 7, BPC NSCLC Cancer
- 9, BPC Pancreas Cancer
- 11, BPC Prostate Cancer

Description

- Identifies the patient cohort for which curation is occurring.
- This field is modifiable by site prior to moving the REDCap project into production.
 - For the BPC project, the cohort is identified by SAGE thorough application of the pre-specified eligibility criteria.

Directives

- Select the cohort for the case being abstracted, which is defined as the group defined by the eligibility criteria for the project.
- Please note, the cohort does not necessarily map to the cancer diagnoses of the patient; for example, a patient identified by a cancer diagnosis on the specimen tested by NGS may have a discordant cancer primary diagnosis.
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Curation Activity

[cur_activity] PN0003

Values

- 1, Curation Case
- 2, Training Curation Full Case
- 3, Dual Curation Full Case
- 4, Other
- 5, Pressure Test Curation
- 6, Pilot Curation

Description

- This field indicates type of curation activity being performed by the curator for a specific case; the curator should choose the most appropriate choice for the specific case.
 - If the curator is being trained on curation, "2, Training Curation Full Case" should be recorded.
 - If the curator is curating for a project, "1, Curation Case" should be recorded.
 - If the curator is curating a case for a second time in the project & the data will be used for QA purpose or inter-rater reliability, "3, Dual Curation Full Case" Should be curated.
- This field will distinguish the curation activity for downstream analyses.
- This field can be modifiable by the site to include other curation activities like modular curation of specific sections.

Directives

- Select the reason for curation of the record. Code “1, Curation case” for cases that are curated during the production phase.
- Code “2, Training Curations Full Case” for curations performed during training.
- Code “3, Dual Curation Full Case” for cases that are curated for either IRR analyses or QA purposes.
- Code “5, Pressure Test Curation” for the first production curations during which the curators are becoming comfortable with a new project.
- Code “6, Pilot Curation” for the first curations of the new project during which a limited number of curators participate.
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

History

06172020: added “5, Pressure Test Curation” and “6, Pilot Curation” choices.

General Patient Characteristics Curation Guidelines

Provenance

Piped structured data from institutional data warehouse, covering the following:

NDI, Institution Tumor Registry (TR) [Vital Status], Internal and External Medical Provider Notes, Internal and External Pathology Reports, Imaging Reports, Laboratory Report Order Dates

- Most data fields on this form are either piped in from either the EHR or data warehouse or are calculated.
- Some of these fields should not be modified, and this is indicated in the Field Note of the data field in the REDCap project.
- Certain date fields may be edited as follows if data is missing or incorrect; this is specified in the Field Note.

Patient Characteristics Piped Data Fields

The patient-related socio-demographic data elements are in large part piped automatically into REDCap from either the institutional data warehouse or electronic medical record (EHR); the data fields are listed below. Do not edit these following fields.

Data Field	HIPAA Identifier	To Be Shared	Data Standard	Source	GENIE Field*	Required
Patient ID [patient_id] PN0006	No	Yes	Randomization Number	SAGE	Yes	No
GENIE Patient ID [genie_patient_id]* PN0006	No	Yes	GENIE		Yes	Yes
Internal MRN [internal_mrn] PN0007	Yes	No	Institutional	EHR		Yes
Last Name [last_nm] PN0008	Yes	No		EHR		No
First Name [first_nm] PN0009	Yes	No		EHR		No
Date of Birth [birth_dt] PN0010	Yes	No		EHR		Yes
Year of Birth [birth_year] PN0011	Yes	Yes		EHR		No
Patient Type: Adult/Pediatric [adult_pedi] PN0012	No	No		EHR		No
Spanish/Hispanic Origin [naaccr_ethnicity_code] PN0013	No	Yes	NAACCR	EHR	Yes	No
First Race Mentioned [naaccr_race_code_primary] PN0014	No	Yes	NAACCR	EHR	Yes	No

Data Field	HIPAA Identifier	To Be Shared	Data Standard	Source	GENIE Field*	Required
Second Race Mentioned [naaccr_race_code_secondary] PN0015	No	Yes	NAACCR	EHR	Yes	No
Third Race Mentioned [naaccr_race_code_tertiary] PN0016	No	Yes	NAACCR	EHR	Yes	No
Sex [naaccr_sex_code] PN0017	No	Yes	NAACCR	EHR	Yes	No
Zip Code at the Time of Curation [zip_code] PN0018	Yes	No		EHR		No
First 3 Digits of Zip Code at the Time of Curation [zip_code_short] PN0019	No	No		EHR		No
Death Indicator at Time of Curation [hybrid_death_ind] PN0020	No	Yes		NDI/EHR		Yes
Date of Death [hybrid_death_dt] PN0021	Yes	No	NDI	NDI		No
Date of Death Source [hybrid_death_source] PN0023	No	Yes	NDI	NDI/EHR		No
NDI Cause of Death ICD-10 Code [ndi_cause_of_death] PN0024	No	No	NDI/ICD-10	NDI		No
NDI Flag: Year of Last Submission [ndi_flag] PN0025	No	Yes	NDI	NDI		No
Date of Last Oncology Visit [last_oncvisit_dt] PN0026	Yes	No***	EHR	EHR		NO/QA: 1 of 4 dates is required
Date of Last Known Alive from Any Source [last_alive_dt] PN0028	Yes	No***		EHR		
Date of Last Visit to Internal Institution [last_anyvisit_dt] PN0030	Yes	No***	EHR	EHR		
Date Transferred to Hospice [enroll_hospice_dt] PN0032	Yes	No***	EHR	EHR		
Other Institution MRN – Institutional Data [oth_mrn] PN0034	Yes	No	EHR	EHR		No
Internal Alternative MRN [mrn_alternative] PN0035	Yes	No	EHR	EHR		No
Primary Medical Oncology Program/Department [primary_program_nm] PN0036	Mo	No	EHR	EHR		No
Flag indicating that patient is on therapy under research protocol [on_therapeutic_prot_ind] PN0037	No	No	Derived from EHR	EHR		Yes

*If applicable

**Suggested

***Interval in days created from the Date of Birth so as to not have a HIPAA identifier

Patient Characteristics Curated Data Fields

Is the patient dead at the time of abstraction?

[hybrid_death_ind] PN0020

Values

- 1. Yes
- 0. No

Directives

- Required.
- Data may be piped into the REDCap project based on source data from the NDI or the site's EHR indicating the patient's date of death.
- At the time of manual curation, if the data piped into this field is incorrect, edit as required.
- If this field is blank and the patient is not known to be dead, code "0, No." This will include patients who are lost to follow-up.
- If this field is blank and the patient is known to be dead, code "1, Yes."
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Provenance

- AllScripts: All document categories- Death notification document/CIS- while searching the patient
- Cerner:
- Epic: Snapshot, PHS viewer, Media, Telephone Encounters

Analytic Notes

- Several data elements are linked to this; if the indicator is yes the following data elements should be populated: Date of Death based on NDI or other source, Source of Death Information, Cause of Death ICD-10 Code, NDI Flag: Year of Last Submission. Otherwise, these data elements should be blank.

Shared Field: Yes

Date of Death, based on NDI, EHR, or curation

[hybrid_death_dt] PN0021

Value

Date field, mmddyyyy

Directives

- Linked to **Is the patient at the time of curation?** If "Yes" is recorded, provide the date of death.
- Data may be piped into the REDCap project based on source data concerning date of death.
- National Death Index (NDI) date of death is considered the gold standard. If this date does not match the date of death in the EHR, **do not** edit it. Instead, contact your project manager.
- If the date entered is incorrect and the Source of Death Information (field below) is **EHR**, **do** enter the correct date and change the Source of Death Information to "1, Curated."
- If the date is blank, record the patient's date of death here. Use date estimation rules if exact date is unknown.

Provenance

- National Death Index Linkage: for research purposes only.

- AllScripts: All document categories- Death notification document/CIS- while searching the patient.
- EPIC: Snapshot or Telephone Encounters

Analytic Notes

- The interval based on the data element, **Is the patient dead at the time of abstraction?** is used in overall survival analyses.
- The data element, **Is the patient dead at the time of abstraction?** and the associated day interval is used in date logic checks.

Shared Field: No, contains PHI. The data element is flagged as an identifier in REDCap. Instead, a calculated interval in days **[hybrid_death_int]** from **Calculated Date of Birth** is generated

Source of Death Information

[hybrid_death_source] PN0023

Values

- 1, Curated
- 2, EHR – for piped data
- 3, NDI – for piped data
- 5, Tumor Registry
- 4, Other – for piped data

Directives

- Linked to **Date of Death, based on NDI, EHR, (tumor registry,) or curation.**
- Data may be piped into the REDCap project based on source data concerning date of death.
 - If structured data from the site's EHR or other clinical system was used, the field should be pre-populated with "2, EHR."
 - If National Death Index data was used for ascertainment of death, the field should be prepopulated with "3, NDI."
 - If the **Date of Death, based on NDI, EHR, (tumor registry,) or curation** is sourced from the Tumor Registry, the field should be prepopulated with "5, Tumor Registry."
 - If any other means were used to pipe in the **Date of Death, based on NDI, EHR, or curation**, select "4, Other" includes obituary, SSDI, and insurance searches.
- If the **Date of Death, based on NDI, EHR, or curation** has not been piped into REDCap, and review of the medical record indicates that the patient has died, record "1, Curated."
- If the **Date of Death, based on NDI, EHR, or curation** was piped into REDCap and the **Source of Death Information** is from the "2, EHR" or "4, Other" review the date of death. If it is incorrect, enter the correct date and change the **Source of Death Information** to "1, Curated."
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Shared Field: Yes

Date of Last Visit to Internal Institution for Oncology Care

[last_oncvisit_dt] PN0026

Value

Date field, mmddyyyy

Directives

- This date may be piped in from data warehouse or EHR, but it may be inaccurate due to the longitudinal nature of data capture. Verify that the date entered is the most recent date that the patient had an in-person visit with

an oncology provider at your institution; these visits may include lab medical oncology, surgical oncology, radiation oncology, palliative care, social work, lab draws, imaging scans, emergency room, or hospital visits.

- The test/procedures done at your institution may include Imaging scans, procedure dates from pathology reports, visit notes with any department, laboratory results. If a lab or procedure is used to document, record the collection date and not the order date.
- Phone calls, emails, or other types of documentation do not count; this date may be recorded in either **Date Last Known to be Alive from any Source or Date**.
- If a hospitalization represents the last known date at the institution for oncology care, use the discharge date.

Analytic Notes

- At least one of the following four data fields is required for the calculation of vital status for each case:
 - **Date of Last Visit to Internal Institution for Oncology Care**
 - **Date of Last Known Visit to Internal Institution for any Care**
 - **Date Last Known to be Alive from any Source**
 - **Date Enrolled in Hospice**
- Logic checks should be implemented to ensure that at least one of these fields is populated.

Shared Field: No, contains PHI. The data element is flagged as an identifier in REDCap. Instead, a calculated interval in days **[last_oncvisit_int]** from **Calculated Date of Birth** is generated.

Date Last Known to be Alive from any Source

[last_alive] PN0028

Value

Date field, mmddyyyy

Directives

- This date is piped in from data warehouse or EHR, but it may be inaccurate due to the longitudinal nature of data. Verify that the date entered is the most recent date that we know the patient was alive.
- The last date for any type of primary care, clinic, or hospital visit to the health care network for oncology or non-oncology-related care should be used.
- Any other confirmation the patient was living on a specific date can be used, such as documentation of a phone call or email exchange with the patient or a family member.
- If a hospitalization represents the last known date alive, use the discharge date.

Analytic Notes

- At least one of the following four data fields is required for the calculation of vital status for each case:
 - **Date of Last Visit to Internal Institution for Oncology Care**
 - **Date of Last Known Visit to Internal Institution for any Care**
 - **Date Last Known to be Alive from any Source**
 - **Date Enrolled in Hospice**
- Logic checks should be implemented to ensure that at least one of these fields is populated.

Shared Field: No, contains PHI. The data element is flagged as an identifier in REDCap. Instead, a calculated interval in days **[last_alive_int]** from **Calculated Date of Birth** is generated.

Date of Last Known Visit to Internal Institution for any Care

[last_anyvisit_dt] PN0030

Value

Date field, mmddyyyy

Directives

- This date is piped in from data warehouse or EHR, but it may be inaccurate due to the longitudinal nature of data. Verify that the date entered is the most recent date that the patient had an in-person visit with any provider or an in-person visit for tests/procedures/therapies at your Institution or health care system.
- Phone calls, emails, or other types of documentation do not count.
- If a hospitalization represents the last known date at the internal institution, use the discharge date.

Analytic Notes

- At least one of the following four data fields is required for the calculation of vital status for each case:
 - **Date of Last Visit to Internal Institution for Oncology Care**
 - **Date of Last Known Visit to Internal Institution for any Care**
 - **Date Last Known to be Alive from any Source**
 - **Date Enrolled in Hospice**
- Logic checks should be implemented to ensure that at least one of these fields is populated.

Shared Field: No, contains PHI. The data element is flagged as an identifier in REDCap. Instead, a calculated interval in days [**last_anyvisit_int**] from **Calculated Date of Birth** is generated.

Date Enrolled in Hospice [enroll_hospice_dt] PN0032

Value

Date field, mmddyyyy

Directives

- This date may be piped in from data warehouse or EHR, but it may be inaccurate due to the longitudinal nature of data. Verify that the date entered is the date that the patient was transferred to hospice care in either the inpatient or outpatient setting. Data can come from email, telephone notifications, medical oncologist progress notes.
- If this date is not available or not relevant, leave this field blank.
- Record the date that the patient is transferred to either inpatient or outpatient hospice care.

Analytic Notes

- At least one of the following four data fields is required for the calculation of vital status for each case:
 - **Date of Last Visit to Internal Institution for Oncology Care**
 - **Date of Last Known Visit to Internal Institution for any Care**
 - **Date Last Known to be Alive from any Source**
 - **Date Enrolled in Hospice**
- Logic checks should be implemented to ensure that at least one of these fields is populated.

Shared Field: No, contains PHI. The data element is flagged as an identifier in REDCap. Instead, a calculated interval in days [**enroll_hospice_int**] from **Calculated Date of Birth** is generated.

Curator Notes Concerning Patient Characteristics and Follow-up Section [pt_notes] PN0038

Value

Free text

Directives

- This note field should be used to document and track any notes concerning the curation of the Patient Characteristics section that will assist in QA review.
- PHI can be entered into this field.

Shared Field: No, contains PHI. The data element is flagged as an identifier in REDCap.

Patient Characteristics Calculated Fields

In order to share data in a de-identified manner, several intervals are calculated in the REDCap project from the anchor date: Date of Birth as follows:

Interval Data Field	Data Element	Calculation
Interval in Days from DOB to Date of Death	hybrid_death_int PN0022	datediff([birth_dt], [hybrid_death_dt], "d", "mdy")
Interval in Days from DOB to Last Visit Date	last_oncvisit_int PN0027	datediff([birth_dt], [last_oncvisit_dt], "d", "mdy")
Interval in Days from DOB to Last Day Known Alive	last_alive_int PN0029	datediff([birth_dt], [last_alive], "d", "mdy")
Interval in Days from DOB to Last Visit to Medical Facility	last_anyvisit_int PN0031	datediff([birth_dt], [last_anyvisit_dt], "d", "mdy")
Interval in Days from DOB to Enrolled in Hospice	enroll_hospice_int PN0033	datediff([birth_dt], [enroll_hospice_dt], "d", "mdy")

Patient Characteristics and Vital Status Data Violations

For QA: Vital Status QA performed?

[vstat_qa] PN0041

Value

- 1, Yes
- 0, No

Directives

- This data element is used to indicate that this REDCap form has undergone data quality assessment with 1, Yes.
- If it has not been reviewed for data quality, it will be blank.

For QA: Were any vital status data quality issues identified?

[vstat_qaissues] PN0042

Value

- 1, Yes
- 0, No

Directives

- Linked to **Vital Status QA performed?** If "1, Yes" this field will need to be completed by the QA manager to indicate whether any data issues were found. If **Vital Status QA performed?** is "0, No" this field will not be visible in REDCap.
- If the no data issues are found, record "0, No."
- If data quality issues are found, record "1, Yes" which will reveal through branching logic the following data elements: For Curator: **Are the QA issues resolved?**, **Types of data quality issues identified?**, **QA issues resolved?**, and **QA Notes**.

For Curator: Are the QA issues resolved?

[vstat_qacurator] PN0040

Value

1, Yes

0, No

Directives

- Linked to QA Manager field, **Were any vital status data quality issues identified?** If 1, Yes this field will need to be completed by the curator following data review and correction to indicate that issue found in QA is addressed.
- If the QA fields are not used or if the QA did not identify any errors, this field will not be visible to the curator.

Type	Description	Location
Major	Incorrect/missing date of death [+/- 1 month]	Patient Characteristics Form
	Incorrect/missing date of death [+/- 1 month]	Patient Characteristics Form
Minor	Missing death information	Patient Characteristics Form
	Missing death indicator	Patient Characteristics Form

General Cancer Diagnosis Curation Guidelines

Provenance

In order of preference: Institutional Tumor Registry (TR), Internal and External Oncology Provider Notes, Internal and External Pathology Reports, Internal and External Imaging Reports

Data Standards

NAACCR with COC, AJCC, SEER

Selecting Cancers to Curate

- This database is designed to collect detailed phenomic data (including the treatments and outcomes) from patients diagnosed with solid tumors with next-generation sequencing data. Each curation effort will be specific to a cancer type; however, if a patient has more than one cancer diagnosis some information will be collected about the other tumors like baseline diagnosis and cancer-directed drugs.
- If the Tumor Registry has collected diagnosis and staging data for a patient's cancer diagnoses, this information will be piped into the REDCap project.
 - Do examine the guidance on whether the cancer diagnosis should be curated at the top of the form.
Curate: **Curation of this Cancer Diagnosis REQUIRED**
No Curation; mark the form complete: **Curation of this Cancer Diagnosis NOT REQUIRED**
 - **Do** review the piped staging data for accuracy.
 - If any there are any concerns about accuracy or completeness of the Tumor Registry data, please contact the curation manager for guidance.
- Many patients have cancer diagnoses that are not evaluated and staged by the Tumor Registry; in these instances, diagnosis data must be curated manually. In REDCap, there are specific data items for curation.
- **Do** curate each cancer diagnosis (invasive and in situ) into a separate instance of the Cancer Diagnosis form.
- **Do** curate each diagnosis of invasive cancer [solid tumors and heme malignancies] and in situ cancer as a separate diagnosis. **SEER Solid tumor rules can be used to determine the number of primary cancer diagnoses:** <https://seer.cancer.gov/tools/solidtumor/> **The**
 - Special directives exist for some cancer types, and REDCap includes branching logic to assist in manual curation.
Skin Cancer: **Do not** curate diagnosis data for Basal Cell skin carcinomas and Squamous Cell skin cancers. **Do** enter melanoma diagnoses.
Hematologic Malignancies: Lymphomas, Leukemias, Multiple Myeloma, etc: **Do** enter the date of diagnosis of these cancers into the Cancer Diagnosis form. **Do not** enter staging information, it is not required.

Tumor Curation Rules

Cancer Type	Cancer and Diagnosis	Cancer-Directed Drugs	PRISMM: Pathology	PRISMM: Imaging	PRISMM: Medical Onc Assessment
<i>Solid tumors (i.e., bladder, breast, colon/rectum, lung, pancreas, prostate, melanoma)</i>	Diagnosis Staging	Yes	Yes	Yes	Yes
<i>Solid tumor non-melanoma skin cancers (squamous cell and basal cell skin cancers)</i>	Do not record	N/A	N/A	N/A	N/A
<i>Hematologic malignancies (i.e., leukemia, lymphoma, multiple myeloma)</i>	Diagnosis	Yes	Yes – after dx of index cancer	No	No

Curating Cancer Diagnosis and Staging Details

- Sources for information concerning the diagnosis of cancers, recurrences, and distant metastases can be found in the native oncology provider notes, pathology reports, and imaging reports. Other sources can include external reports and provider notes.
- The best place to start is the first internal Medical Oncologist’s consult note.
 - Check the “Past Medical History” for additional cancer diagnoses.
 - Next, check the most recent medical oncology note to see if additional diagnoses are mentioned.
 - A provider may not mention a diagnosis of a cancer type outside their specialty in the notes, particularly if it is in the patient’s remote past. You may discover these additional diagnoses while you are curating path reports or med onc assessments (for example, the patient starts seeing an oncologist of a different specialty). **Do not** spend more than a few minutes looking through the medical record for details of cancer diagnoses.
- For the solid tumors that require stage, use the appropriate staging version based on the date of diagnosis.
 - Tumors diagnosed on or after January 1, 2018: Please use the AJCC 8th Edition staging manual. Links are provided in Appendix Two: Data Standards that include the AJCC 8th Edition, Staging Guidelines for the clinical and path staging components.
 - If a MD provides the patient’s stage, record this in the curated stage group data field.
 - Tumors diagnosed prior to January 1, 2018 that require manual curation can be staged using the AJCC, 7th Edition staging manual, even if this staging convention was not used at the time that the patient was diagnosed. Links are provided in Appendix Two: Data Standards that include the AJCC 7th Edition, Staging Guidelines.
- For Solid Tumors that require stage, the data field, **Was the patient's cancer diagnosed as Stage IV?** is critical. For cancers diagnosed in the past at external institutions, explicit details about the stage may not be available; however, using evidence like use of adjuvant chemotherapy to assign early or late stage at diagnosis is permitted.
- For the Solid Tumors with data piped in from the TR, ‘Was the patient's cancer diagnosed as Stage IV?’ is required if the data field “Tumor Registry: Best Group Stage” field is blank/88/99 require curation of the cancer stage in the curating section of the same form.

Piped Cancer Diagnosis Data Fields

Tumor Registry Data Fields

Many patients have at least one diagnosis with information piped into REDCap from the Tumor Registry; the data fields are listed below. Except for **Tumor Registry Best Group Stage**, do not edit the following fields:

Tumor Registry Cancer Sequence Number	Tumor Registry Path T Code
Tumor Registry Diagnosis Date	Tumor Registry Path T Suffix Code
Tumor Registry Accession Number	Tumor Registry Path N Code
Tumor Registry Diagnostic Confirmation Code	Tumor Registry Path N Suffix Code
Tumor Registry Diagnostic Confirmation Description	Tumor Registry Path M Code
Tumor Registry Zip Code at Diagnosis	Tumor Registry Path Group Stage Code
Tumor Registry Zip Code at Diagnosis – 3 digits	Tumor Registry Post Therapy Path T Code
Tumor Registry Date of First Contact at Institution	Tumor Registry Post Therapy Path T Suffix Code
Tumor Registry ICDO3 Site Code	Tumor Registry Post Therapy Path N Code
Tumor Registry ICDO3 Site Description	Tumor Registry Post Therapy Path N Suffix Code
Tumor Registry ICDO3 Histology Code	Tumor Registry Post Therapy Path M Code
Tumor Registry ICDO3 Histology Description	Tumor Registry Post Therapy Path Group Stage Code
Tumor Registry ICDO3 Histology Behavior Code	Tumor Registry Clin T Code
Tumor Registry Laterality Code	Tumor Registry Clin N Code
Tumor Registry Laterality Description	Tumor Registry Clin M Code
Tumor Registry Grade Code	Tumor Registry Clin Group Stage Code
Tumor Registry Post Therapy Grade Code	Tumor Registry General Summary Stage
Tumor Registry Best Group Stage*	Tumor Registry SEER Summary Stage
Tumor Registry Path Descriptor Code	Tumor Registry TNM Edition Number

Tumor Registry Best Group Stage* is a calculated field from the Tumor Registry from the individual staging components (TNM) and will be used in the project as the patient's stage at diagnosis if populated. If data from the TR is present, confirm that the data is correct. If it is not correct, delete the entry and leave the field blank.

Sometimes cancers with behaviors other than 0 or 1 are piped in from the Tumor Registry because these tumors are 'reportable' to the state registry. If this happens, the curator is not required to curate the diagnosis; instead, the curator should mark the form as complete.

Curated Cancer Diagnosis Data Fields

Curated Cancer Sequence Number

[redcap_ca_seq] PN0052

Value

Numeric 1-10

Directives

- Required for every cancer diagnosis.
- Sequentially number each unique invasive or in situ cancer diagnosis. The cancer diagnoses should be ordered based on the date of diagnosis when possible. If a cancer diagnosis is found in the during the course of curation, the Curated Cancer Sequence Number does not need to be modified.

Analytic Notes

- The **CURATED Cancer Sequence Number** will be linked to cancer-directed drug therapies, radiation therapy (if

applicable), and cancer panel testing instances.

Curators will try to apply sequential numbers based on diagnosis date; however, a derived variable should be created that sequences the cancers based on diagnosis date with rules for simultaneously diagnosed tumors.

Shared: Yes

Is this the index cancer

[redcap_ca_index] PN0053

Values

1, Yes
0, No

Directives

- Required for every cancer diagnosis.
- Code “1, Yes” if this diagnosis is the cancer of interest – the cancer that is under evaluation, making the patient eligible for the study requiring curation. In the case of the BPC, this is the cancer that has been sequenced using a multi-gene cancer panel test and submitted to GENIE.
- Code “0, No” if this diagnosis is not the cancer of interest. This cancer could be diagnosed simultaneously to, prior to, or after the cancer of interest.
- In the case of multiple tumors, a patient may have more than one cancer flagged as an index cancer.
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Analytic Notes

- The **Is this the Index Cancer?** flag determines which data is collected for the cancer diagnosis; see Appendix 5.
 - If the value is “0, No” then the PRISMM Data elements may not be collected for the cancer. See Appendix 5: Multiple Tumors and the Index Cancer.
 - If the value is “1, Yes” then the PRISMM Data elements will be collected for the cancer.
 - The **Curated Date of Diagnosis** of the Index Cancer [**Is this the Index Cancer?** =Yes] determines the data curation timeframe of the PRISMM Pathology, Imaging, and Medical Oncology data elements.

Shared: Yes

Curated Date of Diagnosis

[ca_dx_dt] PN0086

Value

Date field, mmddyyyy

Directives

- Required for every cancer diagnosis.
- Regardless of whether data is piped in from the Tumor Registry or not, the **Curated Date of Diagnosis** should be collected.
- The **Curated Date of Initial Diagnosis** should occur prior to the administration of adjuvant, or palliative therapy.
- The **Curated Date of Initial Diagnosis** should be recorded as the first known definitive pathologic evidence of disease; use the date of the procedure. It is possible that a distant metastatic site is found and biopsied before the primary site, in which case the date of this pathologic finding should be recorded as the **Date of Initial Diagnosis**.
- The date of a cytology procedure resulting in “suspicious” of cancer should not be recorded as the date of diagnosis.
- If a pathology report is not confirmatory for the origin of the cancer or mentions possibility of multiple primary cancers but if the corresponding MD notes confirms it to be lung cancer, then the procedure date of pathology

report can be considered for the cancer date of diagnosis.

- If an external path report is positive for cancer but the review report of the same procedure at the internal institution is negative for cancer, then that pathology report cannot be considered for cancer diagnosis date.
- If the diagnosis date based on pathology is not available, use the date of an imaging scan indicating cancer recording evidence of cancer.
- If neither a date based on pathology or an imaging is available, code the date of diagnosis as provided by the medical oncologist.
- If the actual date of diagnosis is not known, use the date estimation rules.

Analytic Notes

- The **Curated Date of Diagnosis** of the Index Cancer [**Is this the Index Cancer?** =Yes] determines the data curation timeframe of the PRISSMM Pathology, Imaging, and Medical Oncology data.
- The creation of the intervals in days for the cancer-related outcomes, including disease-free survival and progression-free survival, should use **Curated Date of Diagnosis, Interval** which is based on the **Curated Date of Diagnosis**.

Data Standard

NAACCR #390/COC

Shared Field: No, contains PHI. The data element is flagged as an identifier in REDCap. Instead, two calculated intervals one in years and one in days [**ca_age** and **ca_dx_int**] from **Calculated Date of Birth** are generated.

Was the date of diagnosis estimated?

[**ca_dx_dt_est**] PN1309

Value

- 1, Yes
- 0, No

Directives

- Indicate whether any part of the date of diagnosis was estimated.

Data Standard

PRISSMM

Shared Field: Yes

History: 11062023 field added to dictionary

Which part of the date was estimated?

[**ca_dx_dt_est_det**] PN1340

Value

- 1, Day only
- 2, Day and month
- 3, Day, month, and year

Directives

- Linked to **Was the date of diagnosis estimated?** If “1, Yes” is recorded, this field is available for curation.
- If the date of diagnosis was estimated, select which part(s) of the date was estimated.

Data Standard

PRISSMM

Shared Field: Yes

History: 11062023 field added to dictionary

Record the source of the curated diagnosis date

[ca_dx_how] PN0089

Values

- 1, Pathology
- 2, Imaging
- 3, Physical Exam
- 4, Other

Directives

- Linked to **Curated Date of Diagnosis**.
- Choose the option that corresponds to the source of the date of diagnosis. Please note, that there is a preference to record the first date that the tumor was pathologically confirmed on a pathology report. If the pathology report is available and reviewed in order to record “1, Pathology.”
- Choose “Other” if the diagnosis date provided in a provider’s clinic note even if the provider refers to source documentation (like a pathology report).
- Complete this field whether or not the diagnosis information is piped in from the Tumor Registry.
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Analytic Notes

- Due to separate data coming from TR and curation, a combined field should be created to facilitate analyses.

Data Standard

Maps to NAACCR #490

Shared Field: Yes

Is this a hematopoietic or lymphoid neoplasm or pre-malignancy?

[ca_heme_malig] PN0090

Values

- 1, Yes
- 0, No

Description

- This field is used to for the identification of tumor types for diagnoses that do not have TR data piped into REDCap.

Directives

- Code “1, Yes” if this diagnosis is a hematologic or lymphoid malignancy such as leukemia, lymphoma, or myeloma, including myeloproliferative neoplasms (MPNs).
- Code “0, No” for solid tumors.
- Do not include monoclonal gammopathy of undetermined significance (MGUS).
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Analytic Notes

- Due to separate data coming from TR and curation, a combined field should be created to facilitate analyses.

Shared Field: Yes

Type of Hematopoietic Lymphoid Neoplasm

[ca_heme_type] PN0091

Values

1, Leukemia NOS | 35, Acute Lymphoblastic Leukemia | 40, Acute Myeloid Leukemia | 45, Chronic Lymphocytic Leukemia | 50, Chronic Myelogenous Leukemia | 5, Non-Hodgkin's Lymphoma | 55, Hodgkin's Lymphoma | 15, MDS Myelodysplastic Syndrome or Myeloproliferative Neoplasm | 20, MGUS Monoclonal Gammopathy of undetermined significance | 25, Multiple Myeloma | 30, Plasmacytoma | 100, Other Hematopoietic or lymphoid neoplasm | 110, Unspecified

Description

- This field is used to for the identification of tumor types for diagnoses that do not have TR data piped into REDCap.

Directives

- If the TR data is piped in for this cancer diagnosis, this data field will not be available for curation.
- Linked to **Is this a hematopoietic or lymphoid neoplasm or pre-malignancy?** If “1, Yes” is recorded, this field is available for curation.
- Choose the response that most specifically describes the diagnosis. Types of each hematopoietic lymphoid neoplasms are listed in the Table below.
- If the type of malignancy is not specified, code “6, Unspecified.”
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Table: Types of Hematopoietic Lymphoid Neoplasms

Choice	Specific Diagnosis
Lymphoma	AIDS-Related Lymphoma
Lymphoma	Chronic Lymphocytic Leukemia/Small Lymphocytic Lymphoma (NHL)
Lymphoma	Burkitt Lymphoma (NHL)
Lymphoma	Cutaneous T-Cell Lymphoma
Lymphoma	Cutaneous T-Cell Lymphoma – Mycosis Fungoides
Lymphoma	Cutaneous T-Cell Lymphoma – Sézary Syndrome
Lymphoma	Hodgkin Lymphoma
Lymphoma	Follicular Lymphoma (NHL)
Lymphoma	Mantle Cell Lymphoma (NHL)
Lymphoma	Marginal Zone Lymphoma (NHL)
Lymphoma	Non-Hodgkin Lymphoma (NHL)
Lymphoma	Primary CNS Lymphoma (Lymphoma)
Myelodysplastic/Myeloproliferative	Myelodysplastic Syndromes
Multiple Myeloma/Plasma Cell Neoplasms	Multiple Myeloma
Plasma Cell/Multiple Myeloma Neoplasms	Plasma Cell Neoplasms
Other hematopoietic or lymphoid neoplasm	Myelodysplastic Syndromes
Other hematopoietic or lymphoid neoplasm	Myeloproliferative Neoplasm (MPN)

Data Standard

Analytic Notes

- Due to separate data coming from TR and curation, a combined field should be created to facilitate analyses.

Shared Field: Yes

Record the Type of Cancer

[ca_type] PN0092

Values

1, Adrenocortical Carcinoma|3, Anal Cancer|5, Appendix Cancer|6, Bile Duct Cancer|7, Bladder Cancer|11, Brain Cancer|13, Breast Cancer|14, Breast Sarcoma|15, NET or Carcinoid|17, Cervical Cancer|97, Corpus Uteri Carcinoma and Carcinosarcoma|99, Corpus Uteri Sarcoma|19, Colon Cancer|20, Colon/Rectum Cancer|21, Esophagus Cancer|201, Ewing Sarcoma|23, Fallopian Tube Cancer|25, Gallbladder Cancer|205, Germ Cell Tumor|27, GIST|29, Head and Neck Cancer|30, Mesothelioma|35, Ill Defined/Cancer of Unknown Primary|39, Liver Cancer|41, Lung Cancer, NOS|45, Melanoma|47, Merkel Cell|211, Neuroblastoma|51, Non Small Cell Lung Cancer|215, Osteosarcoma|55, Ovarian Cancer|57, Pancreatic Cancer|58, Parathyroid Cancer|59, Penis Cancer|61, Peritoneum Cancer|63, Placenta Cancer|65, Prostate Cancer|67, Rectum and Rectosigmoid Cancer|69, Renal Kidney Cancer|71, Renal Pelvis Cancer|221, Retinoblastoma|225, Rhabdomyosarcoma|73, Scrotum Cancer|75, Small Cell Lung Cancer|77, Small Intestine Cancer|95, Soft tissue sarcoma |81, Stomach Cancer|83, Testis Cancer|85, Thymus Cancer|87, Thyroid Cancer|89, Uterus Cancer NOS|91, Vagina Cancer|93, Vulva Cancer|231, Wilms Tumor|301, Leukemia|303, Non Hodgkins Lymphoma |305, Hodgkins Lymphoma|307, Multiple Myeloma |309, Other hematopoietic or lymphoid neoplasm|1000, Other|1010, Not stated|1020, Unknown

Directives

- Linked to **Is this a hematopoietic or lymphoid neoplasm or pre-malignancy?** If “0, No” is recorded, this field is available for curation.
- Choose the cancer type that most specifically describes the cancer diagnosis. This is most often found in the diagnostic pathology report and can be confirmed in a medical oncologist’s note.
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Analytic Notes

- In the final data set, the curated **(Record the) Type of Cancer** and the derived cancer type from the Tumor Registry data based on the ICD-O-3 Site and Histology should be harmonized to create a composite Cancer Type data element.

Data Standard: Mapped to AJCC Collaborative Stage, v2.05, AJCC v8, NCI/SEER

Analytic Notes

- Due to separate data coming from TR and curation, a combined field should be created to facilitate analyses.

Shared Field: Yes

Cancer Type Brain

[ca_type_brain] PN0093

Values

11, Diffuse astrocytic and oligodendroglial tumors | 13, Other astrocytic tumors | 4, Ependymal tumors | 15, Other gliomas | 17, Choroid plexus tumors | 19, Neuronal and mixed neuronal-gilial tumors | 21, Embryonal tumors | 7, Meningiomas | 27, Mesenchymal non meningotheial tumors | 23, Tumors of cranial and paraspinal nerves | 25, Tumors of the pineal region | 29, Tumors of the sellar region | 31, Germ cell tumors | 9, Other | 99, Unknown/Not Stated

Directives

- Linked to **Record the Type of Cancer**. If “11, Brain” was chosen, this field is available for curation.
- Enter the type of brain cancer, if known. If not known, record “99, Unknown/Not stated.”

Data Standard: WHO CNS Tumors

History

02252020: Additional codes added to comply with the WHO brain/CNS cancer types.

Shared Field: Yes

Cancer Type Other

[ca_type_oth] PN0094

Values

Free text

Description

- Unique text entries of data elements will be added to the list of cancers diagnoses in **Record the Type of Cancer**.

Directives

- Linked to **Record the Type of Cancer**. If “1000, Other” was chosen, this field is available for curation.
- Enter the cancer type here.

Data Standard

Shared Field: No

Cancer Primary Site ICD-O-3

[ca_site] PN0095

Values

ICD-O-3 Site/Topography code list, *See Appendix 7a*

Directives

- Linked to **Is this a hematopoietic or lymphoid neoplasm or pre-malignancy?** If “No” is recorded, this field is available for curation.
- Record the primary site of the tumor. For example, if the patient has Colon Cancer, and the tumor is located in the ascending colon, record “C18.2 Ascending colon.” If the specific location is not available, use the more general code; in the case of cancer of the colon, code “C18.9 Colon NOS.”
- Do not record a regional or distant metastatic site. For example, if a liver site was biopsied first that was consistent with colon cancer and later the ascending colon was biopsied, record “C18.2 Ascending colon.”

Analytic Notes

- Due to separate data items coming from TR and curation, a combined data element for the site of primary cancer should be created to facilitate analyses.

Data Standard

NAACCR # 521: ICD-O-3 Primary Site/Topography Codes

History

03032020: removed leading zeros

Shared Field: Yes

Pathologic Grade or Differentiation of Tumor

[ca_grade_diff] PN0096

Values

- 1, I
- 2, II
- 3, III
- 4, IV
- 9, Grade/differentiation unknown, not stated, or not applicable
- 10, Low Grade
- 11, High Grade

Directives

- Linked to **Is this a hematopoietic or lymphoid neoplasm or pre-malignancy?** If “No” is recorded, this field is available for curation.
OR
- Linked to **Tumor Registry Grade Code**. If this field is blank or if “8” or “9” is recorded, this field is available for curation.
- Record the pathologic grade or degree of differentiation of the primary tumor. Grade or Differentiation for solid tumors should be coded using the NAACCR rules:
- Pathologists describe the tumor grade using three systems or formats:
 1. Two levels of similarity; also called a two-grade system
 2. Three levels of similarity; also called a three-grade system (code according to “Coding for solid tumors.”)
 3. Four levels of similarity; also called a four-grade system. The four-grade system describes the tumor as
 - b. Grade I; also called well-differentiated
 - c. Grade II; also called moderately differentiated
 - d. Grade III; also called poorly differentiated
 - e. Grade IV; also called undifferentiated or anaplastic
- Do NOT code grade based on a metastatic site or recurrence.
- Since the tumor’s grade can be changed by systemic therapy, it is important to record grade prior to drug (or radiation) therapy. If grade is unknown prior to neoadjuvant therapy, code ‘Grade/differentiation unknown, not stated, or not applicable’.
- If grade is available in external path report from the time of diagnosis but not mentioned in the internal review of the external pathology, curate the grade from the external path report.
- If grade differs between the biopsy and resection pathology report (and the resection happens prior to systemic therapy, consider the grade from resection report only. The resection pathology report gets preference over a biopsy or a Medical Oncology visit note.
- If two grades are indicated, choose the highest grade.
- Additional information is located in Appendix 6: Coding Tumor Grade.
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Analytic Note

- Due to separate data items coming from TR and curation, a data element with the curated **Grade or Differentiation of the Tumor** should be combined with the **Tumor Registry: Grade** to make a composite Grade at Diagnosis data element.

Data Standard

NAACCR #440

Shared Field: Yes

Was the patient’s cancer diagnosed as stage IV?

[ca_stage_iv] PN0097

Values

- 1, Yes
- 0, No
- 2, Not Applicable
- 9, Unknown

Description

- This field is used to for staging of solid tumors that do not have TR data piped into REDCap. It is required for all solid tumors without piped TR data.

Directives

- Linked to **Is this a hematopoietic or lymphoid neoplasm or pre-malignancy?** If “No” is recorded, this field is available for curation.
OR
- Linked to **TR: AJCC Best Stage**. If this field is blank or if “99”, or “88” is recorded, this field is available for curation.
- Choose “1, Yes” for patients with Stage IV disease. This may also be described as metastatic at diagnosis, extensive stage, or M1 using the AJCC v7 for cases diagnosed prior to January 1, 2018 and AJCC v8 for cases diagnosed on or after January 1, 2018.
- Choose “0, No” for patients with Stage 0, I, II, or III disease. This may also be called early stage, limited stage, or M0 using the AJCC v7 for cases diagnosed prior to January 1, 2018 and AJCC v8 for cases diagnosed on or after January 1, 2018.
- If a cancer is diagnosed in the past at an external institution and no information about stage is available, code “2, Not Applicable/Unknown.”
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Provenance

- Internal or external medical oncologist note with preference for the internal medical oncologist, internal or external pathology reports, and internal and external imaging reports.

Analytic Note

- In the final data set, a derived data element matching this field will be created by harmonizing staging data from the Tumor Registry and curated staging data.

Data Standard

NAACCR #900, 960, 1003, 1013

Shared Field: Yes

Curated Group Stage at Initial Diagnosis

[ca_stage] PN0098

Values

0, 0 | 17, 0A | 18, 0is | 1, I | 2, IA | 19, IA1 | 20, IA2 | 21, IA3 | 3, IB | 4, IC | 5, II | 6, IIA | 7, IIB | 8, IIC | 9, III | 10, IIIA | 11, IIIB | 12, IIIC | 13, IV | 14, IVA | 15, IVB | 16, IVC | 98, Not Applicable | 99, Unknown

Description

- This field is used to for the staging for diagnoses that do not have TR data piped into REDCap.
- If the case has TR data, this field will be blank.

Directives

- If the TR data is piped in for this cancer diagnosis, this data field will not be available for curation.

- Linked to **Is this a hematopoietic or lymphoid neoplasm or pre-malignancy?** If “No” is recorded, this field is available for curation.
OR
- Linked to **TR: AJCC Best Stage**. If this field is blank or if “99”, or “88” is recorded, this field is available for curation.
- Record the group stage at diagnosis based on the medical oncologist’s notes. For patients who are diagnosed at an external institution, use the native medical oncologist’s consult note.
- Choose the stage of the cancer at the time of diagnosis.
- If unknown, code “99, Unknown.”
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Provenance

- Internal or external medical oncologist note with preference for the internal medical oncologist.

Analytic Note

- In the final data set, the **Tumor Registry: AJCC Best Stage** and the **Curated Group Stage at Initial Diagnosis** may be harmonized to create a derived group stage data.

Data Standard

Metriq AJCC Best Stage based on AJCC components

Shared Field: Yes

Clinical Group Stage

[ca_clin_group_stage] PN1305

Values

0, 0 | 17, 0A | 18, 0is | 1, I | 2, IA | 19, IA1 | 20, IA2 | 21, IA3 | 3, IB | 4, IC | 5, II | 6, IIA | 7, IIB | 8, IIC | 9, III | 10, IIIA | 11, IIIB | 12, IIIC | 13, IV | 14, IVA | 15, IVB | 16, IVC | 98, Not Applicable | 99, Unknown

Description

- This field is used to for the staging for diagnoses that do not have TR data piped into REDCap.
- If the case has TR data, this field will be blank.

Directives

- If the TR data is piped in for this cancer diagnosis, this data field will not be available for curation.
- Linked to **Is this a hematopoietic or lymphoid neoplasm or pre-malignancy?** If “No” is recorded, this field is available for curation.
OR
- Linked to **TR: AJCC Best Stage**. If this field is blank or if “99”, or “88” is recorded, this field is available for curation.
- Record the clinical group stage at diagnosis based on the medical oncologist’s notes. For patients who are diagnosed at an external institution, use the native medical oncologist’s consult note.
- Clinical stage is assigned using information gathered during the diagnostic workup from the time of initial diagnosis until the initiation of primary treatment or within 4 months of cancer diagnosis, or until the time of cancer progression.
- If unknown, code “99, Unknown.”
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Provenance

- Internal or external medical oncologist note with preference for the internal medical oncologist.

Analytic Note

- In the final data set, the Tumor Registry and curated clinical staging components may be harmonized to create a derived staging data elements.

Data Standard

NAACCR-Mappable #970

Shared Field: Yes

History: 11062023 field added to dictionary

[Clinical T Stage](#), [Clinical T1 Stage Detail](#), [Clinical T2 Stage Detail](#), [Clinical T3 Stage Detail](#), [Clinical T4 Stage Detail](#)
Values: Clinical T Stage [\[ca_clin_t_stage\]](#) PN0099

1, TX | 2, T0 | 36, Ta | 40, Tis | 8, T1 | 17, T2 | 24, T3 | 29, T4 | 98, Not Applicable | 99, Unknown

Values: Clinical Tis Stage Detail [\[ca_path_tis_det\]](#) PN0980

41, Tis LCIS | 42, Tis DCIS | 43, Tis Paget's

Values: Clinical T1 Stage Detail [\[ca_clin_t1_det\]](#) PN0100

7, T1mic | 9, T1a | 10, T1a1 | 11, T1a2 | 12, T1b | 13, T1b1 | 14, T1b2 | 15, T1c | 16, T1d | 98, Not Applicable | 99, Unknown

Values: Clinical T2 Stage Detail [\[ca_clin_t2_det\]](#) PN0101

18, T2a | 19, T2a1 | 20, T2a2 | 21, T2b | 22, T2c | 23, T2d | 98, Not Applicable | 99, Unknown

Values: Clinical T3 Stage Detail [\[ca_clin_t3_det\]](#) PN0102

25, T3a | 26, T3b | 27, T3c | 28, T3d | 29, T3e | 98, Not Applicable | 99, Unknown

Values: Clinical T4 Stage Detail [\[ca_clin_t4_det\]](#) PN0103

30, T4a | 31, T4b | 32, T4c | 33, T4d | 34, T4e | 98, Not Applicable | 99, Unknown

Description

- These fields are used to for staging of solid tumors that do not have TR data piped into REDCap.
- If the case has TR data, these fields will be blank.

Directives

- If the TR staging data is piped in for this cancer diagnosis, these data fields will not be available for curation.
- Linked to **Was the patient's cancer diagnosed as stage IV?** If "No" was selected, these data fields are available for curation.
- For each level of Clinical T Staging, select the most appropriate T stage using the AJCC v7 for cases diagnosed prior to January 1, 2018 and AJCC v8 for cases diagnosed on or after January 1, 2018.
- If any are not known or unavailable, code "99, Unknown."
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Provenance

- Native and external Imaging reports and biopsy pathology reports and physical exams

Analytic Note

- In the final data set, the Tumor Registry and curated clinical staging components may be harmonized to create a derived staging data elements.

Data Standard

NAACCR-Mappable #940

Shared Fields: Yes

Clinical N Stage, Clinical N1 Stage Detail, Clinical N2 Stage Detail, Clinical N3 Stage Detail

Values: Clinical N Stage [ca_clin_n_stage] PN0104

1, NX | 2, NO | 5, N1 | 9, N2 | 13, N3 | 17, N4 | 18, N+ | 98, Not Applicable | 99, Unknown

Values: Clinical N1 Stage Detail [ca_clin_n1_det] PN0105

6, N1a | 7, N1b | 8, N1c | 98, Not Applicable | 99, Unknown

Values: Clinical N2 Stage Detail [ca_clin_n2_det] PN0106

10, N2a | 11, N2b | 12, N2c | 98, Not Applicable | 99, Unknown

Values: Clinical N3 Stage Detail [ca_clin_n3_det] PN0107

14, N3a | 15, N3b | 16, N3c | 98, Not Applicable | 99, Unknown

Description

- This field is used to for staging of solid tumors that do not have TR data piped into REDCap for the clinical N stage.
- If the case has TR data, these fields will be blank.

Directives

- If the TR staging data is piped in for this cancer diagnosis, these data fields will not be available for curation.
- Linked to **Was the patient's cancer diagnosed as stage IV?** If "No" was selected, these data fields are available for curation.
- For each level of Clinical N Staging, select the most appropriate N (nodal) stage. N+ can be curated if regional lymph node involvement is documented without a specific N stage.
- If unknown, code "99, Unknown."
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Provenance

- Native and external biopsy or imaging reports, if available.

Analytic Note

- In the final data set, the Tumor Registry and curated clinical staging components will be harmonized to create a derived stage at diagnosis data element across all cases.

Data Standard

NAACCR #950

Shared Fields: Yes

Pathologic Staging Data Items

The following data elements are based on a primary tumor resection. If a primary resection has not been performed,

record "Not Applicable."

If it is unknown whether the patient had a resection, record "Unknown."

Did the patient receive neoadjuvant chemotherapy or radiation therapy before pathological stage diagnosis? [ca_tx_pre_path_stage] PN0108

Values

- 1, Yes
- 0, No
- 98, Not Applicable
- 99, Unknown

Description

- This field is used to for staging of solid tumors that do not have TR data piped into REDCap.
- If the case has TR data, this field will be blank.

Directives

- If the TR staging data is piped in for this cancer diagnosis, this data field will not be available for curation.
- Linked to **Was the patient's cancer diagnosed as stage IV?** If "No" was selected, this data field is available for curation.
- Record whether the patient received neoadjuvant chemotherapy, which is therapy received prior to the excision of the primary cancer with purpose of shrinking the tumor.
- If the pathology report for the primary tumor excision contains a 'y' this indicates that the patient received neoadjuvant chemotherapy; code "1, Yes."
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.
- If the patient did not have a resection performed as part of primary treatment for the cancer diagnosis, record "not applicable".

Data Standard

NAACCR-Mappable #920

Shared Field: Yes

Pathologic Group Stage [ca_path_group_stage] PN0109

Values

0, 0 | 1, 0A | 2, 0is | 3, I | 4, IA | 5, IA1 | 6, IA2 | 31, IA3 | 7, IB | 8, IB1 | 9, IB2 | 10, IC | 11, IS | 12, II | 13, IIA | 14, IIA1 | 15, IIA2 | 16, IIB | 17, IIC | 18, III | 19, IIIA | 20, IIIB | 21, IIIC | 22, IIIC1 | 23, IIIC2 | 24, IV | 25, IVA | 26, IVA1 | 27, IVA2 | 28, IVB | 29, IVC | 30, Occult | 98, Not Applicable | 99, Unknown

Description

- This field is used to for staging of solid tumors that do not have TR data piped into REDCap.
- If the case has TR data, this field will be blank.

Directives

- If the TR staging data is piped in for this cancer diagnosis, these data fields will not be available for curation.
- Linked to **Was the patient's cancer diagnosed as stage IV?** If "No" was selected, these data fields are available for curation.
- Select the appropriate Pathologic Group Stage based on surgical evidence.

- If any are unknown, code “99, Unknown.”
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Analytic Note

- In the final data set, the Tumor Registry and curated pathologic staging components will be harmonized to create a derived stage at diagnosis data element across all cases.

Data Standard

NAACCR-Mappable #920

Shared Field: Yes

Pathologic T Stage, Pathologic T1 Stage Detail, Pathologic T2 Stage Detail, Pathologic T3 Stage Detail, Pathologic T4 Stage Detail

Values: Pathologic T Stage [\[ca_path_t_stage\]](#) PN0110

1, TX | 2, T0 | 36, Ta | 40, Tis | 8, T1 | 17, T2 | 24, T3 | 29, T4 | 98, Not Applicable | 99, Unknown

Values: Pathologic Tis Stage Detail [\[ca_path_tis_det\]](#) PN0981

41, Tis LCIS | 42, Tis DCIS | 43, Tis Paget's

Values: Pathologic T1 Stage Detail [\[ca_path_t1_det\]](#) PN0111

7, T1mic | 9, T1a | 10, T1a1 | 11, T1a2 | 12, T1b | 13, T1b1 | 14, T1b2 | 15, T1c | 16, T1d | 98, Not Applicable | 99, Unknown

Values: Pathologic T2 Stage Detail [\[ca_path_t2_det\]](#) PN0112

18, T2a | 19, T2a1 | 20, T2a2 | 21, T2b | 22, T2c | 23, T2d | 98, Not Applicable | 99, Unknown

Values: Pathologic T3 Stage Detail [\[ca_path_t3_det\]](#) PN0113

25, T3a | 26, T3b | 27, T3c | 28, T3d | 29, T4e | 98, Not Applicable | 99, Unknown

Values: Pathologic T4 Stage Detail [\[ca_path_t4_det\]](#) PN0114

30, T4a | 31, T4b | 32, T4c | 33, T4d | 34, T4e | 98, Not Applicable | 99, Unknown

Description

- These fields are used to for staging of solid tumors that do not have TR data piped into REDCap.
- If the case has TR data, these fields will be blank.

Directives

- If the TR staging data is piped in for this cancer diagnosis, these data fields will not be available for curation.
- Linked to **Was the patient's cancer diagnosed as stage IV?** If “No” was selected, these data fields are available for curation.
- Pathologic stage components are often available on the pathology reports from the primary tumor's resection.
- For each level of Pathologic T Staging, select the most appropriate T stage.
- If any are unknown, code “99, Unknown.”
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Provenance

- Native and external biopsy and resection pathology reports, if available.

History: Pathologic Tis Stage Detail added 07062020

Analytic Note

- In the final data set, the Tumor Registry and curated pathologic staging components should be harmonized to create a derived stage at diagnosis data element across all cases.

Data Standard

Mappable #880

Shared Fields: Yes

[Pathologic N Stage](#), [Pathologic N1 Stage Detail](#), [Pathologic N2 Stage Detail](#), [Pathologic N3 Stage Detail](#)

Values: Pathologic N Stage [\[ca_path_n_stage\]](#) PN0115

1, NX | 2, N0 | 5, N1 | 9, N2 | 13, N3 | 17, N4 | 98, Not Applicable | 99, Unknown

Values: Pathologic N0 Stage Detail [\[ca_path_n0_det\]](#) PN1040

21, N0(i-) | 22, N0(i+) | 23, N0(mol-) | 23, N0(mol+) | 98, Not Applicable | 99, Unknown

Values: Pathologic N1 Stage Detail [\[ca_path_n1_det\]](#) PN0116

6, N1a | 7, N1b | 8, N1c | 98, Not Applicable | 99, Unknown

Values: Pathologic N2 Stage Detail [\[ca_path_n2_det\]](#) PN0117

10, N2a | 11, N2b | 12, N2c | 98, Not Applicable | 99, Unknown

Values: Pathologic N3 Stage Detail [\[ca_path_n3_det\]](#) PN0118

14, N3a | 15, N3b | 16, N3c | 98, Not Applicable | 99, Unknown

Description

- These fields are used to for staging of solid tumors that do not have TR data piped into REDCap.
- If the case has TR data, these fields will be blank.

Directives

- If the TR data is piped in for this cancer diagnosis, these data fields will not be available for curation.
- Linked to **Was the patient's cancer diagnosed as stage IV?** If "0, No" was selected, these data field is available for curation.
- For each level of Pathologic N Staging, select the most appropriate N stage.
- If any are unknown, code "99, Unknown."
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Provenance

- Native and external biopsy and resection pathology reports, if available.

History: N0 details added 07062020

Analytic Note

- In the final data set, the Tumor Registry and curated pathologic staging components should be harmonized to create a derived stage at diagnosis data element across all cases.

Data Standard

Mappable #890

Shared Field: Yes

TNM Edition Number

[ca_tnm_edition_num] PN1164

Values

1, First Edition | 2, Second Edition | 3, Third Edition | 4, Fourth Edition recommended for use for cases diagnosed 1993-1997 | 5, Fifth Edition recommended for use for cases diagnosed 1998-2002 | 6, Sixth Edition recommended for use for cases diagnosed 2003-2009 | 7, Seventh Edition recommended for use with cases diagnosed 2010-2017 | 8, Eighth Edition recommended for use with cases diagnosed 2018+ for which a later edition has not yet been released | 9, Ninth Edition recommended for use with chapter-specific cases diagnosed 2021+ | 88, Not applicable | 99, Edition Unknown

Directives

- Linked to **TR: TNM Edition Number**. If this field is blank, this field is available for curation.
- Select the AJCC Edition Number used to stage the cancer. If the edition number is not explicitly stated, check to see if the staging information in the patient's chart is consistent with the AJCC Edition in use at the time of the patient's diagnosis. If so, record the edition number based on the date of diagnosis; if not, curate "99, Edition Unknown".

Provenance

- Native and external resection pathology reports or medical oncologist notes.

Data Standard

NAACCR #1060

History

070402021: Field added to dictionary.

Shared Field: Yes

General Summary Stage

[ca_gen_sum_stage_2] PN0119

Values

9, Unstaged | 0, In situ | 1, Localized | 2, Regional direct extension only | 3, Regional lymph nodes only | 4, Regional direct extension and regional lymph nodes | 5, Regional NOS | 41, Advanced | 7, Distant | 8, Not applicable

Directives

- Linked to Type of Cancer: data field available for the following cancers: 11, Brain Cancer, 57, Pancreatic Cancer, 211, Neuroblastoma, 201, Ewing Sarcoma, 221, Retinoblastoma, 225, Rhabdomyosarcoma, and 231, Wilms Tumor.
- In some instances, the stage at diagnosis may not be recorded using the AJCC staging convention, instead the SEER Summary Stage may be used. If this is the case, record the stage.
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Analytic Note

- In the final data set, the Tumor Registry and curated pathologic staging components should be harmonized to create a derived stage at diagnosis data element across all cases.

Data Standard

SEER Summary Stage 2000, NAACCR #759

Shared Field: Yes

Cancer-Specific Diagnosis Data Fields

These data fields will only be visible for the diagnosis of interest that you are currently curating. The Cancer-Specific Data are to be collected during the initial diagnosis, work up, and first course of treatment. Some of the data fields have specific instructions as to when to be collected (See guidelines below for each cancer type.)

Most of the cancer-specific data elements will be found in pathology reports, especially pathology reports from the resection. Use both native and external reports to complete.

These data fields should be completed; if the data cannot be found in the EHR, code “999, Unknown” or similar field.

Breast Cancer Site-Specific Data Fields

These data fields are specific to breast cancer only. To answer these questions, use information from the diagnosis of the breast cancer through completion of surgery(ies) in the first course of treatment.

BCA: Estrogen Receptor (ER) Status (CS SSF1)

[ca_bca_er] PN0120

Values

Blank for other cancer diagnoses

- 10, Positive/elevated
- 20, Negative/normal
- 30, Borderline
- 988, Not applicable
- 996, Test ordered/results not interpretable
- 997, Test ordered/results not in chart
- 998, Test not done
- 999, Unknown or no information

Directives

- This data field is available in REDCap for invasive breast cancer diagnoses only.
- Record the percent of primary tumor cells staining positive for estrogen receptor. Do not leave blank.
- The result of an ER test on the breast primary is preferred for this field. Results from nodes or metastases may be used ONLY if there is no evidence of primary tumor.
- In the case that ER is reported on more than one specimen from a single primary, and any specimen is positive, record positive.
- In the case where there are multiple tumors with different ER results, code the results from the largest tumor size.
- Record the value for specimens prior to neoadjuvant therapy. If there are no results prior to neoadjuvant treatment, code the results from a post-treatment specimen.
- If ER is negative or percent is less than 1% code ‘20. Negative/Normal’.
- If the ER status is unknown or documentation is not available, code “999, Unknown.”
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Provenance

- Give preference to the native or external pathology report during diagnosis. If these are not available, use the medical oncologist’s report.

Analytic Note

- This data field for other cancer diagnoses will be blank on data export.

Data Standard

NAACCR #2880: For breast cancer only. Conforms to NAACCR site-specific factor #3827.

Shared Field: Yes

BCA: Progesterone Receptor (PR) Status (CS SSF2)

[ca_bca_pr] PN0121

Values

Blank for other cancer diagnoses

- 10, Positive/elevated
- 20, Negative/normal
- 30, Borderline
- 988, Not applicable
- 996, Test ordered/results not interpretable
- 997, Test ordered/results not in chart
- 998, Test not done
- 999, Unknown or no information

Directives

- This data field is available in REDCap for invasive breast cancer diagnoses only.
- Record the percent of primary tumor cells staining positive for progesterone receptor. Do not leave blank.
- The result of an PR test on the breast primary is preferred for this field. Results from nodes or metastases may be used ONLY if there is no evidence of primary tumor.
- In the case that PR is reported on more than one specimen from a single primary, and any specimen is positive, record positive.
- In the case where there are multiple tumors with different PR results, code the results from the largest tumor size.
- Record the value for specimens prior to neoadjuvant therapy. If there are no results prior to neoadjuvant treatment, code the results from a post-treatment resection.
- If PR is negative or percent is less than 1% code '20. Negative/Normal'.
- If the PR status is unknown or documentation is not available, code "999, Unknown."
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Provenance

- Give preference to the native or external pathology report during diagnosis. If these are not available, use the medical oncologist's report.

Analytic Note

- This data field for other cancer diagnoses will be blank on data export.

Data Standard

NAACCR #2890: Conforms to NAACCR site-specific factor.

Shared Field: Yes

BCA: HER2 Immunohistochemistry (IHC) Lab Value (CS SSF 8)

[ca_bca_her2ihc_val] PN0122

Values

Blank | 0, Score of 0 | 10, Score of 1+ | 20, Score of 2+ | 30, Score of 3+ | 988, Not applicable | 997, Test

ordered/results not in chart | 998, Test not done | 999, Unknown or no information

Directives

- This data field is available in REDCap for invasive breast cancer diagnoses only.
- Record the HER2 IHC (immunohistochemistry) values. An immunohistochemistry (IHC) test identifies the protein expressed by the gene (ERBB2). Do not leave blank.
- The HER2 IHC test performed on the primary breast tissue is to be recorded in this data item.
- In the case that HER2 IHC is reported on more than one specimen from a single primary, and any specimen is positive, record the highest value.
- In the case where there are multiple tumors with different HER2 results, code the results from the largest tumor size.
- Results from nodal or metastatic tissue may be used, ONLY when there is no evidence of primary tumor.
- Record the value for specimens prior to neoadjuvant therapy; however, if neoadjuvant therapy is given and there are no HER2 IHC results from pre-treatment specimens, report the findings from post-treatment specimens.
- If the HER2 status by IHC is unknown or documentation is not available, code “999, Unknown.”
- Do not report the results of a HER2 as part of a multigene test such as OncotypeDX or MammaPrint.
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Reporting Results of HER2 Testing by Immunohistochemistry (IHC)

Result	Criteria
Negative (Score 0)	No staining observed or Incomplete, faint/barely perceptible membrane staining in $\leq 10\%$ of invasive tumor cells
Negative (Score 1+)	Incomplete, faint/barely perceptible membrane staining in $>10\%$ of invasive tumor cells*
Equivocal (Score 2+)	Incomplete and/or weak to moderate circumferential membrane staining in $>10\%$ of invasive tumor cells or Complete, intense, circumferential membrane staining in $\leq 10\%$ of invasive tumor cells*
Positive (Score 3+)	Complete, intense, circumferential membrane stain

Source: <https://www.naacccr.org/SSDI/SSDI-Manual.pdf?v=1527608547>

Provenance

- Give preference to the native or external pathology report during diagnosis. If these are not available, use the medical oncologist’s report.

Analytic Note

- This data field for other cancer diagnoses will be blank on data export.

Data Standard

NAACCR #2862: Conforms to NAACCR site-specific factor.

Shared Field: Yes

BCA: HER2 IHC Lab Interpretation (CS SSF 9)
[ca_bca_her2ihc_intp] PN0123

Values

Blank | 10, Positive/elevated | 20, Negative/normal | 30, Borderline; equivocal; indeterminate; undetermined whether positive or negative | 988, Not applicable | 997, Test ordered/results not in chart | 998, Test not done | 999, Unknown or no information

Directives

- This data field is available in REDCap for invasive breast cancer diagnoses only.
- Record the appropriate HER2 Lab Interpretation at diagnosis. Do not leave blank.
- The HER2 test performed on the primary breast tissue is to be recorded in this data item.
- Results from nodal or metastatic tissue may be used, ONLY when there is no evidence of primary tumor.
- Record the value for specimens prior to neoadjuvant therapy; however, if neoadjuvant therapy is given and there are no HER2 IHC results from pre-treatment specimens, report the findings from post-treatment specimens.
- If unknown, code “999, Unknown.”
- Do not report the results of a HER2 as part of a multigene test such as OncotypeDX or MammaPrint.
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Provenance

- Give preference to the native or external pathology report during diagnosis. If these are not available, use the medical oncologist’s report.

Data Standard

NAACCR #2863: Conforms to NAACCR site-specific factor.

Shared Field: Yes

BCA: HER2 ISH Summary (SSDI)

[ca_bca_her2ish] PN0982

Values

Blank for other cancer diagnoses

- 3, Positive amplified
- 0, Negative not amplified
- 2, Borderline/equivocal/indeterminant
- 7, Test ordered results not in chart
- 9, Not documented in medical record

Directives

- This data field is available in REDCap for invasive breast cancer diagnoses only.
- Record the result of ISH (FISH, CISH, SISH) for HER2 protein at diagnosis. Do not leave blank.
- The HER2 ISH test performed on the primary breast tissue with a priority for the invasive component is to be recorded in this data item.
- Results from nodal or metastatic tissue may be used, ONLY when there is no evidence of primary tumor.
- Record the value for specimens prior to neoadjuvant therapy; however, if neoadjuvant therapy is given and there are no HER2 ISH results from pre-treatment specimens, report the findings from post-treatment specimens.
- If the HER2 ISH is unknown or documentation is not available, code “9, Not documented in medical record.”
- Do not report the results of a HER2 as part of a multigene test such as OncotypeDX or MammaPrint.
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Provenance

- Give preference to the native or external pathology report during diagnosis. If these are not available, use the medical oncologist's report.

History: added 07082020 to v345

Analytic Note

- This data field for other cancer diagnoses will be blank on data export.

Data Standard

NAACCR #2869

Shared Field: Yes

BCA: HER2 Summary Result of Testing (CS SSF 15)

[ca_bca_her2ihc_summ] PN0124

Values

Blank for other cancer diagnoses

- 10, Positive/elevated/amplified
- 20, Negative/normal within normal limits
- 30, Borderline/equivocal/indeterminant
- 988, Not applicable
- 997, Test ordered results not in chart
- 998, Tests not done
- 999, Unknown or no information

Directives

- This data field is available in REDCap for invasive breast cancer diagnoses only.
- Record the summary result for HER2. Do not leave blank.
- The HER2 tests performed on the primary breast tissue is to be recorded in this data item.
- Results from nodal or metastatic tissue may be used, ONLY when there is no evidence of primary tumor.
- Record the value for specimens prior to neoadjuvant therapy; however, if neoadjuvant therapy is given and there are no HER2 results from pre-treatment specimens, report the findings from post-treatment specimens.
- If the HER2 summary status is unknown or documentation is not available, code "999, Unknown."
- Do not report the results of a HER2 as part of a multigene test such as OncotypeDX or MammaPrint.
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Provenance

- Give preference to the native or external pathology report during diagnosis. If these are not available, use the medical oncologist's report.

Analytic Note

- This data field for other cancer diagnoses will be blank on data export.

Data Standard

NAACCR #2869

Shared Field: Yes

BCA: Oncotype Dx Recurrence Score-Invasive (SSDI)

[ca_bca_mgene] PN0125

Values

Blank | 0, 0 | 1, 1 | 2, 2 | 3, 3 | 4, 4 | 5, 5 | 6, 6 | 7, 7 | 8, 8 | 9, 9 | 10, 10 | 11, 11 | 12, 12 | 13, 13 | 14, 14 | 15, 15 | 16, 16 | 17, 17 | 18, 18 | 19, 19 | 20, 20 | 021, 21 | 22, 22 | 23, 23 | 024, 24 | 25, 25 | 26, 26 | 27, 27 | 28, 28 | 29, 29 | 30, 30 | 31, 31 | 32, 32 | 33, 33 | 34, 34 | 35, 35 | 36, 36 | 37, 37 | 38, 28 | 39, 39 | 40, 40 | 41, 41 | 42, 42 | 43, 43 | 44, 44 | 45, 45 | 46, 46 | 47, 47 | 48, 48 | 49, 49 | 50, 50 | 51, 51 | 52, 52 | 53, 53 | 54, 54 | 55, 55 | 56, 56 | 57, 57 | 58, 58 | 59, 59 | 60, 60 | 061, 61 | 62, 62 | 63, 63 | 64, 64 | 65, 65 | 66, 66 | 67, 67 | 68, 68 | 69, 69 | 70, 70 | 71, 71 | 72, 72 | 73, 73 | 74, 74 | 75, 75 | 76, 76 | 77, 77 | 78, 78 | 79, 79 | 80, 80 | 81, 81 | 82, 82 | 83, 83 | 84, 84 | 85, 85 | 086, 86 | 87, 87 | 88, 88 | 89, 89 | 90, 90 | 91, 91 | 92, 92 | 93, 93 | 94, 94 | 95, 95 | 96, 96 | 97, 97 | 98, 98 | 99, 99 | 100, 100 | xx4, Stated as less than 11 | xx5, Stated as equal to or greater than 11 | xx6, Not applicable; in situ case | xx7, Test ordered, results not in chart | xx9, Not documented in medical record

Directives

- This data field is available in invasive REDCap for breast cancer diagnoses only.
- Record the OncoTypeDX Recurrence Score for the invasive cancer; the Oncotype Dx-Invasive recurrence score is reported as a whole number 0-100.
- In the case where Oncotype D is reported on more than one breast tumor, code the highest value.
- If the cancer is diagnosed exclusively as a Stage 0, code “xx6, Not applicable; in situ case.”
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Provenance

- Give preference to the native or external pathology report during diagnosis. If these are not available, use the medical oncologist’s report.

Analytic Note

- This data field for other cancer diagnoses will be blank on data export.

Data Standard

NAACCR #3904

History

01152020: removed leading zeros.

Shared Field: Yes

BCA: Multigene Signature Method (SSDI/ CS SSF 22)

[ca_bca_mgene] PN0126

Values

Blank | 1, Mammaprint | 2, PAM50 (Prosigna) | 3, Breast Cancer Index | 4, EndoPredict | 5, Test performed type of test unknown | 6, Multiple tests any tests in codes 1-4 | 7, Test ordered results not in chart | 8, Not applicable

Directives

- This data field is available in REDCap for invasive breast cancer diagnoses only.
- Record the multigene test used.
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Provenance

- Give preference to the native or external pathology report during diagnosis. If these are not available, use the

medical oncologist's report.

Analytic Note

- This data field for other cancer diagnoses will be blank on data export.

Data Standard

NAACCR #3894. Conforms to SSDI #2876

Shared Field: Yes

BCA: Multigene Signature Results (SSDI & CS SSF 23)

[ca_bca_mgeneresult] PN0127

Values

Blank | X1, Score 100 | X2, Low risk | X3, Moderate risk | X4, High risk | X7, Test ordered results no results | X8, Not applicable | X9, Not documented in medical record

Directives

- This data field is available in REDCap for invasive breast cancer diagnoses only.
- Linked to **BCA: Multigene Signature Method**, if this data field is completed, the results will be available.
- Record the score or risk for the test performed. The same test should be used to record Multigene Signature Method.
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Provenance

- Give preference to the native or external pathology report during diagnosis. If these are not available, use the medical oncologist's report.

Analytic Note

- This data field for other cancer diagnoses will be blank on data export.

Data Standard

NAACCR #3895. Conforms to SSDI #2877.

Shared Field: Yes

Colon Rectal Cancer Site-Specific Data Fields

These questions are specific to diagnoses of Colon/Rectal cancer (CRC) only. To answer these questions, use information from the diagnosis of colon/rectal cancer through completion of surgery(ies) in the first course of treatment.

CRC: Tumor Deposits (CS SSF 4)

[ca_crc_td] PN0128

[naaccr_ca_crc_td] PN1222

Values

Blank | 0, 0 | 1, 1 | 2, 2 | 3, 3 | 4, 4 | 5, 5 | 6, 6 | 7, 7 | 8, 8 | 9, 9 | 10, 10 | 11, 11 | 12, 12 | 13, 13 | 14, 14 | 15, 15 | 16, 16 | 17, 17 | 18, 18 | 19, 19 | 20, 20 | 21, 21 | 22, 22 | 23, 23 | 24, 24 | 25, 25 | 26, 26 | 27, 27 | 28, 28 | 29, 29 | 30, 30 | 31, 31 | 32, 32 | 33, 33 | 34, 34 | 35, 35 | 36, 36 | 37, 37 | 38, 38 | 39, 39 | 40, 40 | 41, 41 | 42, 42 | 43, 43 | 44, 44 | 45, 45 | 46, 46 | 47, 47 | 48, 48 | 49, 49 | 50, 50 | 51, 51 | 52, 52 | 53, 53 | 54, 54 | 55, 55 | 56, 56 | 57, 57 | 58, 58 | 59, 59 | 60, 60 | 61, 61 | 62, 62 | 63, 63 | 64, 64 | 65, 65 | 66, 66 | 67, 67 | 68, 68 | 69, 69 | 70, 70 | 71, 71 | 72, 72 | 73, 73 | 74, 74 | 75, 75 | 76, 76 | 77, 77 | 78, 78 | 79, 79 | 80, 80 | 81, 81 | 82, 82 | 83, 83 | 84, 84 | 85, 85 | 86, 86 | 87, 87 |

88, 88 | 89, 89 | 90, 90 | 91, 91 | 92, 92 | 93, 93 | 94, 94 | 95, 95 | 96, 96 | 97, 97 | 98, 98 | 99, 99 | X1, 100 or more | X2, Tumor Deposits identified, number unknown | X8, Not applicable: Information not collected for this case | X9, Not documented in medical record; Cannot be determined by the pathologist; Pathology report does not mention tumor deposits; No surgical resection done; Tumor Deposits not assessed or unknown if assessed

Description

- A tumor deposit is defined as a discrete nodule of cancer in pericolic/perirectal fat or in adjacent mesentery (mesocolic or rectal fat) within the lymph drainage area of the primary carcinoma, without identifiable lymph node tissue or identifiable vascular structure.
- Synonyms include discontinuous extramural extension, malignant tumor foci, malignant peritumoral deposits, satellite nodule

Directives

- This data field is available in REDCap for colon/rectum cancer diagnoses only. Data from tumor registries can be piped in to field [naaccr_ca__crc_td]; curated data should be entered in field [ca_crc_td] if piped-in data is unavailable or incorrect. .
- Record the number of tumor deposits as recorded in the pathology report from surgical resection of the primary site of cancer.
- Do not count all sites of tumor mentioned; this will be reported explicitly as “tumor deposits.”
- Do not count positive lymph nodes.
- If the patient did not have a resection, it is unknown whether the patient had a resection, or a resection was performed and the pathology report is available for review and tumor deposits are not mentioned, record “X9, Not documented in medical record.”

Provenance

- Use the native or external pathology report during diagnosis; give preference to the native pathology report if these are not available, use the medical oncologist’s report.

Analytic Note

- This data field for cancer diagnoses other than colon or rectal will be blank on data export.

Data Standard

NAACCR #2910. Conforms to NAACCR site-specific data item #3934.

Shared Field: Yes

History

02212023: Per SEER standards: cases diagnosed prior to 2018 were curated as “0” if surgical resection of the primary site was performed, the pathology report was available for review, and tumor deposits were not mentioned. This guideline was followed for BPC Phase 1 Colorectal Cancer cohort.

CRC: Circumferential Resection Margin (CS SSF 6)

[ca_crc_crm_2] PN1220

[naaccr_ca_crc_crm] PN1223

Values

Blank | XX0, 100 mm or greater | XX1, Margins clear distance from tumor not stated | XX2, Margins cannot be assessed | XX3, Described as at least 1 mm | XX4, Described as at least 2 mm | XX5, Described as at least 3 mm | XX6, Described as greater than 3 mm | XX7, No resection of primary site | 0, 0 | 1, 1 | 2, 2 | 3, 3 | 4, 4 | 5, 5 | 6, 6 | 7, 7 | 8, 8 | 9, 9 | 10, 10 | 11, 11 | 12, 12 | 13, 13 | 14, 14 | 15, 15 | 16, 16 | 17, 17 | 18, 18 | 19, 19 | 20, 20 | 21, 21 | 22, 22 | 23, 23 | 24, 24 | 25, 25 | 26, 26 | 27, 27 | 28, 28 | 29, 29 | 30, 30 | 31, 31 | 32, 32 | 33, 33 | 34, 34 | 35, 35 | 36, 36 | 37, 37 | 38, 38 | 39, 39 | 40, 40 | 41, 41 | 42, 42 | 43, 43 | 44, 44 | 45, 45 | 46, 46 | 47, 47 | 48, 48 | 49, 49 | 50, 50 | 51, 51 | 52, 52 | 53, 53

| 54, 54 | 55, 55 | 56, 56 | 57, 57 | 58, 58 | 59, 59 | 60, 60 | 61, 61 | 62, 62 | 63, 63 | 64, 64 | 65, 65 | 66, 66 | 67, 67 | 68, 68 | 69, 69 | 70, 70 | 71, 71 | 72, 72 | 73, 73 | 74, 74 | 75, 75 | 76, 76 | 77, 77 | 78, 78 | 79, 79 | 80, 80 | 81, 81 | 82, 82 | 83, 83 | 84, 84 | 85, 85 | 86, 86 | 87, 87 | 88, 88 | 89, 89 | 90, 90 | 91, 91 | 92, 92 | 93, 93 | 94, 94 | 95, 95 | 96, 96 | 97, 97 | 98, 98 | 99, 99 | XX8, Not applicable | XX9, Not documented in medical record

Description

- The Circumferential Resection Margin (CRM), also referred to as the radial margin or the mesenteric resection margin, is the measurement of the distance from the deepest invasion of the tumor to the margin of resection in the retroperitoneum or mesentery.
- The CRM is also be called the radial margin or mesenteric (mesocolon), resection margin, soft tissue margin.

Directives

- This data field is available in REDCap for colon/rectum cancer diagnoses only. Data from tumor registries can be piped in to field [naaccr_ca_crc_crm]; curated data should be entered in field [ca_crc_crm_2] if piped-in data is unavailable or incorrect.
- Record to the nearest tenth in millimeters (mm) the distance between the leading edge of the tumor and the circumferential resection margin.
- If this is reported in centimeters (cm) in the medical record, you will need to convert the measurement to mm by multiplying by 10.
- If the margin is positive record '0'.
- You may use a value from a clinic note if the path report is not in the EHR.
- Do not assume closest margin or only margin reported is the same as the circumferential margin. If the patient had a resection of the primary site, the pathology report is available for review, and CRM is not mentioned, record "XX9, Not documented in medical record".

Provenance

- Use the native or external pathology report during diagnosis; give preference to the native pathology report. If these are not available, use the medical oncologist's report.

Analytic Note

- This data field for cancer diagnoses other than colon or rectal cancer will be blank on data export.

Data Standard

NAACCR #2930. Conforms to NAACCR site-specific item #3823.

Source Document: Internal and External Pathology Report

Shared Field: Yes

History

02212023 ca_crc_crm retired and ca_crc_crm_2 added. See appendix 13 for value mapping.

CRC: Microsatellite Instability (MSI) (CS SSF 7)

[ca_crc_msi] PN1219

[naaccr_ca_crc_msi] PN1225

Values

Blank for other cancer diagnoses

- 0, Microsatellite instability (MSI) stable; microsatellite stable (MSS); negative, NOS AND/OR Mismatch repair (MMR) intact, no loss of nuclear expression of MMR proteins; MMR proficient (pMMR or MMR-P)
- 1, MSI unstable low (MSI-L)
- 2, MSI unstable high (MSI-H) AND/OR MMR deficient (dMMR or MMR-D) loss of nuclear expression of one or

more MMR proteins, MMR protein deficient

8, Not applicable: Information not collected for this case.

9, Not documented in medical record; MSI-indeterminate; MSI-equivocal; Microsatellite instability not assessed or unknown if assessed

Directives

- This data field is available in REDCap for colon/rectum cancer diagnoses only. Data from tumor registries can be piped into field [naaccr_ca_crc_msi]; curated data should be entered in field [ca_crc_msi] if piped-in data is unavailable or incorrect.
- Record the result of testing for MMR or MSI performed from time of diagnosis through first course of treatment. Testing for MSI is done by immunology or genetic testing. You may use the results from tumor sequencing tests reported outside of this period if the specimen sequenced was collected during the appropriate time period.
- Results from nodal or metastatic tissue may be used.
- Code “0, MSI Stable” if the report only specifies a low probability of MSI-H.
- Code “2, MSI unstable High (MSI-H)” if the report states MSI is unstable but does not specify high or low.

Provenance

- Use the native or external pathology report or tumor sequencing report; give preference to the native pathology report.. If these are not available, use the medical oncologist’s report.

Analytic Note

- This data field for cancer diagnoses other than colon or rectal cancer will be blank on data export.

Data Standard

NAACCR # #2861. Conforms to NAACCR site-specific data item #3890.

Shared Field: Yes

CRC: Perineural Invasion (CS SSF 8)

[ca_crc_peri_inv_2] PN1221

[naaccr_ca_crc_peri_inv] PN1225

Values

Blank for other cancer diagnoses

0, Perineural invasion not identified/present

1, Perineural invasion identified/present

8, Not applicable

9, Not documented in medical record; Pathology report does not mention perineural invasion; Cannot be determined by the pathologist; Perineural invasion not assessed or unknown if assessed

Directives

- This data field is available in REDCap for colon/rectum cancer diagnoses only. Data from tumor registries can be piped in to field [naaccr_ca_crc_peri_inv]; curated data should be entered in field [ca_crc_peri_inv_2] if piped-in data is unavailable or incorrect.
- Indicate whether perineural invasion was identified in the pathology report from surgical resection of the primary site of cancer. .
- Absence of perineural invasion can only be taken from a surgical resection pathology report. However, if perineural invasion is not mentioned, do not assume that means it was not present.
- Please note that the codes (numbers) for the values are not visible in REDCap; however, the codes will be available at data export.

Provenance

- Use the native or external pathology report during diagnosis; give preference to the native pathology report. If these are not available, use the medical oncologist's report.

Analytic Note

- This data field for cancer diagnoses other than colon or rectal cancer will be blank on data export.

Data Standard

NAACCR # #2863. Conforms to NAACCR site-specific data item #3909.

History

02212023: Retired field ca_crc_peri_inv and added ca_crc_peri_inv_2 based on updates to NAACCR dictionary. See

Appendix 13 for value mappings.

02252020: removed leading zeros from codes 000/010.

Shared Field: Yes

Risk Factor Data Fields

These questions are specific to cases being curated for Lung and Bladder cancer only. To answer these questions, use information from the diagnosis of lung cancer through completion of surgery(ies) in the first course of treatment.

Cigarette Use at Time of Diagnosis

[ca_lung_cigarette] PN0133

Values

Blank for other cancer diagnoses

- 1, Never used
- 2, Current user
- 3, Former user (quit < 1 year)
- 4, Former user (quit >1 year)
- 5, Former user (unknown time)
- 100, Unknown

Directives

- This data field is available in REDCap for lung cancer and bladder cancer diagnoses.
- Record whether the patient ever smoked and the length of time that the patient smoked cigarettes at the time of diagnosis.
- If the patient discontinued smoking at the time of diagnosis, record "2, Current user."
- Please note that the codes (numbers) for the values are not visible in REDCap; however, the codes will be available at data export.

Provenance

- Use the native or external pathology report during diagnosis; give preference to the native pathology report. If these are not available, use the medical oncologist's report.

Analytic Note

- This data field will be blank on data export for cancer diagnoses other than lung, bladder, or renal kidney cancer.

Data Standard

NPCR (National Program of Cancer Registries)

Shared Field: Yes

Lung Cancer Site-Specific Data Fields

These questions are specific to cases being curated for Lung cancer only. To answer these questions, use information from the diagnosis of lung cancer through completion of surgery(ies) in the first course of treatment.

LUNG: Separate Tumor Nodules – Ipsilateral Lung (CS SSF 1)

[ca_lung_sep_tumor] PN0134

Values

Blank for other cancer diagnoses

- 0, No separate tumor nodules
- 10, Separate tumor nodules of same histologic type in ipsilateral lung same lobe
- 20, Separate tumor nodules of same histologic type in ipsilateral lung, different lobe
- 30, Separate tumor nodules of same histologic type in ipsilateral lung same AND different lobes
- 40, Separate tumor nodules of same histologic type in ipsilateral lung unknown if same or different lobe(s)
- 988, Not applicable
- 999, Unknown if separate tumor nodules

Directives

- This data field is available in REDCap for lung cancer diagnoses only.
- Record the correct choice for whether separate tumor nodules are identified in the ipsilateral (same) lung at the time of diagnosis. Separate tumor nodules can be defined clinically (by imaging) and/or pathologically.
- If separate tumor nodules are not mentioned in imaging and/or path reports, use code “000, No separate nodules.”
- For single tumor nodules, code “000, No separate tumor nodules.”
- Code “999, if there is no relevant imaging or resection of the primary site.
- A statement from a physician note may be used if the path report is not available.
- This data field should not be used to document contralateral tumor nodules.
- Please note that the codes (numbers) for the choices are not visible in REDCap; however, the codes will be available at data export.

Provenance: Native and external imaging reports and pathology reports, and MD report if imaging reports are missing.

Analytic Note

- This data field for cancer diagnoses other than lung cancers will be blank on data export.

Data Standards

NAACCR # #2880 (Site-specific factor). Conforms to NAACCR site-specific data item #3929 though the SSDI has more choices.

History

01152020: removed leading zeros from codes 000/010/020/030/040.

Shared Field: Yes

LUNG: Visceral and Parietal Pleural Invasion (2018 EOD SSDI)

[ca_lung_pl_el_inv_2] PN1175

Values

Blank for other cancer diagnoses | 0, PL 0; No evidence of visceral pleural invasion (PL); Tumor does not completely traverse the elastic layer of the pleura | 4, Invasion of visceral pleural present, NOS; Stated as PL1 or PL2 | 5, Tumor invades into or through the parietal pleural OR chest wall; Stated as PL3 | 6, Tumor extends to pleura, NOS; not stated if visceral or parietal | 8, Not applicable: Information not collected for this case | 9, Not documented in medical record; No surgical resection of primary site is performed; Visceral Pleural Invasion not assessed or unknown

if assessed or cannot be determined

Directives

- This data field is available in REDCap for lung cancer diagnoses only.
- Indicate whether evidence of visceral pleural invasion was identified in the pathology report from surgical resection of the primary site of cancer. Do not use an FNA biopsy to curate data.
- Record “0, PL 0; No evidence of visceral pleural invasion (PL)” if surgical resection of the primary site is performed, the pathology report is available for review, and visceral pleural invasion is not present.
- Code 8 is used by the Tumor Registry and will only apply to data piped in from the TR; if code 8 is piped in then this field should be manually updated by the curator.
- Please note that the numbers (codes) for the choices are not visible in REDCap; however, the codes will be available at data export.

Provenance: The sources for curation should be Internal and External Pathology Reports. However, if these are not available, you may use a statement from a clinic note.

Analytic Note

- This data field for cancer diagnoses other than lung cancers will be blank on data export.
- This field maps to retired field ca_lung_pl_el_inv; see details in Appendix 12.

Data Standards

Conforms to 2018 EOD SSDI NAACCR #3937 v21+.

History

05092022: ca_lung_pl_el_inv retired and ca_lung_pl_el_inv_2 added.

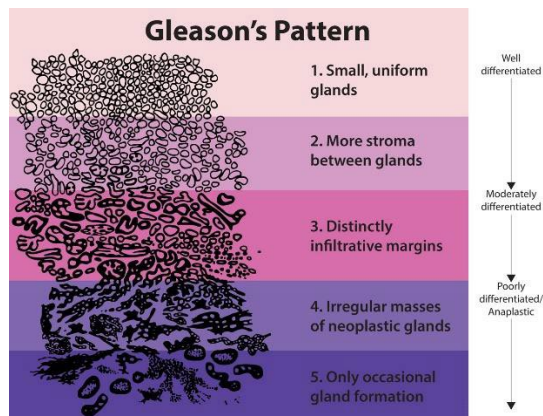
01152020: removed leading zeros.

Shared Field: Yes

Prostate Cancer Site-Specific Data Fields

These questions are specific to cases being curated for prostate cancer only. To answer these questions, use information from the diagnosis of prostate cancer through completion of surgery(ies) in the first course of treatment.

In prostate cancer, the grade of the tumor is very important. Usually prostate cancers are graded using Gleason score or pattern. Gleason grading for prostate primaries is based on a 5-component system (5 histologic patterns).



Source: <https://training.seer.cancer.gov/prostate/abstract-code-stage/morphology.html>

Prostatic cancer generally shows two main histologic patterns. The primary pattern, the pattern occupying greater than 50% of the cancer, is usually indicated by the first number of the Gleason grade, and the secondary pattern is usually indicated by the second number. These two numbers are added together to create a pattern score whose largest value

is 10. A score could be written like this: 3+4=7 or 7/10

- If there are two numbers, assume that they refer to two patterns (the first number being the primary pattern and the second number the secondary pattern), and sum them to obtain the score.
- If only one number is given and it is less than or equal to 5, assume that it describes a pattern (since scores of 5 or less would reflect Primary or Secondary Pattern Scores of 1 or 2). Code the number as the primary pattern and code the secondary pattern as Unknown. For example, if only one number is given and it is a 3, code “39” for Gleason Patterns and “X9” for Gleason Score. If only one number is given and it is greater than 5, assume that it is a score. For example, if only one number is given and it is a 7, code “X6” for Gleason Patterns and “07” for Gleason Score.
- If the pathology report specifies a specific number out of a total of 10, the first number given is the score. For example, if the pathology report says Gleason 7/10, code “07” for Gleason Score and “X6” for Gleason Patterns.

It is important to note that patients with small cell carcinoma of the prostate will most likely not have a Gleason scores or patterns reported. For these patients, use the not applicable option.

PROS: Gleason Patterns Clinical (CS SSF 7) [ca_pros_clin_gpattern2] PN0136

Values

Blank | 11, Primary pattern 1 secondary pattern 1 | 12, Primary pattern 1 secondary pattern 2 | 13, Primary pattern 1 secondary pattern 3 | 14, Primary pattern 1 secondary pattern 4 | 15, Primary pattern 1 secondary pattern 5 | 19, Primary pattern 1 secondary pattern unknown | 21, Primary pattern 2 secondary pattern 1 | 22, Primary pattern 2 secondary pattern 2 | 23, Primary pattern 2 secondary pattern 3 | 24, Primary pattern 2 secondary pattern 4 | 25, Primary pattern 2 secondary pattern 5 | 29, Primary pattern 2 secondary pattern unknown | 31, Primary pattern 3 secondary pattern 1 | 32, Primary pattern 3 secondary pattern 2 | 33, Primary pattern 3 secondary pattern 3 | 34, Primary pattern 3 secondary pattern 4 | 35, Primary pattern 3 secondary pattern 5 | 39, Primary pattern 3 secondary pattern unknown | 41, Primary pattern 4 secondary pattern 1 | 42, Primary pattern 4 secondary pattern 2 | 43, Primary pattern 4 secondary pattern 3 | 44, Primary pattern 4 secondary pattern 4 | 45, Primary pattern 4 secondary pattern 5 | 49, Primary pattern 4 secondary pattern unknown | 51, Primary pattern 5 secondary pattern 1 | 52, Primary pattern 5 secondary pattern 2 | 53, Primary pattern 5 secondary pattern 3 | 54, Primary pattern 5 secondary pattern 4 | 55, Primary pattern 5 secondary pattern 5 | 59, Primary pattern 5 secondary pattern unknown | X6, Primary pattern unknown, secondary pattern unknown | X7, No needle core biopsy/TURP performed | X8, Not applicable information not collected for this case | X9, Not documented in medical record

Directives

- This data field is available in REDCap for prostate cancer diagnoses only.
- Use pathology reports from needle biopsies or transurethral resection of prostate/bladder that contains prostate tissue. A clinician report may be used if no other documentation is available.
- If the prostate cancer is small cell carcinoma, Gleason patterns will not be reported; record “X8, Not applicable information not collected for this case.”
- Please note that the numbers (codes) for the choices are not visible in REDCap; however, the codes will be available at data export.

Analytic Note

- This data field for cancer diagnoses other than prostate cancer will be blank on data export.

Data Standards

NAACCR #2861: Conforms to SSDI #3838.

History

02252020: retired [ca_pros_clin_gpattern] and added [ca_pros_clin_gpattern_2]

Shared Field: Yes

PROS: Gleason Score Clinical (CS SSF 8)

[ca_pros_clin_gscore] PN0137

Values

Blank | 2, Gleason score 2 | 3, Gleason score 3 | 4, Gleason score 4 | 5, Gleason score 5 | 6, Gleason score 6 | 7, Gleason score 7 | 8, Gleason score 8 | 9, Gleason score 9 | 10, Gleason score 10 | X7, No needle core biopsy/TURP performed | X8, Not applicable | X9, Not documented in medical record

Directives

- This data field is available in REDCap for prostate cancer diagnoses only.
- Use pathology reports from needle biopsies or transurethral resection of prostate/bladder that contains prostate tissue. A clinician report may be used if no other documentation is available.
- If the prostate cancer is small cell carcinoma, Gleason scores will not be reported; record “X8, Not applicable.”
- Please note that the numbers (codes) for the choices are not visible in REDCap; however, the codes will be available at data export.

Analytic Note

- This data field for cancer diagnoses other than prostate cancer will be blank on data export.

Data Standards

History

03012020: removed leading zeros.

Shared Field: Yes

PROS: Gleason Patterns Pathological (CS SSF 9)

[ca_pros_path_gpattern_2] PN0138

Values

Blank | 11, Primary pattern 1 secondary pattern 1 | 12, Primary pattern 1 secondary pattern 2 | 13, Primary pattern 1 secondary pattern 3 | 14, Primary pattern 1 secondary pattern 4 | 15, Primary pattern 1 secondary pattern 5 | 19, Primary pattern 1 secondary pattern unknown | 21, Primary pattern 2 secondary pattern 1 | 22, Primary pattern 2 secondary pattern 2 | 23, Primary pattern 2 secondary pattern 3 | 24, Primary pattern 2 secondary pattern 4 | 25, Primary pattern 2 secondary pattern 5 | 29, Primary pattern 2 secondary pattern unknown | 31, Primary pattern 3 secondary pattern 1 | 32, Primary pattern 3 secondary pattern 2 | 33, Primary pattern 3 secondary pattern 3 | 34, Primary pattern 3 secondary pattern 4 | 35, Primary pattern 3 secondary pattern 5 | 39, Primary pattern 3 secondary pattern unknown | 41, Primary pattern 4 secondary pattern 1 | 42, Primary pattern 4 secondary pattern 2 | 43, Primary pattern 4 secondary pattern 3 | 44, Primary pattern 4 secondary pattern 4 | 45, Primary pattern 4 secondary pattern 5 | 49, Primary pattern 4 secondary pattern unknown | 51, Primary pattern 5 secondary pattern 1 | 52, Primary pattern 5 secondary pattern 2 | 53, Primary pattern 5 secondary pattern 3 | 54, Primary pattern 5 secondary pattern 4 | 55, Primary pattern 5 secondary pattern 5 | 59, Primary pattern 5 secondary pattern unknown | X6, Primary pattern unknown, secondary pattern unknown | X7, No needle core biopsy/TURP performed | X8, Not applicable information not collected for this case | X9, Not documented in medical record

Directives

- This data field is available in REDCap for prostate cancer diagnoses only.
- Use pathology reports from needle biopsies or transurethral resection of prostate/bladder that contains prostate tissue or from prostatectomy to curate this field. A clinician report may be used if no other documentation is available.
- If the prostate cancer is small cell carcinoma, Gleason patterns will not be reported; record “X8, Not applicable

information not collected for this case.”

- Please note that the numbers (codes) for the choices are not visible in REDCap; however, the codes will be available at data export.

Analytic Note

- This data field for cancer diagnoses other than prostate cancer will be blank on data export.

Data Standards

NAACCR #2863: Conforms to SSDI #3839.

History

02252020: retired [ca_pros_path_gpattern] and added [ca_pros_path_gpattern_2]

Shared Field: Yes

PROS: Gleason Score Pathological (CS SSF 10)

[ca_pros_path_gscore] PN0139

Values

Blank | 2, Gleason score 2 | 3, Gleason score 3 | 4, Gleason score 4 | 5, Gleason score 5 | 6, Gleason score 6 | 7, Gleason score 7 | 8, Gleason score 8 | 9, Gleason score 9 | 10, Gleason score 10 | X7, No prostatectomy | X8, Not applicable | X9, Not documented in medical record

Directives

- This data field is available in REDCap for prostate cancer diagnoses only.
- Use pathology reports from needle biopsies or transurethral resection of prostate/bladder that contains prostate tissue or from prostatectomy to curate this field. A clinician report may be used if no other documentation is available.
- If the prostate cancer is small cell carcinoma, Gleason scores will not be reported; record “X8, Not applicable.”
- Please note that the numbers (codes) for the choices are not visible in REDCap; however, the codes will be available at data export.

Analytic Note

- This data field for cancer diagnoses other than prostate cancer will be blank on data export.

Data Standards

NAACCR #2864: Conforms to SSDI #3841.

History

03012020: removed leading zeros.

Shared Field: Yes

PROS: Gleason Tertiary Pattern (CS SSF 11)

[ca_pros_path_gtert] PN0140

Values

Blank | 10, Tertiary pattern 1 | 20, Tertiary pattern 2 | 30, Tertiary pattern 3 | 40, Tertiary pattern 4 | 50, Tertiary pattern 5 | X7, No prostatectomy/autopsy performed | X8, Not applicable | X9, Not documented in medical record

Directives

- This data field is available in REDCap for prostate cancer diagnoses only.
- Used to code information on the Gleason tertiary pattern from a prostatectomy.

- If the prostate cancer is small cell carcinoma, Gleason patterns will not be reported; record “X8, Not applicable.”
- Please note that the numbers (codes) for the choices are not visible in REDCap; however, the codes will be available at data export.

Analytic Note

- This data field for cancer diagnoses other than prostate cancer will be blank on data export.

Data Standards

NAACCR #2865: Conforms to SSDI #3842.

Shared Field: Yes

PROS: Number of Cores Positive (CS SSF 12)

[ca_pros_num_corespos] PN0141

Values

Blank | Numeric

Directives

- This data field is available in REDCap for prostate cancer diagnoses only.
- Paired with **Number of Cores Examined**.
- Record the number of cores positive examined from core needle biopsies.
- Code ‘0’ if all examined cores are negative.
- Please note that the numbers (codes) for the choices are not visible in REDCap; however, the codes will be available at data export.

Analytic Note

- This data field for cancer diagnoses other than prostate cancer will be blank on data export.

Data Standards

NAACCR #2866. Conforms to SSDI #3898.

Shared Field: Yes

PROS: Number of Cores Examined (CS SSF 13)

[ca_pros_num_cores] PN0142

Values

Blank | Numeric

Directives

- This data field is available in REDCap for prostate cancer diagnoses only.
- Paired with **Number of Cores Positive**.
- Record the number of cores examined from core needle biopsies.
- Please note that the numbers (codes) for the choices are not visible in REDCap; however, the codes will be available at data export.

Analytic Note

- This data field for cancer diagnoses other than prostate cancer will be blank on data export.

Data Standards

NAACCR #2867. Conforms to SSDI #3897. Paired with Number of Cores Positive. Record the number of cores examined from core needle biopsies.

Shared Field: Yes

Kidney Cancer Site-Specific Data Fields

These questions are specific to cases being curated for kidney cancer only. To answer these questions, use information from the diagnosis of kidney cancer through completion of surgery(ies) in the first course of treatment.

KID: Invasion Beyond Capsule (SEER SSDI, CS SSF 1)

[ca_kid_inv_cap] PN1231

[naaccr_ca_kid_inv_cap] PN1230

[naaccr_ca_kid_inv_cap_2] PN1306

Values

0, Invasion beyond capsule not identified | 1, Perinephric (beyond renal capsule) fat or tissue | 2, Renal sinus | 3, Gerota's fascia | 4, Any combination of codes 1-3 | 5, Invasion beyond capsule, NOS | 8, Not applicable: Information not collected for this case | 9, Not documented in medical record; Invasion beyond capsule not assessed or unknown if assessed; No surgical resection of primary site is performed

Directives

- This data field is available in REDCap for kidney cancer diagnoses only.
- Use the pathology report from resection of the primary kidney tumor to curate this field.
- Data from tumor registries can be piped into field [naaccr_ca_kid_inv_cap] for cases diagnosed 1/1/2018+ and field [naaccr_ca_kid_inv_cap_2] for cases diagnosed prior to 2018; curated data should be entered in field [ca_kid_inv_cap] if piped-in data is unavailable or incorrect.
- For the GENIE BPC: this field was not curated. Only piped-in data from the tumor registry was collected.

Analytic Note

- This data field for cancer diagnoses other than kidney cancer will be blank on data export.

Data Standards

NAACCR #3864: [ca_kid_inv_cap] and [naaccr_ca_kid_inv_cap] conform to SSDI #3864. Tumor registry data for cases diagnosed 1/1/2018+.

NAACCR #2880: [naaccr_ca_kid_inv_cap_2] conforms to NAACCR #2880, CS SSF 1. Tumor registry data for cases diagnosed prior to 2018.

Shared Field: Yes

History: 11062023 field added to dictionary

KID: Major Vein Involvement (SEER SSDI, CS SSF 2)

[ca_kid_maj_vein] PN1233

[naaccr_ca_kid_maj_vein] PN1232

[naaccr_ca_kid_maj_vein_2] PN1307

Values

0, Major vein involvement not present/not identified | 1, Renal vein or its segmental branches | 2, Inferior vena cava (IVC) | 3, Major vein invasion, NOS | 4, Any combination of codes 1-3 | 8, Not applicable: Information not collected for this case | 9, Not documented in medical record; Vein involvement not assessed or unknown if assessed; No surgical resection of primary

site is performed

Directives

- This data field is available in REDCap for kidney cancer diagnoses only.
- Use the pathology report from resection of the primary kidney tumor to curate this field.
- Data from tumor registries can be piped into field [naaccr_ca_kid_maj_vein] for cases diagnosed 1/1/2018+ and field [naaccr_ca_kid_maj_vein_2] for cases diagnosed prior to 2018; curated data should be entered in field [ca_kid_maj_vein] if piped-in data is unavailable or incorrect.
- For the GENIE BPC: this field was not curated. Only piped-in data from the tumor registry was collected.

Analytic Note

- This data field for cancer diagnoses other than kidney cancer will be blank on data export.

Data Standards

NAACCR #3886: [ca_kid_maj_vein] and [naaccr_ca_kid_maj_vein] conform to SSDI #3886. Tumor registry data for cases diagnosed 1/1/2018+.

NAACCR #2880: [naaccr_ca_kid_maj_vein_2] conforms to NAACCR #2890, CS SSF 2. Tumor registry data for cases diagnosed prior to 2018.

Shared Field: Yes

History: 11062023 field added to dictionary

KID: Ipsilateral Adrenal Gland Involvement (SEER SSDI, CS SSF 3)

[ca_kid_ips_adrenal] PN1235

[naaccr_ca_kid_ips_adrenal] PN1234

[naaccr_ca_kid_ips_adren_2] PN1308

Values

0, Ipsilateral adrenal gland involvement not present/not identified | 1, Adrenal gland involvement by direct involvement (contiguous involvement) | 2, Adrenal gland involvement by separate nodule (discontiguous involvement) | 3, Combination of code 1-2 | 4, Ipsilateral adrenal gland involvement, unknown if direct involvement or separate nodule | 8, Not applicable: Information not collected for this case | 9, Not documented in medical record; Ipsilateral adrenal gland not resected; Ipsilateral adrenal gland involvement not assessed or unknown if assessed; No surgical resection of primary site is performed

Directives

- This data field is available in REDCap for kidney cancer diagnoses only.
- Record whether the ipsilateral adrenal gland (located on same side as the primary kidney tumor) is involved by direct extension of the primary tumor (contiguous involvement) or a separate nodule (discontiguous involvement).
- If surgical resection is done and tumor is reported as “confined to kidney” and staging is based on size, curate as “0, Ipsilateral adrenal gland involvement not present/not identified”.
- Data from tumor registries can be piped into field [naaccr_ca_kid_ips_adrenal] for cases diagnosed 1/1/2018+ and field [naaccr_ca_kid_ips_adren_2] for cases diagnosed prior to 2018; curated data should be entered in field [ca_kid_maj_vein] if piped-in data is unavailable or incorrect.
- For the GENIE BPC: this field was not curated. Only piped-in data from the tumor registry was collected.
- Please note that the numbers (codes) for the values are not visible in REDCap, however, the codes will be available at data export.

Analytic Note

- This data field for cancer diagnoses other than kidney cancer will be blank on data export.

Data Standards

NAACCR #3886: [ca_kid_maj_vein] and [naaccr_ca_kid_maj_vein] conform to SSDI #3886. Tumor registry data for cases diagnosed 1/1/2018+.

NAACCR #2880: [naaccr_ca_kid_maj_vein_2] conforms to NAACCR #2890, CS SSF 2. Tumor registry data for cases diagnosed prior to 2018.

Shared Field: Yes

History: 11062023 field added to dictionary

Ovarian Cancer Site-Specific Data Fields

These questions are specific to cases being curated for ovarian, fallopian tube, or primary peritoneal cancer only. To answer these questions, use information from the diagnosis of cancer through completion of surgery(ies) in the first course of treatment.

OVC: FIGO Stage

[ca_oc_figo] PN1341

Values

1, FIGO Stage I | 1A, FIGO Stage IA | 1B, FIGO Stage IB | 1C, FIGO Stage IC | 1C1, FIGO Stage IC1 | 1C2, FIGO Stage IC2 | 1C3, FIGO Stage IC3 | 2, FIGO Stage II | 2A, FIGO Stage IIA | 2B, FIGO Stage IIB | 3, FIGO Stage III | 3A, FIGO Stage IIIA | 3A1, FIGO Stage IIIA1 | 3A11, FIGO Stage IIIA1i | 3A12, FIGO Stage IIIA1ii | 3A2, FIGO Stage IIIA2 | 3B, FIGO Stage IIIB | 3C, FIGO Stage IIIC | 4, FIGO Stage IV | 4A, FIGO Stage IVA | 4B, FIGO Stage IVB | 97, Not applicable: Carcinoma in situ (intraepithelial, noninvasive, preinvasive) | 99, Not documented in medical record; FIGO stage unknown, not assessed or unknown if assessed

Directives

- This data field is available in REDCap for ovarian, fallopian tube, and primary peritoneal cancer diagnoses only.
- Record the highest FIGO staged documented by the managing physician (medical oncologist or gynecologic oncologist) at the time of diagnosis. If a stage group is documented, assume it is the FIGO stage unless otherwise stated.
- If more than one FIGO stage is documented from the clinical and pathological workup, record the higher or more extensive stage.
- Timeframe for collection: date of diagnosis through work up and first course of treatment.
- Provenance: Medical oncologist or gynecologic oncologist notes.

Analytic Note

- This data field for cancer diagnoses other than ovarian, fallopian tube, or primary peritoneal cancer will be blank on data export.

Data Standards

NAACCR #3836: conforms to SSDI #3836.

Shared Field: Yes

History: 07022024 field added to dictionary

OVC: CA-125 Pre-Treatment Interpretation

[ca_oc_ca125] PN1342

Values

- 0, Negative/normal; within normal limits
- 1, Positive/elevated
- 2, Stated as borderline; undetermined whether positive or negative
- 9, Not documented in medical record; CA-125 not assessed or unknown if assessed

Directives

- This data field is available in REDCap for ovarian, fallopian tube, and primary peritoneal cancer diagnoses only.
- Review clinical laboratory reports from the time of initial diagnosis until first treatment begins (including surgery). Record the interpretation of the highest CA-125 values from a blood or serum test based on the laboratory's reference range.
- If a clinical laboratory report is not available, record the clinician's interpretation of the highest value of CA-125 prior to treatment. If the clinician does not provide an interpretation but documents the CA-125 lab value and reference range, interpret the lab value based on the reference range provided. If the clinician provides a lab value without a reference range, curate values over 35 U/mL as Positive/Elevated.
- May also be referred to as: Cancer Antigen 125, CA 125, CA125, Carbohydrate Antigen 125, mucin 16, MUC16, CA 125 II.
- Provenance: Clinical laboratory reports, medical oncology consultation notes.
- Timeframe for collection: Initial diagnosis prior to surgery or other treatment. Do not use documents dated more than 60 days prior to the curated date of diagnosis.

Analytic Note

- This data field for cancer diagnoses other than ovarian, fallopian tube, or primary peritoneal cancer will be blank on data export.

Data Standards

NAACCR #3818: conforms to SSDI #3818.

Shared Field: Yes

History: 07022024 field added to dictionary

OVC: Residual Tumor Volume Post Cytoreduction

[ca_oc_res_tumor_vol] PN1343

Values

0, No gross residual tumor nodules | 50, Residual tumor nodule(s) 1 centimeter (cm) or less | 60, Residual tumor nodule(s) greater than 1 cm | 70, Macroscopic residual tumor, size not stated | 80, Procedure described as optimal debulking and size of residual tumor nodule(s) not given | 97, No cytoreductive surgery performed or Non-invasive neoplasm (behavior /2) | 99, Not documented in medical record or Residual tumor status after cytoreductive surgery not assessed or unknown if assessed

Directives

- This data field is available in REDCap for ovarian, fallopian tube, and primary peritoneal cancer diagnoses only.
- Categorize the gross residual tumor volume following debulking or cytoreduction. This refers to the amount of tumor remaining after the operation and is usually described in the operative report.
- If not explicitly stated, code as 99.
- Data provenance: Operative report from debulking or cytoreductive surgery. If the operative report is not available,

use the medical oncologist's or gynecologic oncologist's report.

- Timeframe for collection: Initial diagnosis through work up and first course of treatment.

Analytic Note

- This data field for cancer diagnoses other than ovarian, fallopian tube, or primary peritoneal cancer will be blank on data export.

Data Standards

NAACCR #3921: conforms to SSDI #3921.

Shared Field: Yes

History: 07022024 field added to dictionary

OVC: Chemotherapy Response Score

[ca_oc_chemo_response] PN1379

Values

- 1, CRS 1: No or minimal tumor response
- 2, CRS 2: Appreciable tumor response amid tumor that is readily identifiable
- 3, CRS 3: Complete or near-complete response with no residual tumor or minimal, irregularly scattered tumor foci seen as individual cells, cell groups or nodules up to 2mm maximum size.
- 4, No resection performed
- 9, Not documented in the record; CRS not assessed or unknown.

Directives

- This data field is available in REDCap for ovarian, fallopian tube, and primary peritoneal cancer diagnoses only.
- If the patient received neoadjuvant therapy prior to resection or debulking surgery, indicate the chemotherapy response score documented in the pathology report for the relevant surgery.
- Timeframe for collection: at completion of surgery as part of first course of treatment, including interval debulking surgery for patients with advanced disease.
- Data provenance: Pathology report from the primary tumor resection or debulking surgery.

Analytic Note

- This data field for cancer diagnoses other than ovarian, fallopian tube, or primary peritoneal cancer will be blank on data export.

Data Standards

College of American Pathologists and the International Collaboration on Cancer Reporting

Shared Field: Yes

History: 07022024 field added to dictionary

OVC: Patients with stage IV cancer at the time of initial diagnosis: record the sites of metastases present during the time of diagnosis. Select all that apply:

[ca_dmets_sites] PN1344

Values

1, Bone | 2, Brain | 10, Diaphragm | 3, Distant lymph nodes | 4, Liver parenchyma (exclude lesions that only involve the liver surface or serosa) | 5, Lung | 11, Peritoneum (including mesentery and omentum) | 6, Pleural effusion with positive cytology | 7, Spleen parenchyma (exclude lesions that only involve the spleen surface or capsule) | 12, Other pelvic structures/organs

| 13, Other abdominal structures/organs | 8, Other distant site(s) outside of the abdomen and pelvis | 9, None of the above

Directives

- This data field is available in REDCap for ovarian, fallopian tube, and primary peritoneal cancer diagnoses only. For patients who are stage IV at diagnosis only.
- Review imaging reports, pathology reports, and medical oncologist notes for sites of metastases from the time of initial diagnosis with ovarian, fallopian tube, or primary peritoneal cancer until initiation of any systemic therapy or signs of progression, or up to 4 months from diagnosis, whichever is sooner. Select all sites that are described as a metastasis or involved with metastatic ovarian, fallopian tube, or primary peritoneal cancer.
- Only include pleural effusion that has been confirmed by cytology.
- Liver and spleen tumors/lesions must involve the parenchyma; tumors/lesions stated only as involving the surface, capsule or serosa should not be included.
- Distant lymph nodes are those located outside of the abdomen and pelvis (see slide 15 for list of regional lymph nodes). The inguinal lymph nodes are a common site of distant metastasis.
- If the patient does not have metastases in any of these sites, select “None of the above”.
- Provenance: Medical oncologist notes, pathology reports, imaging reports

Analytic Note

- This data field for cancer diagnoses other than ovarian, fallopian tube, or primary peritoneal cancer will be blank on data export.

Data Standards

PRISMM

Shared Field: Yes

History: 07022024 field added to dictionary

Sites of Distant Metastasis at Diagnosis Data Fields

For patients diagnosed with a recorded Stage IV cancer diagnosis by through the Tumor Registry: **AJCC Best Stage** and/or manual curation, **Was the patient's cancer diagnosed as Stage IV?**

Does this stage IV patient have sites of distant metastasis at the time of diagnosis?

[ca_dmets_yn] PN0143

Values

- 1, Yes
- 0, No

Directives

- Linked with **Tumor Registry: AJCC Best Stage** or **Was the patient's cancer diagnosed as Stage IV?** If either indicates that the patient's cancer is diagnosed with stage IV cancer, this field is available for curation.
- Record “1, Yes” if the patient's is known to have distant sites of metastasis at the time of diagnosis. Most stage IV cancers will have at least one distant metastasis. Prostate cancer is an example where stage IV does not require a distant site of metastasis.
- Please note that the numbers (codes) for the choices are not visible in REDCap; however, the codes will be available at data export.

Data Standards

NAACCR #

Shared Field: Yes

Sites of Distant Metastases at Diagnosis 10 sites available

[ca_first_dmets1], [ca_first_dmets2], [ca_first_dmets3], [ca_first_dmets4], [ca_first_dmets5], [ca_first_dmets6], [ca_first_dmets7], [ca_first_dmets8], [ca_first_dmets9], [ca_first_dmets10]

PN0144, PN0145, PN0146, PN0147, PN0148, PN0149, PN0150, PN0151, PN0152, PN0153

Values

Blank | ICD-O-3 Site/Topography code list, *See Appendix 7a*

Directives

- Linked with **Does this stage IV patient have sites of distant metastasis at the time of diagnosis?** If “1, Yes” was coded, this field is available for curation. Otherwise, the field will be blank at export.
- Record the site of distant metastasis using the ICD-O-3 Site list. Review imaging reports, pathology reports, and medical oncologist notes from the time of initial diagnosis until initiation of any treatment, or up to 4 months from diagnosis, whichever is sooner. Include all sites described as metastases or metastatic with the exception of regional lymph nodes.
- Please note that the numbers (codes) for the choices are not visible in REDCap; however, the codes will be available at data export.

Data Standards

NAACCR #

History

03032020: removed leading zeros.

Shared Field: Yes

Curator Notes Concerning Cancer Diagnosis Section

[ca_qanotes]

Value

Free text

Directives

- This note field should be used to track any notes concerning the Cancer Diagnosis section that will assist curation or QA review.

Shared Field: No, contains PHI. The data element is flagged as an identifier in REDCap.

Cancer Diagnosis and Staging Calculated Fields

In order to share data in a de-identified manner, several intervals are calculated in the REDCap project from the anchor date: Date of Birth as follows:

Interval Data Field	Data Element	Calculation
Eligible case for curation: triggers one of two description fields indicating whether case should be curated based on the behavior code for cancer diagnosis piped in by the tumor registry. For most projects, behavior codes 2 & 3 are eligible and 0 & 1 are not eligible.	tr_eligible PN0049 PN0050 PN0051	if ([naaccr_behavior_cd]='0' or [naaccr_behavior_cd]='1'), 0, 1)
Patient Age at Diagnosis based on Tumor Registry	naaccr_diagnosis_age PN0056	rounddown(datediff([birth_dt], [naaccr_diagnosis_dt], "y", "mdy"))

Interval in Days from Date of Birth to Tumor Registry Date of Diagnosis	naaccr_diagnosis_int PN0057	<code>datediff([birth_dt], [naaccr_diagnosis_dt], "d", "mdy")</code>
Interval in Days from DOB to First Contact at Institution based on Tumor Registry	naaccr_first_contact_int PN0063	<code>datediff([birth_dt], [naaccr_first_contact_dt], "d", "mdy")</code>
Curated Patient Age at Diagnosis	ca_age PN0087	<code>rounddown(datediff([birth_dt], [ca_dx_dt], "y", "mdy"))</code>
Interval in days between the Date of Birth and the Curated Date of Diagnosis	ca_cadx_int PN0088	<code>datediff([birth_dt], [ca_dx_dt], "d", "mdy")</code>

Cancer Diagnosis Data Violations

Type	Description	Location
Major	Incorrect/missing cancer type	CA & Diagnosis Form
	Incorrect/missing diagnosis date [+/- 1 month]	CA & Diagnosis Form
	Incorrect/missing stage [AJCC metastatic/Stage IV v. other]	CA & Diagnosis Form
	Missed cancer diagnosis	QA Assurance
Minor	Incorrect/missing sites of distant metastases at diagnosis	CA & Diagnosis Form
	Missed/Incorrect Cancer Type-Specific questions	CA & Diagnosis Form
Other	Incorrect/missing source of diagnosis date	CA & Diagnosis Form
	Other minor mistake/omission	CA & Diagnosis Form

General Cancer-Directed Drug Therapy Guidelines

Provenance

Pharmacy Systems, Flow Charts, Administered Medications, Internal and External Medical Oncology Notes

Data Sources to be interrogated include:

Data Source	Description	Attributes
Pharmacy records from native site	All stop/start/treatment dates for antineoplastic therapies	Mostly structured
Prescription and flow sheets from native site	Captures some oral/hormonal therapies not dispensed by pharmacy	Semi-structured
Initial oncology visit notes at native site	Chemotherapy exposures prior to 1 st presentation at internal site	Entirely unstructured
Follow-up progress notes from native site when gap in treatment is >2 months	Captures chemotherapy exposures at external sites for patients with fragmented care	Entirely unstructured

Data Standard

NCIt Antineoplastic List: in the case of two codes for one drug, the code with the experimental name(s) is chosen.

Future: HemOnc

Selecting Cancer-Directed Drugs to Curate

- **Do** curate all drugs that are used to treat **all the patient's cancer diagnoses**, not just the cancer of interest. Anti-neoplastics, hormone therapy, hormone therapy, immunotherapy, are examples of the cancer-directed drug therapy.
- **Do not** include drugs used to support the patient's health during cancer treatment like growth factors, anti-emetics, and bisphosphonates (e.g. pegfilgrastim (Neulasta)).
- **Do not** include the following drugs that have both anti-neoplastic and other activities: denosumab, Pamidronate Disodium, steroids.
- **Begin** by reviewing the **Cancer Diagnosis** forms. **Do** note the instance number of each diagnosis, the date of diagnosis, and the cancer type.
 - You will need this information later to link each drug regimen to a specific cancer diagnosis.
 - This will also give you an idea of the date range you need to review.
- Next, review the most recent note from a medical oncologist. Scan quickly to see if there is a summary of the patient's treatment history.
 - This may fall under the "Oncology History" section, "History of Present Illness (HPI)" or similar section.
- If a detailed history is not present, **do** check a couple more notes, such as the initial internal consult note or a note from the middle of the treatment course. If the patient saw multiple oncologists, check a note from a different provider to see if their documentation is more detailed.
 - **Do** read approximately 3 notes looking for this information.
 - **Do not** immediately curate treatment information directly from the provider notes. These provide a good outline of the regimens to curate and can be used if no other information is present, but often contain typos or other errors that may be significant.
- Once you have the outline of the patient's treatment course, look at EHR treatment administration records (e.g. infusion flowsheets) for details.

- **Do** curate regimens that were administered at outside institutions, but **do not** spend more than a couple minutes looking for details.
 - If scanned notes exist from that institution and time period, check one or two of these notes to confirm details of the regimen(s) in question.
 - If these notes cannot be easily located, use information from DFCI notes and estimate dates if needed.
- **Do** start a new instance for each regimen. See directives below for more detail on determining what constitutes one regimen.
- **Oral Chemotherapy:** Since the curation of oral chemotherapy may not have actual administrative data in the EHR, some business rules are in place for these curations.
 - Oral chemotherapy start date= prescription date unless otherwise specified
 - If no end date information is available from the medical oncologist, the duration on oral/hormonal chemotherapy is estimated based on # of pills dispensed at last prescription
- For the purpose of this project, regimens are defined as the drugs that are given together.
 - Regimens will have between 1 to 5 antineoplastic drugs given together.
 - Drugs within a regimen may or may not have the same start or end dates.
 - A drug within a regimen can be held for up to 8 weeks; if the drug is restarted after 8 weeks, a new regimen will need to be entered that includes all the active drugs that were administered.
 - If the pausing of drug 1 is longer than 8 weeks, this should be two regimens with:
Regimen 1: Drug 1 first start date, Drug 1 first end date; Drug 2 start date, Drug 2 end date = the drug's administration before Drug 1 is restarted
Regimen 2: (highlighted portion above): Drug 1 second start date, Drug 1 second end date; Drug 2 second start date (on or after the second start date for Drug 1), Drug 2 second end date
 There are internet sites like: <https://www.timeanddate.com/date/duration.html> that you can use to determine whether a drug/regimen is paused for 8 weeks or more.
- **Timing is critical!** Many of these patients received care at more than one institution; many have more than one cancer diagnosis. We may not have complete records from their entire course of cancer care. Even when records are available, they may not contain as much detail as we want. Try not to spend more than 15 minutes on cancer-directed therapy for a case. **Do** try to avoid major violations, but **do not** worry about avoiding minor violations if that information is not readily available.

REDCap Form
ca_directed_drugs

Cancer-Directed Drug Therapy Directives

Cancer diagnosis associated with the regimen

[drugs_ca] PN0206

Values

Multiselect checkboxes; whole number 1-10 representing cancer diagnosis and “Unknown”

Directives

- Required. Every antineoplastic therapy instance should be associated with one or more cancer diagnosis **Cancer Sequence Number**.
- Link the treatment regimen with the cancer diagnosis or the cancer diagnoses that it was intended to treat by checking the appropriate Curated **Cancer Sequence Number** of that cancer diagnosis corresponding to the **Cancer Sequence Number** found in the Cancer Staging and Diagnosis section. Summary information about the cancer diagnoses is provided in REDCap to assist in selection.
- Do not use the value from the **Tumor Registry: Cancer sequence number of this cancer populated from the Tumor Registry** field.
- A regimen will rarely be linked to more than one cancer diagnosis, but this is possible. For example, if a patient has two different primary cancers that require treatment at the same time, a provider may prescribe chemotherapy intended to treat both. In this case, select the **Curated Cancer Sequence Number** of each cancer.

Shared Field: Yes

Where were the drugs for this line of therapy administered/ordered for the patient?

[drugs_inst] PN0207

Values

- 1, At the internal/native institution only
- 2, Split across internal and external institution
- 3, At external institution only

Directives

- Choose the response that best reflects where the patient received any treatment that is part of this regimen. For oral drugs, only consider institutions that are prescribing those drugs.
- Please note that the codes (numbers) for the choices are not visible in REDCap; however, the codes will be available at data export.

Shared Field: Yes

Institution that the cancer-directed drug therapy was first ordered

[drugs_firstinst] PN0208

Values

- 1, Internal institution
- 2, External institution

Directives

- Linked to **Where were the drugs for this line of therapy administered/ordered for the patient?** If “2, Split across internal and external institution” was selected, the data field is available for curation.
- Select where the institution in which the physician first orders the drug therapy.
- Please note that the codes (numbers) for the choices are not visible in REDCap; however, the codes will be available at data export.

Shared Field: Yes

Is the cancer-directed drug therapy part of a clinical trial?

[drugs_ct_yn] PN0209

Values

- 1, Yes
- 0, No

Directives

- Record whether the drug therapy was part of a clinical trial, meaning one or more drugs in the regimen are being tested as part of an interventional clinical trial.
- The provider may refer to a trial, research study, or protocol. In some institutions, drug administration logs may indicate if a drug is part of a study by using the prefix “ID-” or “investigational drug” before the drug name.
- Select “1, Yes” if this drug therapy was part of an interventional blinded, placebo, SOC, or multiple alternative treatment options clinical trial.
- Select “1, Yes” if this drug therapy was part of an open-label trial.
- Select “1, Yes” if the type of trial is not explicitly mentioned.
- Select “0, No” for standard of care drug therapy. If a clinical trial or research study is not mentioned, assume the response is “0, No”.
- Please note that the codes (numbers) for the choices are not visible in REDCap; however, the codes will be available at data export.

Analytic Note

- This data element will be used to mask the names and codes of investigational drugs prior to data submission to GENIE or other external data sharing projects. If Is the cancer-directed drug therapy part of a clinical trial is “1, Yes” at least one of the drugs should be masked with the code/label, 49, New Investigational Drug for data submission.

Shared Field: Yes

(If an investigational drug therapy) Protocol Number

[drugs_prot_num] PN0210

Value

Free text

Directives

- Linked to **Is the cancer-directed drug therapy part of a clinical trial?** If any “Yes” choice is selected, the data field is available for curation.
- If the any drug in the regimen was part of a clinical trial, record the institutional protocol number for the study.
- If the patient participated in this clinical trial at an outside institution, enter whatever information is available (trial name, local protocol number, NCI study number), or leave blank if not readily available.

Shared Field: No, for institutional use only

How many cancer-directed drugs were received together?

[drugs_num] PN0211

Values

Numeric 1 – 5

Directives

- Count all drugs meant to directly treat the cancer, including chemotherapy, anti-neoplastic, immunotherapy, biologics, and hormone therapy. Do not include supportive care drugs or drugs meant to treat symptoms (e.g. ondansetron, dexamethasone).
- Choose the number of drugs that were included in the regimen. Cancer-directed drugs that were initiated within 30 days of each other are considered one regimen.
- If a medical oncologist states that a patient will receive FOLFOX (5-FU/Leucovorin/Oxaliplatin), but holds the oxaliplatin for several cycles, consider that the patient received 3 drugs together.
- If the drug(s) are paused for 8 or more weeks, the regimen should be discontinued. Start a new instance when the drug(s) are restarted.
- If a new drug is added to an existing regimen, the regimen should be discontinued. Start a new instance beginning with the initiation date of the new drug.
- If a drug is replaced in the regimen, then the regimen should be discontinued; a new regimen should be started.

Shared Field: Yes

For Non-Standard Drug Administration Only: If the drug/drugs are not administered as either oral or as IV or infused, select the route of administration.

[drugs_admin] PN0978

Values

- 1, Intraperitoneal (direct into the abdominal cavity)
- 2, Intrapleural (direct into the pleural space around the lung)
- 3, Intralesional (direct into a lesion)
- 4, Intrathecal (by spinal catheter)
- 5, Intravesical (bladder installation)
- 6, Other

Directives

- If the anti-neoplastic drugs are not administered as iv/infused or oral, indicate the method of administration.
- For HIPEC, record “1, Intraperitoneal.”
- For hepatic arterial infusion (HAI), record “6, Other.”
- For intravitreal administration, record “6, Other.”

History

042020: Added to capture non-standard drug administration to v344.

Shared Field: Yes

Has this cancer-directed drug therapy been discontinued?

[drugs_dc_yn] PN0212

Values

- 1, Yes
- 2, No
- 3, Unknown, no documentation found

Directives

- Record “Yes” if the regimen has been discontinued. A regimen has been discontinued if:
 - All drugs included have been stopped.
 - OR
 - A new drug has been initiated.

OR

- There was a treatment break or hold that lasted for 8 or more weeks.

Answer “No” if:

- The regimen is continuing at time of curation.

Answer “Unknown” if:

- The regimen was ongoing at time of most recent documentation and there is no further follow-up.
- This includes patients we know are deceased; although we know treatment is not ongoing, only choose “Yes” if there is explicit documentation that meets the above criteria.

Shared Field: Yes

Coded Name of Drug, Drugs 1 through 5

[drugs_drug_1], [drugs_drug_2], [drugs_drug_3], [drugs_drug_4], [drugs_drug_5] PN0214, PN0222, PN0230, PN0238, PN0246

Value: Dropdown list cancer-directed drugs and synonyms with NCI codes

Data Standard: NCI Thesaurus: List of Antineoplastic Drugs

Directives

- The number of drug data fields is controlled by the data field: **How many cancer-directed drugs were received together?** The number of Coded Name of Drug fields available for curation will depend on the number entered.
- Select the name of the drug given from the existing list whenever possible. The list of cancer-directed drugs includes the generic names, experimental names, and trade name synonyms. REDCap has a type ahead functionality for ease of data entry.
- If the drug is not listed in the pull-down list record, “Other NOS”, ‘Other Antineoplastic’, ‘Other Hormone’, or ‘Clinical Trial Drug’ and record the name of the drug in the field **Name of Cancer-Directed Drug, Other**.
- If the drug is part of clinical trial that is Drug(s) vs. Placebo, enter the coded name of the drug (not placebo). Even if the patient is later unblinded and was on placebo, still always use enter the drug name. Indicate in the “Notes” box at the bottom of the form that the trial was drug vs. placebo.

Analytic Note

- The cancer drug label contains the generic/ingredient name with the synonyms in parentheses: Nivolumab(BMS936558,MDX1106,NIVO,ONO4538,Opdivo)
For result display the content in parentheses can be removed.
- To create a regimen, the names of the drugs can be mapped using HemOnc.org
- Linked to **Is the cancer-directed drug therapy part of a clinical trial?** if “1, Yes” is recorded, the code of this drug must be masked prior to data submission to GENIE or similar data sharing projects with the code/label, “49135, Investigational New Drug” for data submission.

History

02252020: Updated the antineoplastic drug list.

Shared Field: Yes; however, if a drug is part of an investigational drug trial, the code and name must be masked with 49135, Investigational New Drug.

Name of Cancer-Directed Drug, 1 through 5, Not Listed

[drugs_drug_oth_1], [drugs_drug_oth_2], [drugs_drug_oth_3], [drugs_drug_oth_4], [drugs_drug_oth_5] PN0215, PN0223, PN0231, PN0239, PN0247

Value

Free text

Directives

- Linked to **Is the cancer-directed drug therapy part of a clinical trial?** If any “Other” is selected, the data field is available for curation.
- If you chose “Other” above, type the name of the drug here. Use the generic name instead of a brand name if possible (e.g. “paclitaxel” instead of “Taxol”).

Shared Field: No

Start Date of Drug, Drugs 1 through 5

[drugs_startdt_drug_1], [drugs_startdt_drug_2], [drugs_startdt_drug_3], [drugs_startdt_drug_4], [drugs_startdt_drug_5] PN0216, PN0224, PN0232, PN0240, PN0248

Value

Date field, mmddyyyy

Directives

- Enter the first date of dosing for each cancer-directed drug. Unless no other information is available, do not use the date the drug was ordered or prescribed since this does not indicate the patient actually received that drug.
- If this drug was continued from the prior regimen, use the first date it was administered/taken in combination with the new drug(s) as the start date in the new regimen.

Shared Field: No, contains PHI. The data elements are flagged as an identifier in REDCap. Instead, calculated intervals in days [drugs_startdt_int_1, drugs_startdt_int_2, drugs_startdt_int_3, drugs_startdt_int_4, drugs_startdt_int_5] from **Calculated Date of Birth** are generated for each start date.

End Date of Drug, Drugs 1 through 5

[drugs_enddt_drug_1], [drugs_enddt_drug_2], [drugs_enddt_drug_3], [drugs_enddt_drug_4], [drugs_enddt_drug_5] PN0218, PN0226, PN0234, PN0242, PN0250

Value

Date field, mmddyyyy

Directives

- Linked to **Has this cancer-directed drug therapy been discontinued?** If “Yes” is selected, this field will only be available for curation.
- Enter the last date the patient received/took this drug. For oral medications, this can be long after the last time it was prescribed, so you are more likely to find the date in clinic notes. If you cannot find an exact date of discontinuation after looking at notes for a couple minutes, make an estimate using the information you have.
- If you are discontinuing the regimen because a new drug was added, but this drug is continuing, use the last date these drugs were administered without the new drug as the discontinuation date in this regimen.

Shared Field: No, contains PHI. The data elements are flagged as an identifier in REDCap. Instead, calculated intervals in days [drugs_enddt_int_1, drugs_enddt_int_2, drugs_enddt_int_3, drugs_enddt_int_4, drugs_enddt_int_5] from **Calculated Date of Birth** are generated for each end date.

Last Known Date for Administration/Order of Cancer-Directed Drug, Drugs 1 through 5

[drugs_lastdt_drug_1], [drugs_lastdt_drug_2], [drugs_lastdt_drug_3], [drugs_lastdt_drug_4], [drugs_lastdt_drug_5] PN0220, PN0228, PN0236, PN0244, PN0252

Value

Date field, mmddyyyy

Directives

- Linked to **Has this cancer-directed drug therapy been discontinued?** If “Unknown” is selected, this field is available for curation.
- Record the last date the patient received the drug. If that is unknown, use the last date it was ordered or prescribed.

Shared Field: No, contains PHI. The data elements are flagged as an identifier in REDCap. Instead, a calculated intervals in days [**drugs_lastdt_int_1, drugs_lastdt_int_2, drugs_lastdt_int_3, drugs_lastdt_int_4, drugs_lastdt_int_5**] from **Calculated Date of Birth** are generated for each last known date of administration.

Cancer-Specific Cancer-Directed Drug Therapy Data Fields

The following data fields are specific to Renal Cell Carcinoma. You do not need to collect the following fields if Renal Cell Carcinoma is not one of the cancers of interest for your project, and they should be hidden from curator view in your database.

Was this regimen administered as first line systemic treatment for recurrent or metastatic renal cell carcinoma?
[drugs_kid_first_line_yn] PN1236

Values

- 1, Yes
- 0, No

Directives

- This field should only be collected for kidney renal cell carcinoma cases. If your project does not include renal cell carcinoma, this field may be hidden by using the @HIDDEN action tag in the Action Tag box in REDCap.
- If the patient was Stage IV at RCC diagnosis and this is the first systemic therapy they received to treat their disease, choose “Yes”.
- If the patient did not have stage IV RCC at diagnosis, review 1-2 medical oncologist notes from around the time of treatment start to determine whether the patient had developed distant metastases or recurrence following nephrectomy prior to treatment initiation (may be described as Stage IV, metastatic/locoregional recurrence, or description of involvement of various distant anatomic sites).
- If the patient has evidence of distant metastases or locoregional recurrence, and this is the first systemic therapy they received since developing metastatic disease/recurrence, answer yes to the question “Was this regimen administered as first line systemic treatment for metastatic renal cell carcinoma?”
- N.B.: Do not answer yes if this treatment was administered as part of adjuvant therapy plan for patients with stage I-III- disease at diagnosis.

Shared Field: Yes

History: 11062023 field added to dictionary

What was the patient's KPS prior to the earliest drug start date?
[drugs_imdc_kps] PN1237

Values

100, 100: 100 Normal no complaints; no evidence of disease. | 90, 90: Able to carry on normal activity; minor signs or symptoms of disease. Able to carry on normal activity and to work; no special care needed. | 80, 80: Normal activity with effort; some signs or symptoms of disease. | 70, 70: Cares for self; unable to carry on normal activity or to do active work. | 60, 60: Requires occasional assistance, but is able to care for most of his personal needs. Unable to work; able to live at

home and care for most personal needs; varying amount of assistance needed. | 50, 50: Requires considerable assistance and frequent medical care. | 40, 40: Disabled; requires special care and assistance. | 30, 30: Severely disabled; hospital admission is indicated although death not imminent. | 20, 20: Very sick; hospital admission necessary; active; supportive treatment necessary. | 10, 10: Moribund; fatal processes progressing rapidly | 9, Not documented in the note

Directives

- Linked to **“Was this regimen administered as first line systemic treatment for recurrent or metastatic renal cell carcinoma?”** If “1, Yes” is answered, then this field will be available for curation.
- Review the medical oncology note just prior to the date of treatment start and record the KPS score documented in the chart. If KPS was not documented, continue reviewing the next most recent medical oncologist note until you find a score, looking no further than 30 days prior to treatment initiation.
- If no KPS score is documented within 30 days prior to treatment start, use the ECOG score, following the same directives above. If ECOG is also unavailable, record “unknown”.
- Prioritize KPS score over ECOG if both are available within the 30-day timeframe.

Analytic Note

This field can be used to calculate the IMDC (International Metastatic RCC Database Consortium) risk score in the final dataset.

Shared Field: Yes

History: 11062023 field added to dictionary

What was the patient’s ECOG prior to the earliest drug start date?

[drugs_imdc_ecog] PN1238

Values

0, 0: Fully active, able to carry on all pre-disease performance without restriction | 1, 1: Restricted in physically strenuous activity but ambulatory and able to carry out work of a light or sedentary nature, e.g., light house work, office work | 2, 2: Ambulatory and capable of all selfcare but unable to carry out any work activities; up and about more than 50% of waking hours | 3, 3: Capable of only limited selfcare; confined to bed or chair more than 50% of waking hours | 4, 4: Completely disabled; cannot carry on any selfcare; totally confined to bed or chair | 9, Not documented in note

Directives

- Linked to **“Was this regimen administered as first line systemic treatment for recurrent or metastatic renal cell carcinoma?”** If “9, Not documented in the note” is answered, then this field will be available for curation.
- Review the medical oncology note just prior to the date of treatment start and record the ECOG score documented in the chart. If ECOG was not documented, continue reviewing the next most recent medical oncologist note until you find a score, looking no further than 30 days prior to treatment initiation.
- If ECOG is also unavailable, record “Not documented in the note”.
- Prioritize KPS score over ECOG if both are available within the 30-day timeframe.

Analytic Note

This field can be used to calculate the IMDC (International Metastatic RCC Database Consortium) risk score in the final dataset.

Shared Field: Yes

History: 11062023 field added to dictionary

Comprehensive Metabolic Panel Results

Calcium [drugs_imdc_cal] PN1239

Albumin [drugs_imdc_alb] PN1240

Albumin corrected calcium [drugs_imdc_correct_cal] PN1241

Values

Numeric

Directives

- Linked to “**Was this regimen administered as first line systemic treatment for recurrent or metastatic renal cell carcinoma?**” If “1, Yes” is answered, then this field will be available for curation.
- Open the laboratory report section of your EMR and find the first comprehensive metabolic panel test results prior to the first drug start date for this regimen, looking up to 30 days prior to treatment start, if needed.
- If multiple tests were performed during this period, use the results closest to the date of treatment start. Tests performed on the day of treatment administration can be used as long as blood was drawn prior to treatment start.
- Enter the calcium result in mg/dl. Convert the values prior to entering them if reported using other units.
- Enter the albumin result in g/dl. Convert the values prior to entering them if reported using other units.
- **Albumin corrected calcium** is a calculated field.

Analytic Note

These fields can be used to calculate the IMDC (International Metastatic RCC Database Consortium) risk score in the final dataset.

Shared Field: Yes

History: 11062023 field added to dictionary

Complete Blood Count Results

Hemoglobin [drugs_imdc_heme] PN1242

Platelets [drugs_imdc_platelet] PN1243

Absolute Neutrophils [drugs_imdc_neutro] PN1244

Values

Numeric

Directives

- Linked to “**Was this regimen administered as first line systemic treatment for recurrent or metastatic renal cell carcinoma?**” If “1, Yes” is answered, then this field will be available for curation.
- Open the laboratory report section of your EMR and find the first complete blood count test results prior to the first drug start date for this regimen, looking up to 30 days prior to treatment start, if needed.
- If multiple tests were performed during this period, use the results closest to the date of treatment start. Tests performed on the day of treatment administration can be used if blood was drawn prior to treatment start.
- Enter the hemoglobin in g/dl, and platelets and absolute neutrophils counts in 1000 cells/μL. Convert the values

prior to entering them if reported using other units.

Analytic Note

These fields can be used to calculate the IMDC (International Metastatic RCC Database Consortium) risk score in the final dataset.

Shared Field: Yes

History: 11062023 field added to dictionary

Cancer-Directed Drug Therapy Calculated Fields

In order to share data in a de-identified manner, several intervals are calculated in the REDCap project from the anchor date: Date of Birth as follows:

Interval Data Field	Data Element	Calculation
Interval in Days from DOB to Start Date Drug 1	drugs_startdt_int_1 PN0217	datediff([birth_dt], [drugs_startdt_drug_1],"d","mdy")
Interval in Days from DOB to End Date Drug 1	drugs_enddt_int_1 PN0219	datediff([birth_dt], [drugs_enddt_drug_1],"d","mdy")
Interval in Days from DOB to Last Known Date Drug 1	drugs_lastdt_int_1 PN0221	datediff([birth_dt], [drugs_lastdt_drug_1],"d","mdy")
Interval in Days from DOB to Start Date Drug 2	drugs_startdt_int_2 PN0225	datediff([birth_dt], [drugs_startdt_drug_2],"d","mdy")
Interval in Days from DOB to End Date Drug 2	drugs_enddt_int_2 PN0227	datediff([birth_dt], [drugs_enddt_drug_2],"d","mdy")
Interval in Days from DOB to Last Known Date Drug 2	drugs_lastdt_int_2 PN0229	datediff([birth_dt], [drugs_lastdt_drug_2],"d","mdy")
Interval in Days from DOB to Start Date Drug 3	drugs_startdt_int_3 PN0233	datediff([birth_dt], [drugs_startdt_drug_3],"d","mdy")
Interval in Days from DOB to End Date Drug 3	drugs_enddt_int_3 PN0235	datediff([birth_dt], [drugs_enddt_drug_3],"d","mdy")
Interval in Days from DOB to Last Known Date Drug 3	drugs_lastdt_int_3 PN0237	datediff([birth_dt], [drugs_lastdt_drug_3],"d","mdy")
Interval in Days from DOB to Start Date Drug 4	drugs_startdt_int_4 PN0241	datediff([birth_dt], [drugs_startdt_drug_4],"d","mdy")
Interval in Days from DOB to End Date Drug 4	drugs_enddt_int_4 PN0243	datediff([birth_dt], [drugs_enddt_drug_4],"d","mdy")
Interval in Days from DOB to Last Known Date Drug 4	drugs_lastdt_int_4 PN0245	datediff([birth_dt], [drugs_lastdt_drug_4],"d","mdy")
Interval in Days from DOB to Start Date Drug 5	drugs_startdt_int_5 PN0249	datediff([birth_dt], [drugs_startdt_drug_5],"d","mdy")
Interval in Days from DOB to End Date Drug 5	drugs_enddt_int_5 PN0251	datediff([birth_dt], [drugs_enddt_drug_5],"d","mdy")
Interval in Days from DOB to Last Known Date Drug 5	drugs_lastdt_int_5 PN0253	datediff([birth_dt], [drugs_lastdt_drug_5],"d","mdy")

Cancer-Directed Therapy Data Violations

Major	Missing Regimen (if more than 1 drug in a regimen is missed, count this as a missing regimen).
	Incorrect overall start date [+/- 1 month]
	Regimen associated with wrong diagnosis
Minor	Missing/Incorrect drug in regimen
	Incorrect end date [+/- 1 month]
	Missing/Incorrect clinical trial information
	Drugs grouped as regimen incorrectly
	Missing/Incorrect discontinuation status
Other	Missing/Incorrect administration institution
	Other minor mistake/omission

NB: Radiation Therapy may not be active in all PRISMM systems.

General Cancer-Directed Radiation Therapy Guidelines

Provenance

Internal and External Radiation Oncology Notes, Internal and External Medical Oncology Notes

Data Sources to be interrogated include:

Data Source	Description	Attributes
Radiation Therapy Note	All stop/start/treatment dates for antineoplastic therapies	Semi-structured
Prescription and flow sheets from native site	Captures some oral/hormonal therapies not dispensed by pharmacy	Semi-structured
Initial medical oncology visit notes at native site	Radiation therapy exposures prior to 1 st presentation at internal site	Entirely unstructured
Follow-up progress notes from native site	Captures chemotherapy exposures at external sites for patients with fragmented care	Entirely unstructured

Data Standard

NAACCR: Commission on Cancer (COC)

General Directives

- **Do** curate all radiation therapy that the patient has received starting at the time of the diagnosis of the index cancer, using the curated date of diagnosis. This may include radiation therapy for other cancers.
- **Do not** include radiation therapy received prior to the curated date of diagnosis.
- **Begin** by reviewing the **Cancer Diagnosis** forms to find the curated date of diagnosis for the index cancer.
- **Look** for radiation therapy received in the internal institution on or after this date.
- Next, review the most recent note from a medical oncologist. Scan quickly to see if there is a summary of the patient's treatment history to find a summary of radiation therapy received by the patient at an external institution.
 - This may fall under the "Oncology History" section, "History of Present Illness (HPI)" or similar section..
 - **Do not** immediately curate treatment information directly from the provider notes. These provide a good outline of the regimens to curate and can be used if no other information is present, but often contain typos or other errors that may be significant.
- If scanned notes exist from an external institution and time period, review for radiation oncology.
 - If these notes cannot be easily located, use information from DFCI notes and estimate dates if needed.
 - **Do** start a new instance for each regimen.

- **Timing is critical!** Many of these patients received care at more than one institution; many have more than one cancer diagnosis. We may not have complete records from their entire course of cancer care. Even when records are available, they may not contain as much detail as we want. Try not to spend more than 15 minutes on cancer-directed therapy for a case. **Do** try to avoid major violations, but **do not** worry about avoiding minor violations if that information is not readily available.

REDCap Form
ca_directed_radt

Cancer Directed Radiation Therapy Directives

Cancer diagnosis associated with the radiation therapy
[rt_ca] PN0815

Values

Multiselect checkboxes; whole number 1-10 representing cancer diagnosis and “Unknown”

Directives

- Required. Every radiation therapy instance should be associated with one or more cancer diagnosis **Cancer Sequence Number**.
- Link the treatment regimen with the cancer diagnosis or the cancer diagnoses that it was intended to treat by checking the appropriate Curated **Cancer Sequence Number** of that cancer diagnosis corresponding to the **Cancer Sequence Number** found in the Cancer Staging and Diagnosis section. Summary information about the cancer diagnoses is provided in REDCap to assist in selection.
- Do not use the value from the **Tumor Registry: Cancer sequence number of this cancer populated from the Tumor Registry** field.
- Radiation therapy will rarely be linked to more than one cancer diagnosis, but this is possible. For example, if a patient has two different primary cancers that require treatment at the same time, a provider may initiate radiation therapy intended to treat both, or it may not be possible to determine which cancer is being targeted. In these situations, select the **Curated Cancer Sequence Number** of each cancer.

Shared Field: Yes

Is a summary of the radiation therapy available?
[rt_summ] PN0816

Values

1, Yes, from the Radiation Oncologist | 2, Yes, from the Medical Oncologist | 0, No

Directives

- If a summary of the radiation therapy from either the radiation oncologist or the medical oncologist is available in the EHR, please record the appropriate response. If the details are available from both, choose “1, Yes, from the Radiation Oncologist.” The summary should include the dates, modality, planning technique, doses, and type of radiation therapy. A notation from a provider that includes the dates does not constitute a summary of the radiation therapy.
- If a summary of the therapy is not available from either the radiation oncologist or the medical oncologist, record), “No.”
- Please note that the codes (numbers) for the choices are not visible in REDCap; however, the codes will be available at data export.

Shared Field: Yes

Where was the radiation therapy administered?[rt_inst] PN0817

Values

1, At the internal/native institution only|2, Split across internal and external institution|3, At external institution only

Directives

- Record where the radiation therapy was administered.
- Please note that the codes (numbers) for the choices are not visible in REDCap; however, the codes will be available at data export.

Shared Field: Yes

Start Date of Radiation Therapy

[rt_startdt] PN0820

Value

Date field, mmddyyyy

Directives

- Enter the first date of radiation therapy. Estimate the date if it is not provided.

Shared Field: No, contains PHI. The data element is flagged as an identifier in REDCap. Instead, the calculated intervals in days [rt_startdt_int] from **Calculated Date of Birth** are generated for each start date.

End Date of Radiation Therapy

[rt_enddt] PN0821

Value

Date field, mmddyyyy

Directives

- Enter the last date of radiation therapy. If there is no information is available, leave blank.

Shared Field: No, contains PHI. The data element is flagged as an identifier in REDCap. Instead, the calculated intervals in days [rt_startdt_int] from **Calculated Date of Birth** are generated for each start date.

Radiation Treatment Modality

[rt_type] PN0818

Values

1, External beam, NOS|2, External beam, photons|3, External beam, protons|4, External beam, electrons|5, External beam, neutrons|6, External beam, carbon ions|7, Brachytherapy, NOS|8, Brachytherapy, intracavitary, LDR|9, Brachytherapy, intracavitary, HDR|10, Brachytherapy, Interstitial, LDR|11, Brachytherapy, Interstitial, HDR|12, Brachytherapy, electronic|13, Radioisotopes, NOS|15, Radioisotopes, Strontium-89|16, Radioisotopes, Strontium-90|17, Radioembolization, Yttrium-90|99, Treatment radiation modality unknown

Directives

- Record the treatment modality for the radiation therapy.
- Please note that the codes (numbers) for the choices are not visible in REDCap; however, the codes will be available at data export.

Data Standards

NAACCR # 1506, 1516, 1526

Shared Field: Yes

Radiation External Beam Planning Technique [rt_planning] PN0819

Values

- 1, External beam, NOS
- 2, Low energy x-ray/photon therapy
- 3, 2-D therapy
- 4, Conformal or 3-D conformal therapy
- 5, Intensity modulated therapy
- 6, Stereotactic radiotherapy or radiosurgery, NOS
- 7, Stereotactic radiotherapy or radiosurgery, robotic
- 8, Stereotactic radiotherapy or radiosurgery, Gamma Knife
- 9, CT-guided adaptive therapy
- 10, MR-guided online adaptive therapy
- 88, Not Applicable
- 98, Other, NOS
- 99, Unknown

Directives

- Record the external beam planning technique for the radiation therapy. If **Radiation Treatment Modality** for the specific phase is external beam (choices 1, 2, 3, 4, 5, 6) then this field is available for curation.
- Curators can refer to the [2018 Standards for Oncology Registry Entry \(STORE\) manual](#) for additional guidance on how to curate planning technique (pages 288-289).
- Please note that the codes (numbers) for the choices are not visible in REDCap; however, the codes will be available at data export.

Shared Field: Yes

Anatomic Sites Treated with Radiation Therapy [rt_site] PN0824

Values - multiselect

- 1, Neck lymph node regions
- 2, Thoracic lymph node regions
- 3, Neck and thoracic lymph node regions
- 4, Breast/ Chest wall lymph node regions
- 5, Abdominal lymph nodes
- 6, Pelvic lymph nodes
- 7, Abdominal and pelvic lymph nodes
- 9, Lymph node region, NOS
- 10, Eye/orbit/optic nerve
- 11, Pituitary
- 12, Brain
- 13, Brain (Limited)
- 14, Spinal cord
- 20, Nasopharynx
- 21, Oral Cavity
- 22, Oropharynx
- 23, Larynx (glottis) or hypopharynx
- 24, Sinuses/Nasal tract
- 25, Parotid or other salivary glands

26, Thyroid
 29, Head and neck (NOS)
 30, Lung or bronchus
 31, Mesothelium
 32, Thymus
 39, Chest/lung (NOS)
 40, Breast - whole
 41, Breast - partial
 42, Chest wall
 50, Esophagus
 51, Stomach
 52, Small bowel
 53, Colon
 54, Rectum
 55, Anus
 56, Liver
 57, Biliary tree or gallbladder
 58, Pancreas or hepatopancreatic ampulla
 59, Abdomen (NOS)
 60, Bladder - whole
 61, Bladder - partial
 62, Kidney
 63, Ureter
 64, Prostate - whole
 65, Prostate - partial
 66, Urethra
 67, Penis
 68, Testicle or scrotum
 70, Ovaries or fallopian tubes
 71, Uterus or Cervix
 72, Vagina
 73, Vulva
 80, Skull
 81, Spine/vertebral bodies
 82, Shoulder
 83, Ribs
 84, Hip
 85, Pelvic bones
 86, Pelvis (NOS, non-visceral)
 88, Extremity bone, NOS
 90, Skin
 91, Soft tissue
 92, Hemibody
 93, Whole body
 94, Mantle, mini-mantle (obsolete after 2017)
 95, Lower extended field (obsolete after 2017)
 96, Inverted Y (obsolete after 2017)

Directives

- Record the site/sites that was treated. Curators can refer to the [2018 Standards for Oncology Registry Entry \(STORE\) manual](#) for additional guidance on how to curate anatomic site (pages 297-300).
- Please note that the codes (numbers) for the choices are not visible in REDCap; however, the codes will be

available at data export.

Data Standards

NAACCR # 1504, 1514, 1524

Shared Field: Yes

Radiation Dose per Fraction (cGy)

[rt_dose] PN0825

Values

Numeric

Directives

- Record the dose per fraction in Centigray (cGy).

Shared Field: Yes

Total Number of Fractions Received

[rt_fractions] PN0826

Values

Numeric

Directives

- Record the total number of fractions received.
- Enter the total number of fractions of radiation therapy that were administered. This may be summarized in the text of the completion note. If not, find the number of fractions delivered (not planned) to each site and add them to get the total.

Data Standards

NAACCR #1503

History

12162020: Field added to dictionary.

Shared Field: Yes

Radiation Therapy Total Dose Received

[rt_total_dose] PN0825

Values

Numeric

Directives

- Record the total dose received in Centigray (cGy).
- Occasionally dose is reported in Gray (Gy), in which case you need to multiply by 100 to convert to cGy. The total dose is usually reported in a completion note; if not, find the total delivered dose for each site and add them to get the total.

Data Standards

NAACCR #1501

Shared Field: Yes

Cancer-Directed Radiation Therapy Calculated Fields

In order to share data in a de-identified manner, several intervals are calculated in the REDCap project from the anchor date: Date of Birth as follows:

Interval Data Field	Data Element	Calculation
Interval in Days from DOB to Start Date RT	rt_startdt_int PN	datediff([birth_dt], [rt_startdt],"d","mdy")
Interval in Days from DOB to End Date RT	rt_enddt_int PN	datediff([birth_dt], [rt_enddt],"d","mdy")

Cancer-Directed Radiation Therapy Data Violations

Major	Missing RT
	Incorrect overall start date [+/- 1 month]
	RT associated with wrong diagnosis
Minor	Missing modality
	Incorrect end date [+/- 1 month]
	Incorrect dose
	Incorrect site
Other	Missing/Incorrect administration institution
	Other minor mistake/omission

General Pathology Curation Guidelines

PRISMM Pathology (P)

Pathologic evidence of locoregional or distant tumor

Provenance

- Internal (native) and External Pathology Reports, including result reports from external 3rd party laboratories
- Full P curation includes review of both native and external pathology reports

Selecting Pathology Reports to Curate

- Find the date of diagnosis of the index cancer – cancer of interest.
- **Do** review and enter all pathology reports for cytology and surgery procedures performed on or after date of diagnosis of the cancer being curated.
- **Do** curate all pathology and cytology reports from procedures that were performed on or after the earliest diagnosis date. These are most often labeled “Surgical Pathology” or “Cytology.”
 - Reports on specimens from many types of procedures are eligible, including (but not limited to) core needle biopsies, excisional biopsies, aspirations, fine needle aspirations, surgical procedures (such as colectomy or lobectomy), brushings/washings, bone reamings, cytology, etc.
 - Include all sites of invasive cancer, including the primary tumor, loco-regional recurrence, and distant metastases.
 - If patients have more than one cancer diagnosis, include reports relating to all sites of solid tumor disease, regardless of primary invasive cancer.
- There may be multiple procedures reviewed in the same report. **Do** curate each procedure as a separate form in REDCap.
- In the case of multiple tumors of the same type, NSCLC for example, **do** curate all pathology reports for these cancers if they occur prior to the diagnosis date of the index cancer.

Pathology Reports Not to Curate

- **Do not** curate pathology reports for cytogenetics. These are most often labeled “Interpretive Lab Tests,” “Genetics,” or Cytogenetics.”
- **Do not** curate pathology for skin specimens with basal cell or squamous cell carcinoma.
 - This means you will rarely need to curate skin punch biopsies or shave biopsies. **Do** review them to ensure they are not melanoma, site of metastatic disease from a different primary (such as breast cancer that has invaded the skin), or other cancer.
- **Do not** curate cytology for pap smears unless required for the cancer diagnosis that you are curating.
- **Do not** curate autopsy reports.
- **Do not** curate pathology reports labeled as gross description only, unsatisfactory specimen, non-diagnostic, or stent removals are an example of this type of report. If the internal review says a specimen was not reviewed “due to lack of medical necessity,” you do not need to look for additional information about that specimen in an external report.

Pathology Reports Requiring Review

- **Rarely** curate multiple reports about the same procedure/specimen. There may be multiple pathology reports for the same procedure or the same specimen because we have a report from the hospital where the procedure was originally performed and a report from when the specimen was reviewed at BWH, or because more tests were performed on the specimen and a new report is issued with updated information.
 - Verify that reports are about the same procedure/specimen by verifying the procedure date and original accession number.

- If all reports relating to that procedure/specimen include the same number/type of sites reviewed, the same primary cancer, and the same histology, **do not** curate all of them. **Do** curate only one and **do** give preference to internal institution review over an external review.
- If a duplicate external report includes additional pertinent information to the internal report, **do** curate it as a separate pathology instance. Relevant information includes:
 - Additional distinct sites of disease reviewed (for example, internal review includes only lung, but an external report about the same procedure includes lung and lymph nodes)
 - Type of primary cancer, if not stated on internal report
 - Type of histology, if not stated on internal report
- There is often a Surgical Pathology and a Cytology report from the same part of the body on the same day. These are still separate specimens and should be included as separate instances.

REDCap Form

- prissmm_pathology

PRISSMM Pathology Piped Data Elements

In the case that an institution can pipe in the meta data (data associated with the pathology report, like description and ids) for their pathology reports that will assist in the curation effort, the REDCap project contains several fields that are available to hold this data. If the institution does not choose to pipe-in the meta data, then these fields will not be visible to the curators.

Data Element Label	Data Element
Pathology Report Encounter ID	[path_enc_id]
Pathology Report Order ID	[path_ord_id]
Pathology Report Order Date	[path_ord_dt]
Pathology Report Description	[path_desc]
Is this Pathology Report appropriate for PRISSMM	[path_prissmm_yn]

PRISSMM Pathology Data Directives

Pathology Type: Cytology or Surgical Pathology

[path_proc_type] PN0272

Value

- 1, Cytology
- 2, Surgical pathology
- 4, Hematopathology
- 5, Flow Cytometry
- 3, Other

Directives

- Record the type of pathology report is being curated.
- Record “1, Cytology” to indicate pathology reports with cytologic based samples from aspirates, washings, or fine needle aspiration.
- Record “2, Surgical pathology” for reports that are labeled “surgical” or “anatomic” or from the following types of procedures: core needle biopsies, resections, and lobectomies.

History: 02212023 Added hematopathology and flow cytometry to the dropdown. Prior to this update, bone marrow biopsies were curated as “3, Other”.

Shared Field: Yes

Pathology Procedure

[path_proc] PN1080

Value

- 1, Biopsy
- 2, Surgical excision
- 3, Other
- 4, Unknown

Directives

- Only curated for surgical pathology reports.
- Indicate the surgical procedure. If the report is from a tumor resection, select Surgical excision.
- If there are separate specimens listed on the pathology report, one or more from a resection **and** one or more from a biopsy, then curate resection specimens on one instance and biopsy specimens on a separate instance.

History:

12162020: Added field to dictionary.

Shared Field: Yes

Does the excision have positive margins?

[path_proc_margins] PN1087

Value

- 0, No
- 1, Yes
- 10, Indeterminate/Uncertain

Directives

- Only curated for surgical excisions.
- Curate “Yes” if the surgical resection report indicates positive margins. Curate “No” if the report states margins are negative or uninvolved by tumor.
- If the pathology report does not comment on margins, curate “Indeterminate/Uncertain”.

History:

03262021: Added field to dictionary.

Shared Field: Yes

Pathology Report Date

[path_rep_dt] PN0273

Value

Date field, mmddyyyy

Directives

- Record the date that the report is “filed under” in the medical record. This is the date that is seen in the list of all pathology reports in the EHR. It is the date that the report is accessioned.

Shared Field: No, contains PHI. The data element is flagged as an identifier in REDCap. Instead, a calculated interval in

days **[path_rep_int]** from **Calculated Date of Birth** is generated.

Cancer-Directed Procedure Date

[path_proc_dt] PN0275

Value

Date field, mmddyyyy

Directives

- Record the date that the cancer-directed procedure (biopsy, surgery, etc) was performed.
- Sometimes for internal reports, this is referred to as the ‘collection’ date.
- Make sure you are entering the date that the tissue sample was removed from the patient. This may be different than the date the report is accessioned, resulted, signed, and/or the date the report is logged under in the EHR.
- If a procedure date is unavailable do not curate the report date as procedure date, instead leave it blank.

Analytic Note

- The **Cancer-Directed Procedure Date** or its associated interval is used to create the interval for PRISMM:P based on the **Curated Date of Diagnosis** for pathology reports with **Was invasive cancer identified in the specimen? [1-45]** = “1, Yes.”

Shared Field: No, contains PHI. The data element is flagged as an identifier in REDCap. Instead, a calculated interval in days **[path_proc_int]** from **Calculated Date of Birth** is generated.

Institution Where the Pathology was Reviewed

[path_rep_inst] PN0278

Values

- 1, Internal (native) Institution
- 2, External Institution

Directives

- Choose whether the original procedure was performed at an internal institution or an external institution, regardless of which institution issued the pathology report.
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Shared Field: Yes

Institution Where the Procedure was Performed

[path_proc_inst] PN0279

Values

- 1, Internal (native) Institution
- 2, External Institution

Directives

- Choose whether the original procedure was performed at an internal institution or an external institution, regardless of which institution issued the pathology report.
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Shared Field: Yes

How many specimens are included on the pathology report?

[path_num_spec] PN0280

Values

Select single dropdown, whole numbers 1-30, representing up to 30 specimens

Directives

- Choose the number of distinct specimens from one procedure that are included in the report up to 25 specimens that can be curated from each pathology report.
- In general, each specimen identified in the report should be curated.
 - If a specimen contains more than one site, like an organ and a lymph node, curate each as separate specimens. A common example is a lung primary and a positive lymph node listed on the pathology report as one specimen; in REDCap, these should be two specimens.
 - If a single specimen contains more than one invasive histology, curated each as a separate specimen; however, this should not be confused with mixed histologies.
- This number selected will create the same number of sub data entry sections, one for each specimen with the following data fields:
 - Specimen: Anatomic Site
 - Specimen: Was in situ cancer identified in the specimen
 - Specimen: Was invasive cancer identified in the specimen
 - * Specimen with Invasive Cancer: Invasive Cancer Type
 - * Specimen with Invasive Cancer: Histology
 - * Specimen with Invasive Cancer: Other Histology

Shared Field: Yes

Specimen: Anatomic Site (for each specimen 1-30)

[path_site 1-30] PN0281, PN0288, PN0295, PN0302, PN0309, PN0316, PN0323, PN0330, PN0337, PN0344, PN0351, PN0358, PN0365, PN0372, PN0379, PN0386, PN0393, PN0400, PN0407, PN0414, PN0421, PN0428, PN0435, PN0442, PN0449, PN0838, PN0845, PN0852, PN0859, PN0866

Values

ICD-O-3 SEER Site/Topography Codes *See Appendix 7a*

Directives

- Choose the option that best fits the site from which the specimen was removed from the patient.
- If more than one choice is accurate, select the option is most specific. For example, a specimen from the chest wall could be called “Connective Subcutaneous & other soft tissues of thorax” or “Other Thorax NOS” but “Connective Subcutaneous & other soft tissues of thorax” is the best choice.
- It is best to select the more specific choice for sites. For example, if hilar lymph nodes are resected during a thoracotomy, it is preferred to code as C77.1 Intrathoracic lymph nodes, not C77.9 Lymph Nodes, NOS.
- The anatomic site of the specimen is often different than the type of invasive cancer; for example, a specimen from any body part may still be lung cancer.

Analytic Note

- If at least one of the specimens in the instance has **Was invasive cancer identified in the specimen? [1-45]** = “1, Yes” the anatomic sites can be grouped into distant or local sites of cancer.

Data Standard

NAACCR/SEER ICD-O-3 Site with additional values for fluids (e.g., pleural effusion, ascites, and urine) added

History

03032020: removed leading zeros.

04022020: added ability to record 20 more specimens, up to 45.

12162020: reduced number of specimens from 45 to 30.

Shared Field: Yes

Specimen: Was in situ cancer identified in the specimen? (for each specimen 1-30)

[path_insitu 1-30] PN0282, PN0289, PN0303, PN0310, PN0317, PN0324, PN0331, PN0338, PN0345, PN0352, PN0359, PN0366, PN0373, PN0380, PN0387, PN0394, PN0401, PN0408, PN0415, PN0422, PN0429, PN0436, PN0443, PN0450, PN0839, PN0846, PN0853, PN0860, PN0867

Values

1, Yes

0, No

10, Indeterminate/uncertain

Directives

- Indicate whether any in situ (ICD-O-3 Behavior Code 2) cancer is present in the specimen. Breast, bladder, cervical, and lung cancers are examples of cancer types that have in situ disease more commonly noted.
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

History

04022020: added ability to record 20 more specimens, up to 45.

12162020: reduced number of specimens from 45 to 30.

Shared Field: Yes

Specimen with In situ Cancer: What's the cancer's histology? (for each specimen 1-30)

[path_ca_ishist 1-30] PN0983, PN0984, PN0985, PN0986, PN0987, PN0988, PN0989, PN0990, PN0991, PN0992, PN0992, PN0994, PN0995, PN0996, PN0997, PN0998, PN0999, PN1000, PN1001, PN1002, PN1003, PN1004, PN1005, PN1006, PN1007, PN1008, PN1009, PN1010, PN1011, PN1012, PN1013, PN1014, PN1015, PN1016, PN1017, PN1018, PN1019, PN1020, PN1021, PN1022, PN1023, PN1024, PN1025, PN1026, PN1027

Values

ICD-O-3 SEER Histology/Morphology Codes for Behavior 2/Noninvasive (see Appendix 8)

Directives

- Choose the best histology for each specimen.
- This is normally one of the first things stated under "Pathologic Diagnosis" for each specimen. You may need to read the notes section of the path report as well. Be careful to review the noninvasive component.
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Data Standard

NAACCR: SEER ICD-O-3 Histology/Morphology

History

07062020: data fields added, one for each specimen in v345 explicitly for BPC breast cancer project

12162020: reduced number of specimens from 45 to 30.

Shared: Yes

Specimen: Was invasive cancer identified in the specimen? (for each specimen 1-30)

[path_ca 1-30] PN0283, PN0346, PN0353, PN0360, PN0367, PN0374, PN0381, PN0388, PN0395, PN0402, PN0409, PN0290, PN0416, PN0423, PN0430, PN0437, PN0444, PN0451, PN0840, PN0847, PN0854, PN0861, PN0297, PN0868, PN0304, PN0311, PN0318, PN0325, PN0332, PN0339

Values

- 1, Yes
- 0, No
- 10, Indeterminate/uncertain

Directives

- Indicate whether any invasive cancer is present in the specimen.
- Often this includes the word “carcinoma.” There are many similar words that end in -oma that are not invasive cancer. “Carcinoma **in situ**” is also not invasive cancer. Use the linked resources if you are not sure.
- Pathologists often use ambiguous terms when describing findings that are not totally conclusive. Use the table below to make your determination. That said, a cytology report recorded as suspicious is not considered as diagnostic of invasive cancer.
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Analytic Note

- If at least one of the specimens in the instance has **Was invasive cancer identified in the specimen? [1-25]** = “1, Yes” then PRISMM:P is positive for the time interval as defined by **Cancer-Directed Procedure Date** or its associated interval and the **Curated Date of Diagnosis** or its associated interval.

History

04022020: added ability to record 20 more specimens, up to 45

12162020: reduced number of specimens from 45 to 30.

Shared Field: Yes

Pathology Report: Ambiguous Terms

Yes, invasive cancer was identified	No, invasive cancer was not identified
Apparent(ly)	Cannot be ruled out
Appears (to)	Equivocal
Comparable with	Possible
Compatible with	Potentially
Consistent with	Questionable
Favor(s)	Rule out
Likely/most likely	Suggests
Malignant appearing	Worrisome
Presumed	Questionable
Probable	Rule out
Suspect(ed)	Suggests
Suspicious (for)	Worrisome
Typical (of)	

Specimen with Invasive Cancer: Invasive Cancer Type (for each specimen 1-30)

[path_ca_type 1-30] PN0284, PN0291, PN0298, PN0305, PN0312, PN0319, PN0326, PN0333, PN0340, PN0347, PN0354, PN0361, PN0368, PN0375, PN0382, PN0389, PN0396, PN0403, PN0410, PN0417, PN0424, PN0431, PN0438, PN0445, PN0452, PN0841, PN0848, PN0855, PN0862, PN0869

Values

1, Adrenocortical Carcinoma|3, Anal Cancer|5, Appendix Cancer|6, Bile Duct Cancer|7, Bladder Cancer|11, Brain Cancer|13, Breast Cancer|14, Breast Sarcoma|15, NET or Carcinoid|17, Cervical Cancer|97, Corpus Uteri Carcinoma and Carcinosarcoma|99, Corpus Uteri Sarcoma|19, Colon Cancer|20, Colon/Rectum Cancer|21, Esophagus Cancer|201, Ewing Sarcoma|23, Fallopian Tube Cancer|25, Gallbladder Cancer|205, Germ Cell Tumor|27, GIST|29, Head and Neck Cancer|30, Mesothelioma|35, Ill Defined/Cancer of Unknown Primary|39, Liver Cancer|41, Lung Cancer, NOS|45, Melanoma|47, Merkel Cell|211, Neuroblastoma|51, Non Small Cell Lung Cancer|215, Osteosarcoma|55, Ovarian Cancer|57, Pancreatic Cancer|58, Parathyroid Cancer|59, Penis Cancer|61, Peritoneum Cancer|63, Placenta Cancer|65, Prostate Cancer|67, Rectum and Rectosigmoid Cancer|69, Renal Kidney Cancer|71, Renal Pelvis Cancer|221, Retinoblastoma|225, Rhabdomyosarcoma|73, Scrotum Cancer|75, Small Cell Lung Cancer|77, Small Intestine Cancer|95, Soft tissue sarcoma |81, Stomach Cancer|83, Testis Cancer|85, Thymus Cancer|87, Thyroid Cancer|89, Uterus Cancer NOS|91, Vagina Cancer|93, Vulva Cancer|231, Wilms Tumor|301, Leukemia NOS |315, Acute Lymphoblastic Leukemia |317, Acute Myeloid Leukemia |319, Chronic Lymphocytic Leukemia |321, Chronic Myelogenous Leukemia |303, Non Hodgkin's Lymphoma |305, Hodgkin's Lymphoma |307, Multiple Myeloma |323, Plasmacytoma |311, MDS Myelodysplastic syndrome or Myeloproliferative Neoplasm |313, MGUS Monoclonal gammopathy of undetermined significance |309, Other hematopoietic or lymphoid neoplasm|1000, Other|1010, Not stated|1020, Unknown

Directives

- This field will only be available if you chose “1, Yes” for **Was invasive cancer identified in the specimen?**
- Choose the primary invasive cancer type, which for most solid tumors is a combination of the primary site, histology, and behavior codes that map to AJCC and CS.
- The cancer type is normally one of the first things stated under “Pathologic Diagnosis” for each specimen. You may need to read the notes section of the path report as well.
- The report may not definitively state the cancer type – if it says, “consistent with XXX primary” or “favors YYY carcinoma,” select that cancer type. If the report says that the primary cancer type cannot be determined, even if one or two types are favored, you must select “Not Stated.”
- Many pathology reports do not explicitly state what the invasive cancer type is, especially when it is the same as the site in the body the specimen is taken from. If a lung specimen just says “adenocarcinoma” and the patient is known to have lung cancer, it is usually safe to assume that the invasive cancer type is “Non Small Cell Lung

Cancer.” Similarly, if a breast biopsy just indicates, the histology “invasive ductal carcinoma” and the patient is known to have a diagnosis of breast cancer, assume the invasive cancer type is “Breast Cancer.” **However**, if a patient has more than one cancer diagnosis, you may need to read more of the notes, description, or clinical history of the pathology report to verify the invasive cancer type. For example, if a patient with known lung cancer had a colon biopsy with a diagnosis of “adenocarcinoma,” this could be a lung cancer or colon cancer. The pathology report may clarify or you may use implicit evidence from the patient’s medical history to determine the type of cancer. It is especially important to carefully verify invasive cancer type for cytology, lymph node biopsies, or other procedures that tend to be less specific (“malignant cells” for example).

- If the appropriate invasive cancer type is not an available choice, select “Other.”
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Data Standard: Mapped to AJCC Collaborative Stage, v2.05, AJCC v8, NCI/SEER

History

04022020: added ability to record 20 more specimens, up to 45

12162020: reduced number of specimens from 45 to 30.

Shared: Yes

Specimen with Invasive Cancer: Does the tumor invade the muscularis propria? (for each specimen 1-30)

[path_muscprop 1-30] PN0285, PN0292, PN0299, PN0306, PN0313, PN0320, PN0327, PN0334, PN0341, PN0348, PN0355, PN0362, PN0369, PN0376, PN0383, PN0390, PN0397, PN0404, PN0411, PN0418, PN0425, PN0432, PN0439, PN0446, PN0453, PN0842, PN0849, PN0856, PN0863, PN0870

Values

1, Yes
0, No
10, Unknown

Directives

- For Bladder Cancer Only.
- Indicate whether the pathology report indicates whether the tumor in the bladder invades the muscularis propria.
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Data Standard

History

04022020: added ability to record 20 more specimens, up to 45

12162020: reduced number of specimens from 45 to 30.

Shared: Yes

Specimen with Invasive Cancer: What is the cancer’s histology? (for each specimen 1-30)

[path_ca_hist 1-30] PN0286, PN0293, PN0300, PN0307, PN0314, PN0321, PN0328, PN0335, PN0342, PN0349, PN0356, PN0363, PN0370, PN0377, PN0384, PN0391, PN0398, PN0405, PN0412, PN0419, PN0426, PN0433, PN0440, PN0447, PN0454, PN0843, PN0850, PN0857, PN0864, PN0871

Values

ICD-O-3 SEER Histology/Morphology Codes

Directives

- Choose the best histology for each specimen.
- This is normally one of the first things stated under “Pathologic Diagnosis” for each specimen. You may need to read the notes section of the path report as well. Be as specific as possible. For example, “invasive ductal carcinoma” could accurately be called “Carcinoma” but “Duct Carcinoma” is the best choice.
- If the appropriate histology is not an available choice, select “Other histology” and write the histology in the “Other histology not listed” field below.
- The report may not definitively state the histology – if it says, “consistent with XXX primary,” then assume the histology for this specimen is the same as the primary cancer and enter that. For example, if a patient had a previous lung biopsy that showed squamous cell lung cancer and a later lymph node FNA that indicates “consistent with known lung primary,” then the histology for the lymph node would also be Squamous Cell.
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Data Standard

NAACCR: SEER ICD-O-3 Histology/Morphology

History

04022020: added ability to record 20 more specimens, up to 45

12162020: reduced number of specimens from 45 to 30.

Shared: Yes

Specimen with Invasive Cancer: What is the hematologic or lymphoid malignancy's histology? (for each specimen 1-30)

[path_ca_heme_hist 1-30] PN1189, PN1190, PN1191, PN1192, PN1193, PN1194, PN1195, PN1196, PN1197, PN1198, PN1199, PN1200, PN1201, PN1202, PN1203, PN1204, PN1205, PN1206, PN1207, PN1208, PN1209, PN1210, PN1211, PN1212, PN1213, PN1214, PN1215, PN1216, PN1217, PN1218

Values

ICD-O-3 SEER Histology/Morphology Codes

Directives

- Choose the best histology for each specimen.
- This is normally one of the first things stated under “Pathologic Diagnosis” for each specimen. You may need to read the notes section of the path report as well. Be as specific as possible.
- The report may not definitively state the histology – if it says, “consistent with XXX primary,” then assume the histology for this specimen is the same as the primary cancer and enter that.
- If the appropriate histology is not an available choice, select “Other histology” and write the histology in the “Other histology not listed” field below.

Data Standard

NAACCR: SEER ICD-O-3 Histology/Morphology

History

02212023: added field.

Specimen with Invasive Cancer: Other histology not listed (for each specimen 1-30)

[path_oth_hist 1-30] PN0287, PN0294, PN0301, PN0308, PN0315, PN0322, PN0329, PN0336, PN0343, PN0350, PN0357, PN0364, PN0371, PN0378, PN0385, PN0392, PN0399, PN0406, PN0413, PN0420, PN0427, PN0434, PN0441, PN0448,

PN0455, PN0844, PN0851, PN0858, PN0865, PN0872

Values

Free text

Directives

- Linked to **Specimen with Invasive Cancer: What is the cancer's histology?** And **Specimen with Invasive Cancer: What is the hematologic or lymphoid malignancy's histology?** If "Other histology" is selected for either of these two fields, record the histology as text.
- Data Standard

NAACCR: SEER ICD-O-3 Histology/Morphology

History

02252020: Additional codes added to comply with AJCC v8 acceptable ICD-O-3 Histology codes.

04022020: added ability to record 20 more specimens, up to 45

12162020: reduced number of specimens from 45 to 30.

Shared: Yes

Specimen: Was a cancer identified but its invasiveness is unclear? (for each specimen 1-30)

[path_cauncert 1-30] PN1104, PN1106, PN1108, PN1110, PN1112, PN1114, PN1116, PN1118, PN1120, PN1122, PN1124, PN1126, PN1128, PN1120, PN1132, PN1134, PN1136, PN1138, PN1140, PN1142, PN1144, PN1146, PN1148, PN1150, PN1152, PN1154, PN1156, PN1158, PN1160, PN1162

Values

1, Yes
0, No

Directives

- For Urothelial Carcinomas, we need explicit evidence of the cancer's invasiveness; however, it is not always possible for pathologists to determine, particularly for urine and other fluid specimens. If urothelial carcinoma is present but is not clearly identified as invasive or in situ, curate "Yes" for this field.
- Other cancers are assumed to be invasive unless stated otherwise by the pathologist.

History

07232021: field added to dictionary.

Specimen with Invasive Cancer: What is the uncertain cancer's histology? (for each specimen 1-30)

[path_cauncert_hist 1-30] PN1105, PN1107, PN1109, PN1111, PN1113, PN1115, PN1117, PN1119, PN1121, PN1123, PN1125, PN1127, PN1129, PN1131, PN1133, PN1135, PN1137, PN1139, PN1141, PN1143, PN1145, PN1147, PN1149, PN1151, PN1153, PN1155, PN1157, PN1159, PN1161, PN1163

Values

ICD-O-3 SEER Histology/Morphology Codes

Directives

- Choose the best histology for each specimen.
- This is normally one of the first things stated under "Pathologic Diagnosis" for each specimen. You may need to read the notes section of the path report as well. Be as specific as possible.
- The report may not definitively state the histology – if it says, "consistent with XXX primary," then assume the

histology for this specimen is the same as the primary cancer and enter that.

- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Data Standard

NAACCR: SEER ICD-O-3 Histology/Morphology

History

07232021: added field to dictionary.

Specimen with Invasive Cancer: Were sarcomatoid features identified in the specimen? (for each specimen 1-30)

[path_ca_sarc_feat1-30] PN1245-PN1274

Values

- 1, Sarcomatoid features identified
- 0, Sarcomatoid features not identified
- 10, Presence or absence of sarcomatoid features not reported or unknown

Directives

- For Renal Kidney Cancer only. Linked to field **“Cancer type – invasive”**. If Renal Kidney Cancer is selected, this field will be available for curation.
- Indicate whether sarcomatoid features were identified in the specimen.
- Curate “Sarcomatoid features identified” when sarcomatoid features are reported as 1% or higher.
- Only select “sarcomatoid features not identified” when the pathologist explicitly states that these features are absent or not identified.
- If the report does not mention sarcomatoid features, select the last option.

History

11062023: field added to dictionary.

Specimen with Invasive Cancer: Were rhabdoid features identified in the specimen? (for each specimen 1-30)

[path_ca_rhab_feat1-30] PN1310-PN1339

Values

- 1, Rhabdoid features identified
- 0, Rhabdoid features not identified
- 10, Presence or absence of rhabdoid features not reported or unknown

Directives

- For Renal Kidney Cancer only. Linked to field **“Cancer type – invasive”**. If Renal Kidney Cancer is selected, this field will be available for curation.
- Indicate whether rhabdoid features were identified in the specimen.
- Curate “Rhabdoid features identified” when rhabdoid features are reported as 1% or higher.
- Only select “Rhabdoid features not identified” when the pathologist explicitly states that these features are absent or not identified.
- If the report does not mention rhabdoid features, select the last option.

History

11062023: field added to dictionary.

Specimen with Invasive Cancer: Papillary type 1 or type 2? (for each specimen 1-30)

[path_ca_papillary_type1-30] PN1275-PN1304

Values

- 1, Type 1
- 2, Type 2
- 10, Not specified/Unknown

Directives

- For Renal Kidney Cancer only. Linked to field **“Cancer type – invasive”** and **“What is the cancer’s histology?”**. If Renal Kidney Cancer is selected as the cancer type and the histology is either 8260 Papillary Renal Cell Carcinoma or 8260 Papillary Carcinoma, this field will be available for curation.
- For kidney specimens with invasive papillary RCC (histology code 8260): curate whether Type 1 or Type 2 is specified in the pathology report.

History

11062023: field added to dictionary.

Specimen with Invasive Cancer: Was invasive cancer identified in both paired (left and right) organs? (for each specimen 1-30)

[path_ca_paired1-30] PN1345-PN1374

Values

- 0, No
- 1, Yes
- 2, Only 1 organ examined
- 9, Unknown

Directives

- Linked to field **“Anatomic site”** and **“Was invasive cancer identified in the specimen?”**. If ovary or fallopian tube is selected as the anatomic site and invasive cancer is present, this field will be available for curation.
- Only collected on surgical excision pathology reports.
- For ovary and fallopian tube anatomic sites only: indicate whether both (left and right) ovaries or both (left and right) fallopian tubes were involved with cancer.
- If both left and right organs in the pair were resected and both were involved with invasive cancer, choose ‘Yes’. If both organs in the pair were resected and one was involved and the other benign, choose ‘No’.
- Do not add an additional specimen for the second organ in the pair. You only need to curate the paired organ once (e.g., one specimen for left and right ovaries, one specimen for left and right fallopian tubes). If both ovaries or both fallopian tubes are benign, you only need to curate 1 specimen for each organ.

History

07022024: field added to dictionary.

Considering all the prostate specimens, please indicate whether a specialized histologic pattern is noted.

[path_pros_pat] PN1085

Values

- 4, Neuroendocrine/small cell | 3, Sarcomatoid | 1, Ductal | 2, Intraductal | 6, Cribriform | 7, Mucinous | 5, Other | 0, None noted

Directives

- Only curated for prostate specimens.

- Indicate whether any specialized histologic patterns are noted in the pathology report.

History

03262021: added field to dictionary.

Considering all prostate specimens, please indicate the highest Gleason score recorded on this pathology report.

[path_pc_gleason] PN1086

Values

2, Gleason score 2 | 3, Gleason score 3 | 4, Gleason score 4 | 5, Gleason score 5 | 6, Gleason score 6 | 7, Gleason score 7 | 8, Gleason score 8 | 9, Gleason score 9 | 10, Gleason score 10 | X8, Not applicable, no invasive cancer | X9, Not documented in medical record

Directives

- Only curated for prostate specimens.
- Indicate the highest Gleason score recorded in the pathology report.

History

03262021: added field to dictionary.

General Pathology Special Features/Biomarkers Curation Guidelines

Provenance

Internal (native) and External Pathology Reports and External Lab Test Result Reports from 3rd party commercial vendors like Tempus and Foundation

Data Standards

- In general, the College of American Pathologist are followed when available.

Pan-Cancer Path Special Features/Biomarkers

- PRISMM collects data on three pan-cancer biomarkers that are not part of the NGS panel test; these include PD-L, MSI, and MMR.
- Up to three test results for each of the biomarkers can be recorded.
- These tests need not be associated with the cancer of interest, instead, they are associated with the pathology report.

Finding Pathology Biomarkers Results

- Within each pathology report recorded for PRISMM-P, use the “Search” function in EHR to find applicable results; please note that this feature may not work on .pdf reports.
- **Do not** use test results from NGS panel reports or test results found in medical oncology progress notes.
- **Do** pay attention to test type when applicable.

PRISMM Pathology Biomarker Hormone Receptor and HER2 Data Directives

Was Hormone Receptor or HER2 testing reported?

[path_erprher_yn_1, path_erprher_yn_2, path_erprher_yn_3; path_erprher_yn_4; path_erprher_yn_5]
PN1028, PN1032, PN1036, PN1050, PN1057

Values

- 1, Yes
- 0, No

Directives

- For any of the specimens documented in the pathology report to have **invasive cancer**, was at least one of the following test results reported: estrogen receptors, progesterone receptors, HER2 IHC, or HER2 ISH.
- The data element can be modified by augmenting branching logic to restrict to certain cancer types.
- There are three sets of results that can be added per pathology report.

Provenance

Pathology report or internal/external lab report.

Data Standard

- NAACCR

History

07062020: Data fields added for v345.

Shared Field: Yes

HR Testing

[path_er_1], [path_pr_1] PN1029, PN1030

[path_er_2], [path_pr_2] PN1033, PN1034

[path_er_3], [path_pr_3] PN1037, PN1038

[path_er_4], [path_pr_4] PN1051, PN1052

[path_er_5], [path_pr_5] PN1058, PN1059

Matrix Values

- ER:
 - Intact expression/present
 - Loss of Expression/absent
 - Unknown
- PR:
 - Intact expression/present
 - Loss of Expression/absent
 - Unknown

Directives

- For each, chose the result.

Provenance

Pathology report or internal/external lab report.

Data Standard

- NAACCR

History

07062020: Data fields added for v345.

Shared Field: Yes

HER2 Testing

[path_herihc_1], [path_her2ihc_2], [path_her2ihc_3], [path_her2ihc_4], [path_her2ihc_5]

PN1031, PN1039, PN1048, PN1053, PN1060

[path_herish_1] [path_her2ish_2]], [path_her2ish_3] [path_her2ish_4] [path_her2ish_5]

PN1035, PN1047, PN1049, PN1054, PN1061

Matrix Values

- HER2 IHC:
 - Intact expression/present
 - Loss of Expression/absent
 - Unknown
- HER2 ISH:
 - Intact expression/present
 - Loss of Expression/absent
 - Unknown

Directives

- For each, chose the result.

Provenance

Pathology report or internal/external lab report.

Data Standard

- NAACCR

History

07062020: Data fields added for v345.

Shared Field: Yes

Addendum Date of Pathology Report HR/HER2 test result

[path_erprher_add1_dt], [path_erprher_add2_dt], [path_erprher_add3_dt], [path_erprher_add4_dt],
[path_erprher_add5_dt] PN1041, PN1043, PN1045, PN1055, PN1062

Value

Date field, mmddyyyy

Directives

- Linked to **Was HR, or HER2 testing reported?** If “Yes” was selected, this field is available for curation.
- Enter the addendum date of the pathology report from which these results are taken. Note that this may be different from the procedure date or pathology report date.
- If the test result is recorded in a separate pathology report, the date of this report can be entered as the addendum date.
- If the addendum date is not noted in the pathology report, leave blank.

Shared Field: No, contains PHI. The data element is flagged as an identifier in REDCap. Instead, a calculated interval in days from **Calculated Date of Birth** is generated.

PRISMM Pathology Biomarker PD-L1 Data Directives

Was PD-L1 testing reported on this pathology report? (1-3)

[pdl1_yn] PN0457 [pdl1_yn_2] PN0472 [pdl1_yn_3] PN0487

Values

- 1, Yes
- 0, No

Directives

- Search “PD” for applicable results in internal pathology reports. If there is a path report stating that a PD-L1 test was ordered, choose “Yes.” Often this information is located at the bottom of the report or in an addendum.
- If there is no evidence of a PD-L1 test, choose “No.”
- If there were insufficient cells for testing, code “No.”
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Shared Field: Yes

Addendum Date of Pathology Report PD-L1 test result (1-3)

[pd1_prepadt] PN0458 [pd1_prepadt_2] PN0473 [pd1_prepadt_3] PN0488

Value

Date field, mmddyyyy

Directives

- Linked to **Was PD-L1 testing reported on this pathology report?** If “Yes” was selected, this field is available for curation.
- Enter the addendum date of the pathology report from which this result is taken. Note that this may be different from the procedure date or pathology report date.
- If the test result is recorded in a separate pathology report, the date of this report can be entered here.
- If the addendum date is not noted in the pathology report, leave blank.

Shared Field: No, contains PHI. The data element is flagged as an identifier in REDCap. Instead, a calculated interval in days [pd1_prepaint] from **Calculated Date of Birth** is generated.

PD-L1 Antibody Test Type (1-3)

[pd1_test] PN0460

Values

- 1, 22C3
- 2, 28-8
- 3, E1L3N
- 4, SP263
- 5, SP142
- 10, Other
- 11, Unknown

Directives

- Linked to **Was PD-L1 testing reported on this pathology report?** If “Yes” is selected, this field is available for curation.
- Choose the PD-L1 test identified in the pathology report. If the type is not identified, record “10, Other.”
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Shared Field: Yes

How are the PD-L1 results presented in the pathology report? (1-3)

[pd1_type] PN0461 [pd1_type_2] PN0476 [pd1_type_3] PN0491

Values

Multi-select

- ☐ Percentage or percentage range of tumor cells
- ☐ Percentage or percentage range for infiltrating immune cells
- ☐ Numeric CPS (combined probability score) or range for CPS
- ☐ Summary Assessment

Directives

- Linked to **Was PD-L1 testing reported on this pathology report?** If “Yes” is selected, this field is available for curation.
- Select all the responses that describe how the results are presented in the pathology report for PD-L1; this is a multi-select list. Each response will trigger additional data elements for curation.

Shared Field: Yes

Record the percentage of tumor cells positive for PD-L1. (1-3)

[pdl1_perc] PN0462 [pdl1_perc_2] PN0477 [pdl1_perc_3] PN0492

Values

Numeric 1-100

Directives

- Linked to **Was PD-L1 testing reported on any pathology reports for this cancer?** If “Percentage or Percentage Range Tumor Cells” is selected, this field is available for curation.
- From information stated in the pathology report, record the percentage of tumor cells positive for PD-L1.
- If the percentage is not available, leave blank.

Shared Field: Yes

Record the lower value of the percentage of tumor cells positive for PD-L1. (1-3)

[pdl1_tclrange] PN0463 [pdl1_tclrange_2] PN0478 [pdl1_tclrange_3] PN0493

Values

102, 0 | 101, < 1 | 1, 1 | 2, 2 | 3, 3 | 4, 4 | 5, 5 | 6, 6 | 7, 7 | 8, 8 | 9, 9 | 10, 10 | 11, 11 | 12, 12 | 13, 13 | 14, 14 | 15, 15 | 16, 16 | 17, 17 | 18, 18 | 19, 19 | 20, 20 | 21, 21 | 22, 22 | 23, 23 | 24, 24 | 25, 25 | 26, 26 | 27, 27 | 28, 28 | 29, 29 | 30, 30 | 31, 31 | 32, 32 | 33, 33 | 34, 34 | 35, 35 | 36, 36 | 37, 37 | 38, 38 | 39, 39 | 40, 40 | 41, 41 | 42, 42 | 43, 43 | 44, 44 | 45, 45 | 46, 46 | 47, 47 | 48, 48 | 49, 49 | 50, 50 | 51, 51 | 52, 52 | 53, 53 | 54, 54 | 55, 55 | 56, 56 | 57, 57 | 58, 58 | 59, 59 | 60, 60 | 61, 61 | 62, 62 | 63, 63 | 64, 64 | 65, 65 | 66, 66 | 67, 67 | 68, 68 | 69, 69 | 70, 70 | 71, 71 | 72, 72 | 73, 73 | 74, 74 | 75, 75 | 76, 76 | 77, 77 | 78, 78 | 79, 79 | 80, 80 | 81, 81 | 82, 82 | 83, 83 | 84, 84 | 85, 85 | 86, 86 | 87, 87 | 88, 88 | 89, 89 | 90, 90 | 91, 91 | 92, 92 | 93, 93 | 94, 94 | 95, 95 | 96, 96 | 97, 97 | 98, 98 | 99, 99 | 100, 100 | 110, Other | 111, Not Applicable

Directives

- Linked to **Was PD-L1 testing reported on any pathology reports for this cancer?** If “Percentage or Percentage Range Tumor Cells” is selected, this field is available for curation.
- From information stated in the pathology report, record the percentage at the lower range of tumor cells positive for PD-L1.
- If the lower value percentage is stated, but not listed, record “Other.”
- If a range is not presented in the path report, leave blank.
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Shared Field: Yes

Record the higher value of the percentage of tumor cells positive for PD-L1. (1-3)

[pd11_tcurange] PN0464 [pd11_tcurange_2] PN0479 [pd11_tcurange_3] PN0494

Values

102, 0 | 101, < 1 | 1, 1 | 2, 2 | 3, 3 | 4, 4 | 5, 5 | 6, 6 | 7, 7 | 8, 8 | 9, 9 | 10, 10 | 11, 11 | 12, 12 | 13, 13 | 14, 14 | 15, 15 | 16, 16 | 17, 17 | 18, 18 | 19, 19 | 20, 20 | 21, 21 | 22, 22 | 23, 23 | 24, 24 | 25, 25 | 26, 26 | 27, 27 | 28, 28 | 29, 29 | 30, 30 | 31, 31 | 32, 32 | 33, 33 | 34, 34 | 35, 35 | 36, 36 | 37, 37 | 38, 38 | 39, 39 | 40, 40 | 41, 41 | 42, 42 | 43, 43 | 44, 44 | 45, 45 | 46, 46 | 47, 47 | 48, 48 | 49, 49 | 50, 50 | 51, 51 | 52, 52 | 53, 53 | 54, 54 | 55, 55 | 56, 56 | 57, 57 | 58, 58 | 59, 59 | 60, 60 | 61, 61 | 62, 62 | 63, 63 | 64, 64 | 65, 65 | 66, 66 | 67, 67 | 68, 68 | 69, 69 | 70, 70 | 71, 71 | 72, 72 | 73, 73 | 74, 74 | 75, 75 | 76, 76 | 77, 77 | 78, 78 | 79, 79 | 80, 80 | 81, 81 | 82, 82 | 83, 83 | 84, 84 | 85, 85 | 86, 86 | 87, 87 | 88, 88 | 89, 89 | 90, 90 | 91, 91 | 92, 92 | 93, 93 | 94, 94 | 95, 95 | 96, 96 | 97, 97 | 98, 98 | 99, 99 | 100, 100 | 110, Other | 111, Not Applicable

Directives

- Linked to **How are the PD-L1 results presented in the pathology report?** If “Percentage or Percentage Range of Tumor Cells” is selected, this data field is available for curation.
- From information stated in the pathology report, record the percentage at the upper range of tumor cells positive for PD-L1.
- If the upper value percentage is stated, but not listed, record “Other.”
- If a range is not presented in the path report, leave blank.
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Shared Field: Yes

Record the percentage of infiltrating immune cells positive for PD-L1. (1-3)

[pd11_icperc] PN0465 [pd11_icperc_2] PN0480 [pd11_icperc_3] PN0495

Values

Numeric 1-100

Directives

- Linked to **How are the PD-L1 results presented in the pathology report?** If “Percentage or Percentage Range of Infiltrating Immune Cells” is selected, this data field is available for curation.
- From information stated in the pathology report or results report, record the percentage of infiltrating immune cells positive for PD-L1; infiltrating immune cells are found in the tumor micro-environment and are stromal tumor-associated macrophages and lymphocytes.
- From information stated in the pathology report, record the percentage of infiltrating immune cells positive for PD-L1.
- If the percentage for the infiltrating immune cells is not available, leave blank.

Shared Field: Yes

Record the lower value of the percentage of infiltrating immune cells positive for PD-L1. (1-3)

[pd11_iclrange] PN0466 [pd11_iclrange_2] PN0481 [pd11_iclrange_3] PN0496

Values

102, 0 | 101, < 1 | 1, 1 | 2, 2 | 3, 3 | 4, 4 | 5, 5 | 6, 6 | 7, 7 | 8, 8 | 9, 9 | 10, 10 | 11, 11 | 12, 12 | 13, 13 | 14, 14 | 15, 15 | 16, 16 | 17, 17 | 18, 18 | 19, 19 | 20, 20 | 21, 21 | 22, 22 | 23, 23 | 24, 24 | 25, 25 | 26, 26 | 27, 27 | 28, 28 | 29, 29 | 30, 30 | 31, 31 | 32, 32 | 33, 33 | 34, 34 | 35, 35 | 36, 36 | 37, 37 | 38, 28 | 39, 39 | 40, 40 | 41, 41 | 42, 42 | 43, 43 | 44,

44 | 45, 45 | 46, 46 | 47, 47 | 48, 48 | 49, 49 | 50, 50 | 51, 51 | 52, 52 | 53, 53 | 54, 54 | 55, 55 | 56, 56 | 57, 57 | 58, 58 | 59, 59 | 60, 60 | 61, 61 | 62, 62 | 63, 63 | 64, 64 | 65, 65 | 66, 66 | 67, 67 | 68, 68 | 69, 69 | 70, 70 | 71, 71 | 72, 72 | 73, 73 | 74, 74 | 75, 75 | 76, 76 | 77, 77 | 78, 78 | 79, 79 | 80, 80 | 81, 81 | 82, 82 | 83, 83 | 84, 84 | 85, 85 | 86, 86 | 87, 87 | 88, 88 | 89, 89 | 90, 90 | 91, 91 | 92, 92 | 93, 93 | 94, 94 | 95, 95 | 96, 96 | 97, 97 | 98, 98 | 99, 99 | 100, 100 | 110, Other | 111, Not Applicable

Directives

- Linked to **Was PD-L1 testing reported on any pathology reports for this cancer?** if “Percentage or Percentage Range Infiltrating Immune Cells” is selected, this field is available for curation.
- From information stated in the pathology report or results report, record the percentage at the lower range of infiltrating immune cells positive for PD-L1.
- If the lower value percentage is stated, but not listed, record “Other.”
- If a range is not presented in the path report, leave blank.
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Shared Field: Yes

Record the higher value of the percentage infiltrating immune cells positive for PD-L1. (1-3)

[pdl1_icurange] PN0467 [pdl1_icurange_2] PN0482 [pdl1_icurange_3] PN0497

Values

102, 0 | 101, < 1 | 1, 1 | 2, 2 | 3, 3 | 4, 4 | 5, 5 | 6, 6 | 7, 7 | 8, 8 | 9, 9 | 10, 10 | 11, 11 | 12, 12 | 13, 13 | 14, 14 | 15, 15 | 16, 16 | 17, 17 | 18, 18 | 19, 19 | 20, 20 | 21, 21 | 22, 22 | 23, 23 | 24, 24 | 25, 25 | 26, 26 | 27, 27 | 28, 28 | 29, 29 | 30, 30 | 31, 31 | 32, 32 | 33, 33 | 34, 34 | 35, 35 | 36, 36 | 37, 37 | 38, 28 | 39, 39 | 40, 40 | 41, 41 | 42, 42 | 43, 43 | 44, 44 | 45, 45 | 46, 46 | 47, 47 | 48, 48 | 49, 49 | 50, 50 | 51, 51 | 52, 52 | 53, 53 | 54, 54 | 55, 55 | 56, 56 | 57, 57 | 58, 58 | 59, 59 | 60, 60 | 61, 61 | 62, 62 | 63, 63 | 64, 64 | 65, 65 | 66, 66 | 67, 67 | 68, 68 | 69, 69 | 70, 70 | 71, 71 | 72, 72 | 73, 73 | 74, 74 | 75, 75 | 76, 76 | 77, 77 | 78, 78 | 79, 79 | 80, 80 | 81, 81 | 82, 82 | 83, 83 | 84, 84 | 85, 85 | 86, 86 | 87, 87 | 88, 88 | 89, 89 | 90, 90 | 91, 91 | 92, 92 | 93, 93 | 94, 94 | 95, 95 | 96, 96 | 97, 97 | 98, 98 | 99, 99 | 100, 100 | 110, Other | 111, Not Applicable

Directives

- Linked to **Was PD-L1 testing reported on any pathology reports for this cancer?** if “Percentage or Percentage Range Infiltrating Immune Cells” is selected, this field is available for curation.
- From information stated in the pathology report or results report, record the percentage at the upper range of infiltrating immune cells positive for PD-L1.
- If the upper value percentage is stated, but not listed, record “Other.”
- If a range is not presented in the path report, leave blank.
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Shared Field: Yes

Record the Combined Probability Score (CPS) numeric value for PD-L1 expression of the tumor cells. (1-3)

[pdl1_num] PN0468 [pdl1_num_2] PN0483 [pdl1_num_3] PN0498

Values

Numeric

Directives

- Linked to **How are the PD-L1 results presented in the pathology report?** If “Numeric (combined probability score)” is selected, field is available for curation.
- From information stated in the pathology report or results report. record the CPS value.

Shared Field: Yes

Record the lower number in the CPS range

[pd1_lcpsrange] PN0469 [pd1_lcpsrange_2] PN0484 [pd1_lcpsrange_3] PN0499

Values

102, 0 | 101, < 1 | 1, 1 | 2, 2 | 3, 3 | 4, 4 | 5, 5 | 6, 6 | 7, 7 | 8, 8 | 9, 9 | 10, 10 | 11, 11 | 12, 12 | 13, 13 | 14, 14 | 15, 15 | 16, 16 | 17, 17 | 18, 18 | 19, 19 | 20, 20 | 21, 21 | 22, 22 | 23, 23 | 24, 24 | 25, 25 | 26, 26 | 27, 27 | 28, 28 | 29, 29 | 30, 30 | 31, 31 | 32, 32 | 33, 33 | 34, 34 | 35, 35 | 36, 36 | 37, 37 | 38, 38 | 39, 39 | 40, 40 | 41, 41 | 42, 42 | 43, 43 | 44, 44 | 45, 45 | 46, 46 | 47, 47 | 48, 48 | 49, 49 | 50, 50 | 51, 51 | 52, 52 | 53, 53 | 54, 54 | 55, 55 | 56, 56 | 57, 57 | 58, 58 | 59, 59 | 60, 60 | 61, 61 | 62, 62 | 63, 63 | 64, 64 | 65, 65 | 66, 66 | 67, 67 | 68, 68 | 69, 69 | 70, 70 | 71, 71 | 72, 72 | 73, 73 | 74, 74 | 75, 75 | 76, 76 | 77, 77 | 78, 78 | 79, 79 | 80, 80 | 81, 81 | 82, 82 | 83, 83 | 84, 84 | 85, 85 | 86, 86 | 87, 87 | 88, 88 | 89, 89 | 90, 90 | 91, 91 | 92, 92 | 93, 93 | 94, 94 | 95, 95 | 96, 96 | 97, 97 | 98, 98 | 99, 99 | 100, 100 | 110, Other | 111, Not Applicable

Directives

- Linked to **How are the PD-L1 results presented in the pathology report?** Data field available for curation if 'Numeric (combined probability score)' is selected.
- From information stated in the pathology report or results report. record the lower number in the range of the CPS value.

Shared Field: Yes

Record the upper number in the CPS range

[pd1_ucpsrange] PN0470 [pd1_ucpsrange_2] PN0485 [pd1_ucpsrange_3] PN0500

Values

102, 0 | 101, < 1 | 1, 1 | 2, 2 | 3, 3 | 4, 4 | 5, 5 | 6, 6 | 7, 7 | 8, 8 | 9, 9 | 10, 10 | 11, 11 | 12, 12 | 13, 13 | 14, 14 | 15, 15 | 16, 16 | 17, 17 | 18, 18 | 19, 19 | 20, 20 | 21, 21 | 22, 22 | 23, 23 | 24, 24 | 25, 25 | 26, 26 | 27, 27 | 28, 28 | 29, 29 | 30, 30 | 31, 31 | 32, 32 | 33, 33 | 34, 34 | 35, 35 | 36, 36 | 37, 37 | 38, 38 | 39, 39 | 40, 40 | 41, 41 | 42, 42 | 43, 43 | 44, 44 | 45, 45 | 46, 46 | 47, 47 | 48, 48 | 49, 49 | 50, 50 | 51, 51 | 52, 52 | 53, 53 | 54, 54 | 55, 55 | 56, 56 | 57, 57 | 58, 58 | 59, 59 | 60, 60 | 61, 61 | 62, 62 | 63, 63 | 64, 64 | 65, 65 | 66, 66 | 67, 67 | 68, 68 | 69, 69 | 70, 70 | 71, 71 | 72, 72 | 73, 73 | 74, 74 | 75, 75 | 76, 76 | 77, 77 | 78, 78 | 79, 79 | 80, 80 | 81, 81 | 82, 82 | 83, 83 | 84, 84 | 85, 85 | 86, 86 | 87, 87 | 88, 88 | 89, 89 | 90, 90 | 91, 91 | 92, 92 | 93, 93 | 94, 94 | 95, 95 | 96, 96 | 97, 97 | 98, 98 | 99, 99 | 100, 100 | 110, Other | 111, Not Applicable

Directives

- Linked to **How are the PD-L1 results presented in the pathology report?** Data field available for curation if 'Numeric (combined probability score)' is selected.
- From information stated in the pathology report or results report. record the upper number in the range of the CPS value.

Shared Field: Yes

Record the text summary of the PD-L1 result stated in the report.

[pd1_sum] PN0471 [pd1_sum_2] PN0486 [pd1_sum_3] PN0501

Values

- 1, High
- 2, Low
- 3, Positive
- 4, Negative
- 5, Indeterminate/Not stated

Directives

- Linked to **How are the PD-L1 results presented in the pathology report?** If “Summary Assessment” is selected, field is available for curation.
- From information stated in the pathology report. record the overall result as stated in the report.
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Shared Field: Yes

PRISMM Pathology Biomarker Microsatellite Instability (MSI) Data Directives (three sets)

Was microsatellite instability (MSI) testing using PCR reported?

[msi_yn] PN0502 [msi_yn_2] PN0508 [msi_yn_3] PN0514

Values

- 1, Yes
- 0, No

Directives

- Search “microsat” for applicable results in internal pathology reports. If there is a path report stating that a microsatellite instability test using PCR was ordered. choose “Yes.” Often this information is located at the bottom of the report or in an addendum.
- If there is no evidence of a microsatellite instability test, choose “No.”
- Please note that the codes (numbers) for the responses are not visible in REDCap; however, the codes will be available at data export.

Data Standard

- CAP

Addendum Date for Pathology Report MSI test result

[msi_prepadt] PN0503 [msi_prepadt_2] PN0509 [msi_prepadt_3] PN0515

Value

Date field. mmddyyyy

Directives

- Linked to **Was microsatellite instability (MSI) testing using PCR reported on any path specimen?** If “Yes” is selected, the field is available for curation.
- Enter the addendum date of the pathology report this result is taken from. Note that this may be different from the procedure date or pathology report date.
- If the test result is recorded in a separate pathology report, the date of this report can be entered here.
- If the addendum date is not noted in the pathology report, leave blank.

Data Standard

- CAP

Shared Field: No. a calculated interval from date of birth will be created

Select the stated MSI result

[msi_result] PN0505 [msi_result_2] PN0511 [msi_result_3] PN0517

Values

- 1, MSS: STABLE

- 2, MSI-H: HIGH
- 3, MSI-L: LOW or instability in < 30% of microsatellite markers
- 4, Indeterminate/Not stated

Directives

- Linked to **Was microsatellite instability (MSI) testing using PCR reported?** If “Yes” is selected, the field is available for curation.
- From information stated in the pathology report, record the overall MSI result as stated in the report.
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Data Standard

- CAP

Shared Field: Yes

MSI-H: High Details

[msi_high] PN0506 [msi_high_2] PN0512 [msi_high_3] PN0518

Values

- 1, >=30% of the markers exhibit instability
- 2, 2 or more of the 5 markers exhibit instability
- 3, Other

Directives

- Linked to **Select the stated MSI result.** If “MSI-H: HIGH” is selected, the field is available for curation.
- From information stated in the pathology report, record the overall additional details about the MSI-H result.
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Data Standard

- CAP

Shared Field: Yes

MSI-L: Low Details

[msi_low] PN0507 [msi_low_2] PN0513 [msi_low_3] PN0519

Values

- 1, 1 - 29% of the markers exhibit instability
- 2, 1 of the 5 markers exhibit instability
- 3, Other

Directives

- Linked to **Select the stated MSI result.** If “MSI-L: LOW” is selected, the field is available for curation.
- From information stated in the pathology report, record the overall additional details about the MSI-L result.
- Please note that the codes (numbers) for the responses are not visible in REDCap; however, the codes will be available at data export.

Data Standard

- CAP

Shared Field: Yes

PRISSMM Pathology Biomarker MMR Data Directives (three sets)

Was testing for Mismatch Repair or expression of MLH1 or MSH2 or MSH6 or PMS2 proteins using IHC reported?
[mmr_yn] PN0520 [mmr_yn_2] PN0530 [mmr_yn_3] PN0540

Values

- 1, Yes
- 0, No

Directives

- Search “mismatch” or one of the protein names (e.g. “MLH”) for applicable results in internal pathology reports. Only include testing done by immunohistochemistry (IHC) to determine whether MMR is documented in the pathology report.
- If testing for MMR is documented, select “Yes.” Often this information is located at the bottom of the report or in an addendum.
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Data Standard

- CAP

Shared Field: Yes

Addendum Date for Pathology Report MMR test result

[mmr_prepadt] PN0521 [mmr_prepadt_2] PN0531 [mmr_prepadt_3] PN0541

Value

Date field. mmddyyyy

Directives

- Linked to **Was testing done for Mismatch Repair or expression/presence of MLH1 or MSH2 or MSH6 or PMS2?** If “Yes” is selected, the field is available for curation.
- Enter the addendum date of the pathology report this result is taken from. Note that this may be different from the procedure date or pathology report date.
- If the test result is recorded in a separate pathology report, the date of this report can be entered here.
- If the addendum date is not noted in the pathology report, leave blank.

Data Standard

- CAP

Shared Field: No. a calculated interval from date of birth will be created

MMR Test Using Immunohistochemistry (IHC) Staining

Matrix Values

MLH1: [mmr_mlh1] PN0523, [mmr_mlh1_2] PN0533, [mmr_mlh1_3] PN0543

- Intact expression/present
- Loss of Expression/absent
- Unknown

MSH2: [mmr_msh2] PN0524, [mmr_msh2_2] PN0534, [mmr_msh2_3] PN0544

- Intact expression/present
- Loss of Expression/absent
- Unknown

MSH6: [mmr_msh2] PN0525, [mmr_msh2_2] PN0535, [mmr_msh2_3] PN0545

- Intact expression/present
- Loss of Expression/absent
- Unknown

PMS2: [mmr_pms2] PN0526, [mmr_pms2_2] PN0536, [mmr_pms2_3] PN0546

- Intact expression/present
- Loss of Expression/absent
- Unknown

Directives

- For each, chose the staining result.

Data Standard

- CAP

Shared Field: Yes

Record the overall MMR interpretation as stated in the report.

[mmr_result] PN0527 [mmr_result_2] PN0537 [mmr_result_3] PN0547

Values

1, No loss of nuclear expression of MMR proteins: No evidence of deficient mismatch repair (low probability of MSI-H): proficient | 5, Loss of nuclear expression of one or more MMR proteins; deficient mismatch repair | 8, Indeterminate/Not stated

Directives

- Search “mismatch” or one of the protein names (e.g. “MLH”) for applicable results in internal pathology reports. Only include testing done by immunohistochemistry (IHC) to determine whether MMR is documented in the pathology report.
- If testing for MMR is documented, record the best option summarizing the assessment of MMR protein.
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Data Standard

- CAP

Shared Field: Yes

If the interpretation was MMR Proficient (MMR-p), record the details provided in the report

[mmrp_det] PN0528 [mmrp_det_2] PN0538 [mmrp_det_3] PN0548

Values

1, No loss of nuclear expression of MLH1, MSH2, MSH6 or PMS2 proteins

2, Described as MMR intact, preserved, normal

Directives

- Linked to **Record the overall MMR interpretation as stated in the report.** If “1, No loss of nuclear expression of MMR proteins: No evidence of deficient mismatch repair (low probability of MSI-H): proficient” is recorded, then record the details of the test result.
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Data Standard

- CAP

Shared Field: Yes

If the interpretation was MMR Deficient (MMR-d), record the details provided in the report
[mmrp_det] PN0529 [mmrp_det_2] PN0539 [mmrp_det_3] PN0549

Values

- 1, Loss of nuclear expression (or absence of) one or more of MLH1, MSH2, MSH6 or PMS2 proteins
- 2, Described as abnormal

Directives

- Linked to **Record the overall MMR interpretation as stated in the report.** If “5, Loss of nuclear expression of one or more MMR proteins; deficient mismatch repair” is recorded, then record the details of the test result.
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Data Standard

- CAP

Shared Field: Yes

PRISMM Pathology Calculated Fields

In order to share data in a de-identified manner, several intervals are calculated in the REDCap project from the anchor date: Date of Birth as follows:

Interval Data Field	Data Element	Calculation
Interval in Days from DOB to Pathology Report Date	path_rep_int PN0274	datediff([birth_dt], [path_rep_dt], "d", "mdy")
Interval in Days from DOB to Pathology Procedure Date	path_proc_int PN0276	datediff([birth_dt], [path_proc_dt], "d", "mdy")
Interval in Days from DOB to Pathology Addendum Date: PD-L1 (1-3)	path_erprher_add1_int PN1042 path_erprher_add1_int PN1044 path_erprher_add1_int PN1046	datediff([birth_dt], [pdl1_prepadt], "d", "mdy")
Interval in Days from DOB to Pathology Addendum Date: PD-L1 (1-3)	pdl1_prepaint PN0459 pdl1_prepaint_2 PN0474 pdl1_prepaint_3 PN0489	datediff([birth_dt], [pdl1_prepadt], "d", "mdy")
Interval in Days from DOB to Pathology Addendum Date: MSI	msi_prepaint PN0504 msi_prepaint_2 PN0510 msi_prepaint_3 PN0516	datediff([birth_dt], [msi_prepadt], "d", "mdy")
Interval in Days from DOB to Pathology Addendum Date: MMR	mmr_prepaint PN0522 mmr_prepaint_2 PN0532 mmr_prepaint_3 PN0542	datediff([birth_dt], [mmr_prepadt], "d", "mdy")

PRISMM Pathology and Pathology Special Features Data Violations

Major	Incorrect/missing procedure date [+/- 1 month]
	Incorrect/Missing cancer type
	Missed/incorrectly identified invasive cancer
	Missed eligible pathology report(s)
Minor	Incorrect/missing histology
	Incorrect/missing specimens
	Incorrect report date [+/- 1 month]
	Ineligible report included
Other	Incorrect/missing procedure institution
	Missing/incorrect "during diagnosis" status

Major	Missed PD-L1, MSI, or MMR testing
	Incorrect result: PD-L1
	Incorrect result: MSI [Stable v. High/Low]
Minor	Incorrect/missing histology
	Ineligible report included
Other	Incorrect/missing procedure institution

General PRISSMM Radiology/Imaging Curation Guidelines

REDCap Form

prissmm_imaging

Provenance

- Internal (native) and External Imaging Reports: Bone Scans, CTs, MRIs, PET/PET CTS, NM Scans
- Full P curation includes review of both native and external imaging/radiology reports

Selecting Imaging Reports to Curate

- Find the date of diagnosis of the cancer of interest.
- **Do** review and enter data for all imaging reports for scans that occur in the month of the diagnosis and after.
 - **Do** abstract data from all bone scans, CTs, MRIs, PET and PET/CTs, and nuclear medicine scans.
 - **Do not** review or curate mammograms, unless curating for breast cancer.
 - **Do not** review or curate ultrasounds. or x-rays. even if they are related to cancer.
 - **Do not** include any scans that are performed for purposes other than monitoring cancer status. including MUGA, HIDA scans, bone density, etc.
 - **Do not** review any fluoroscopy, IR biopsy, or other procedure images.
- **Do** review images in chronological order when possible starting with the oldest scan.
 - Use what you know from prior scans to interpret a report. For example. if a lymph node was categorized as cancer on one scan and the next scan says “lymph node unchanged” you can assume this is still cancer. Do not assume that lack of mentioning a site that was included on a previous scan means it has resolved.
- **Do** only read the “Indication” and “Interpretation/Impression” section.
 - Synonyms for Indication include: Opinion, Conclusion, Integrated Summary, Reason for Exam
- Even when ambiguous, **do not** seek clarification in “Findings” or other earlier parts of the report.
- **Do** enter each Radiology/Imaging report as its own instance.
 - **Exception:** If there are multiple reports for the same type of scan on the same date (e.g. a CT Chest report and a CT Abdomen/Pelvis report) with the same information in the Impression. curate these as one instance (otherwise if the Impressions are different. record separately).
 - **Exception:** Scans performed at external sites may be reviewed again by internal radiologists. Only enter one instance for each scan. Give preference to internal review.

PRISSMM Imaging Report Piped Data Elements

In the case that an institution can pipe in the meta data (data associated with the imaging report, like description and id) for their imaging reports, which will assist in the curation effort, the REDCap project contains several fields that are available to hold this data. If the institution does not choose to pipe-in the meta data, then these fields will not be visible to the curators.

Data Element Label	Data Element
Imaging Report Encounter ID	[image_enc_id]
Imaging Report Order ID	[image_ord_id]
Imaging Report Description	[image_desc]
Is this Imaging Report appropriate for PRISSMM	[image_prissmm_yn]

PRISSMM Radiology/Imaging Data Directives

Institution Where Scan was Performed

[image_inst_perf] PN0567

Values

- 1, Internal institution
- 2, External institution

Directives

- Record the institution in which the scan was performed.

Data Standard

- PRISSMM

Shared Field: No, a calculated interval in days from **Calculated Date of Birth** is generated

Institution Where Scan was Interpreted

[image_inst_perf] PN0568

Values

- 1, Internal institution
- 2, External institution

Directives

- Record the institution in which the scan was interpreted. For scans performed at an external institution, the internal (native) institution's interpretations are preferred.

Data Standard

- PRISSMM

Shared Field: No, a calculated interval in days from **Calculated Date of Birth** is generated

Imaging Scan Date

[image_scan_dt] PN0569

Value

Date field. mmddyyyy

Directives

- It is best to take the date from the header of the report itself.

Data Standard

- PRISSMM

Shared Field: No, a calculated interval in days from **Calculated Date of Birth** is generated

Imaging Report Date

[image_report_dt] PN0571

Value

Date field. mmddyyyy

Directives

- Record the date that the report is “filed under” in the medical record. This is the date that is seen in the list of all radiology/imaging reports in the EHR. usually under a column header like “Performed” or “Date of Service.”
- Leave blank if this is the same as the scan date.

Data Standard

- PRISSMM

Shared Field: No, a calculated interval in days from **Calculated Date of Birth** is generated

Imaging Scan Type

[image_scan_type] PN0573

Values

- 1, CT
- 3, MRI
- 5, PET or PET-CT
- 13, PET MRI
- 7, Bone Scan
- 9, Other Nuclear Medicine Scan
- 11, Mammogram – Breast Cancer only
- 20, Other CA-specific scan

Directives

- Record the type of scan was performed.
- Mammograms should be recorded for breast cancer diagnoses beginning in the month and year of the Index cancer diagnosis.
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Data Standard

- PRISSMM

Shared Field: Yes

Imaging Scan Type - Other

[oth_scan] PN0574

Values

Free text

Directives

- Linked to **Imaging Scan Type**.
- If imaging scan type is “20, Other CA Specific Scan” record the scan type.

Shared Field: Yes

Imaging Scan Site

[image_scansite] PN0575

Values

Multi-select

- ☐ Brain/Head
- ☐ Spine
- ☐ Neck
- ☐ Chest
- ☐ Abdomen
- ☐ Pelvis
- ☐ Extremity
- ☐ Full Body

Directives

- Select all sites of the body that the scan evaluated. This is a multi-select item.
- For scans that cover the entire body, choose “Full Body” like PET or bone scan.
 - Do not check each individual site.
 - “Full Body” should be used for PET or PET/CT even though the scan covers the skull base to thigh.

Data Standard

- PRISMM

Shared Field: Yes

Is there any evidence of cancer on this imaging report?

[image_ca] PN0576

Values

- 1, Yes, the Impression states or implies there is evidence of cancer
- 2, No, the Impression states or implies there is no evidence of cancer
- 3, The Impression is uncertain, indeterminate, or equivocal
- 4, The Impression does not mention cancer

Directives

- Report whether there is **any** evidence of cancer on the scan per the radiologist’s interpretation. Use the “Impression” section of radiology/imaging report to choose the best response from the options below. “Indication” section may also be used for context.
- **Evidence of cancer may be explicit or implicit.**
- **Explicit evidence of cancer includes radiologist mention of:**
 - Tumor or neoplasm (unless specifically stated to be benign or indeterminate).
 - An abnormality such as a mass, lesion, nodule, opacity, or enlarged lymph node (aka “adenopathy” or “lymphadenopathy”), etc, that is described as malignant, metastatic, or as cancer.
 - A nodule, lesion, or mass that is ≥ 1 cm unless otherwise noted to be benign. This does not apply to lymph nodes.
 - Lymph node ≥ 1 cm that the radiologist is following for change in size.
 - Any finding that is described as recurrence, disease progression, or change in disease/tumor burden.
 - On PET, FDG-avid/uptake/increased activity or hypermetabolic activity = cancer

Implicit evidence of cancer includes:

- Context from “Indication” section of report, such as stating that the patient has a known active cancer.
 - Definite statements like “patient with known liver metastases” or “patient with Stage IV cancer. on treatment” are implicit evidence of cancer.
 - Statements like “history of breast cancer” or “lung cancer restaging” do not necessarily indicate presence of cancer on this scan. Do not count this as evidence of cancer.
- Context from previous scans. If a site is previously described as cancer or concerning for cancer, continue to consider it cancer unless otherwise stated.

- Radiologist mention of treatment effect or response to therapy.

Radiologists often describe the images they see in ambiguous terms that may represent cancer, a benign finding, or a different condition or illness.

Interpreting terms describing new abnormalities (i.e., findings that have not been included on previous reports):

- If the report's impression describes a new abnormality with no other information. consider this **no evidence of cancer**.
- If the report's impression describes a new abnormality and says a non-cancerous condition is favored/likely. consider this **no evidence of cancer**.
- If the report's impression describes a new abnormality and says a non-cancerous condition is favored/likely. but cancer is a possibility. consider this **uncertain/indeterminate**.
- If the report's impression describes a new abnormality and says a non-cancerous condition and cancer are both possible (neither is favored), consider this **uncertain/indeterminate**.
- If the report's impression describes a new abnormality and says cancer is favored/likely, but a non-cancerous condition is a possibility. consider this **yes. evidence of cancer**.

For abnormalities that have been noted in previous reports:

- If the abnormality was previously described as cancer, assume it is still cancer until stated otherwise.
- If the abnormality was previously noted to be suspicious/concerning for cancer, assume it is still cancer until stated otherwise.
- Sites previously identified as cancer that are stated to be resolved should generally be considered to have no evidence of disease. If there are remaining sites of cancer, consider the resolved sites as a sign of improvement and assess overall change accordingly. **Do not** record resolved sites as anatomic sites. This does not apply to statements about resolution of FDG avidity if the tumor/mass/lesion is still present.
- If a cancer site was surgically removed: In general, no evidence of disease when post-surgical/post-resection changes are mentioned unless residual disease is noted
- Post-radiation and post-treatment changes:
 - If post-treatment or post-radiation changes are mentioned, including scarring/fibrosis/pneumonitis related to treatment, **and** the radiologist explicitly states no evidence of disease/no residual disease/no recurrence: No evidence of cancer
 - If a residual abnormality is mentioned and attributed to post-treatment or post-radiation-treatment changes, including scarring/fibrosis/pneumonitis related to treatment, **and no** statement about residual disease or recurrence: Indeterminate for cancer
 - If a residual abnormality is mentioned **without** attribution to post-treatment/post-radiation changes or other benign causes such as scarring/fibrosis/pneumonitis **and no** statement about residual disease or recurrence: Yes, evidence of cancer

Specific Site Guidance

- In order for a pleural effusion, pericardial effusion, ascites, or other fluid collection to be considered as a cancer site, it must explicitly be described as malignant within the report or there must be implicit evidence from a prior report. However, for patients with an ovarian, fallopian tube, or primary peritoneal cancer diagnosis, abdominopelvic ascites should be considered evidence of cancer even if the radiologist does not explicitly describe it as malignant.
- In order for sclerosis of the bone collection to be considered as a cancer site, it must explicitly be described as malignant within the report or there must be implicit evidence from a prior report. Otherwise, If the Impression only says sclerotic bone and there is no prior report that suggests that this is a metastatic site, then this is not evidence of disease.
- If the imaging scan's impression states that foci are likely treated bone mets, this is evidence of cancer. This can

be implicit evidence of cancer for future scans.

- For Mammograms/MRIs of breast cancer, BI-RADS (*Breast Imaging Reporting and Data System*) should be used if recorded on the report. Scores of 4C, 5, and 6 indicate cancer.
 - 4: Suspicious abnormality – Biopsy should be considered Findings do not definitely look like cancer but could be cancer. The radiologist is concerned enough to recommend a biopsy. The findings in this category can have a wide range of suspicion levels. For this reason, this category is often divided further:
 - 4A: Finding with a low likelihood of being cancer (more than 2% but no more than 10%)
 - 4B: Finding with a moderate likelihood of being cancer (more than 10% but no more than 50%)
 - 4C: Finding with a high likelihood of being cancer (more than 50% but less than 95%), but not as high as Category 5
 - 5: Highly suggestive of malignancy – Appropriate action should be taken The findings look like cancer and have a high chance (at least 95%) of being cancer. Biopsy is very strongly recommended.
 - 6: Known biopsy-proven malignancy – Appropriate action should be taken This category is only used for findings on a mammogram that have already been shown to be cancer by a previous biopsy. Mammograms may be used in this way to see how well the cancer is responding to treatment.

Interpreting ambiguous terms (i.e. words describing likelihood or suspicion of cancer):

Yes. the report states or implies there is evidence of cancer if any of the below terms are used to describe possibility of cancer:	The report is uncertain. indeterminate. or equivocal if any of the below terms are used to describe possibility of cancer:	No. the report states or implies there is no evidence of cancer if any of the below terms are used to describe possibility of cancer:
Concerning (for) <i>[cancer/malignancy/metastases/ recurrence/ relapse/ progression]</i>	Bears watching (with mention of cancer)	Bears watching (with no mention of cancer)
Comparable with	Cannot be ruled out	Follow up recommended (with no mention of cancer)
Compatible with	Follow up recommended (with mention of cancer)	
Consistent with	Equivocal	
Favors	Possible	
Likely/most likely	Potentially	
Presumed	Questionable	
Probable	Rule out	
Suggests	May/Could (with mention of cancer)	
Suspect(ed)	Other words that indicate ≤ 50 % chance the finding is cancer	
Suspicious (for)		
Typical of		
Worrisome (for)		
Or other words that indicate >50% chance the finding is cancer		

- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Data Standard

- PRISSMM

Which best describes the radiologist's assessment of the change in the patient's cancer status recorded in the

Impression?

[image_overall] PN0577

Values

- 1, Improving/Responding
- 2, Stable/No change
- 3, Mixed
- 4, Progressing/Worsening/Enlarging
- 5, Not stated/Indeterminate

Directives

- This field will only be available if you chose “Yes” for evidence of cancer. When cancer is present, report what the radiologist says the general or overall cancer status is. Use the “Impression” section of radiology/imaging report to choose the best response from the options below. “Indication” section may also be used for context.
- When no overall interpretation or summary statement is explicitly given, use the description of the sites of cancer to interpret as below.

General guidelines:

- Do not include changes in size that are specifically attributed to something other than cancer directed therapy in your interpretation (for example, “opacity has decreased consistent with resolving pneumonia”).
- Do not include body parts or abnormalities that are not known to be involved with cancer in making your determination.
- Do not use changes in pleural/pericardial/other effusions, ascites, or other fluid-based abnormalities in determining response, even if these are known to be malignant.
- Only use terms that describe cancer status (e.g. progressing, responding, relapse, recurrence) and terms that describe changes in size (e.g. increase, decrease, larger, smaller) to evaluate response. Do not use terms that describe clarity or visualization on the scan (e.g. more prominent, conspicuous, increased avidity).
- If measurements are included in the impression section of the report you are curating and in the impression section of the reference scan, you can compare these measurements to evaluate response. However, preference is still given to overall/summary statement.
- Only record comparison to reference date scan. If a report says something like, “stable since last scan, but progressing over several months,” choose stable. The interpretation/impression section often does not specifically mention the reference scan date; in this case, assume everything noted in the interpretation section represents a comparison to the reference scan.
- “Treatment effect” is an indication of improvement/response.
- Interval increase or decrease are indications of progression or improvement.
- “Sclerosis” is an indication of status change only in bone lesions. It can represent cancer growth or cancer responding to treatment, so more information is needed.
- “Reactive” usually indicates a healing process that is not necessarily related to cancer and should generally be ignored.

Guidelines for specific response:

- **Responding:** Cancer is responding to treatment, decreasing in size, improving, resolved. This includes mild/minimal/slight response.
 - Choose this if all sites of cancer noted are decreasing.
 - Choose this if some sites are stable/unchanged and some are decreasing.
 - A site of disease that is noted to be gone, resolved, or no longer visualized means that site is **responding**.
 - If a site previously known to be cancer is not mentioned on the report, do not assume this means it has responded or resolved.
- **Stable/No Change:** Choose this option if no change is mentioned for any site of disease.
- **Mixed:** This indicates that some sites of disease are new or increasing/progressing and some are

decreasing/responding or resolved.

- **Progressing:** Cancer is growing, increasing in size, larger tumor/disease burden. progressing. This includes mild/minimal/slight progression/increase. Choose this if:
 - All sites of cancer are growing/increasing.
 - Some sites are stable/unchanged and some are increasing.
 - The report states that **there are new sites of disease**. Do not assume a site is a new lesion simply because you have not seen it in the impression previously; some radiologists write more detailed reports than others. However, if you can reasonably assume from context that the site of cancer is new, choose “progressing.”
 - Patient was previously free of cancer (NED/no evidence of disease) and the report indicates or implies there is evidence of cancer.
- **Not stated/indeterminate:** Avoid whenever possible. Choose this if radiologist does not state response or status for any site or if entire scan is indeterminate. If some sites are indeterminate and others are stable/responding/progressing, choose the option that is more definite.
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.
- If the imaging scan’s impression states that foci are likely treated bone mets, this is evidence of cancer. Determining the radiologist’s impression is tough between Improving and indeterminate. Treated implies improving but doesn’t say so explicitly. Code this as Indeterminate since most curators won’t be able to tell about this and most will curate it this way.

Data Standard

- PRISSMM

Shared: Yes

Where (in the body) is the cancer located? 15 sites available

[image_casite1-15] PN0578, PN0579, PN0580, PN0581, PN0582, PN0583, PN0584, PN0585, PN0586, PN0587, PN0588, PN0589, PN0590, PN0591, PN0592

Values

SEER ICD-O-3 Site/Topography Code List (See Appendix 7a)

Directives

- Record each site of the body in which cancer was present as reported in the Impression of the imaging report in a separate data field; there are 15 fields that can be used to collect this data. Assume that sites previously known to be cancer are still cancer until otherwise stated. See “General Guidelines” below for more detail on when to consider something a site of cancer. This includes locations that are explicitly stated to be cancerous or malignant as well as sites that are noted to be likely/probably/suspicious/concerning for cancer. See “Evidence of Cancer” directives for more guidance.

General Guidelines

- Assume that sites previously known to be cancer are still cancer until otherwise stated.
- Choose the most specific option available. For example, if hilar lymph nodes are resected during a thoracotomy, code the them as “C77.1 Intrathoracic lymph nodes” not “C77.9 Lymph Nodes, NOS.”
- Only select sites of disease mentioned on that report. Do not include sites previously known to be cancer if they are not mentioned in the report being curated.
- Do not include sites that are possibly, but not favored to be cancer. Use the ambiguous terms list above for guidance. “Indeterminate” sites should not be selected.
- Do not include sites stated to be resolved.
- “Pathologic fracture” is an indication that this was (at some point) a site of cancer, but more is needed to consider it a site.

- “Sclerotic bone” must have either explicit or implicit evidence of malignancy to be considered a site of cancer.
- “Adenopathy” and “lymphadenopathy” may not necessarily mean cancer. Count this as a site of disease if you already know there is cancer in the lymph nodes or if the radiologist refers to change in size as a response to therapy.
- Pleural effusions, pericardial effusions, ascites, or other fluid collection should be explicitly described as malignant within the report or have implicit evidence from a prior report to be considered a site.

Analytic Note

- If at least one of the specimens in the instance has **Where (in the body) is the cancer located? 15 sites available** is not blank, the anatomic sites can be grouped into distant or local sites of cancer.

Data Standard

- NAACCR/ICD-O-3 Site/Topography with additional values for fluids

History

- 03032020: removed leading zeros from codes.

Shared Field: Yes

Reference Scan Date

[image_ref_scan_dt] PN0593

Value

Date field. mmddyyyy

Directives

- Record the date of the scan that this one is being compared to; it is often referred to as the comparison scan. Leave blank if no reference or comparison scan date is mentioned in the report.
- If the report includes more than one reference scan, choose the date mentioned in the Impression; if there is no date mentioned in the Impression. Record the **most recent** date, regardless of modality, including ultrasounds.
EXCEPTION: do not enter comparisons to X-Rays.

Data Standard

- PRISMM

Shared Field: No, a calculated interval in days from **Calculated Date of Birth** is generated

PRISMM Radiology Imaging Calculated Fields

In order to share data in a de-identified manner, several intervals are calculated in the REDCap project from the anchor date: Date of Birth as follows:

Interval Data Field	Interval	Calculation
Interval from Date of Birth to Date of Imaging Scan	[image_scan_int]	datediff([birth_dt], [image_scan_dt], "d", "mdy")
Interval from Date of Birth to Date of Imaging Report	[image_report_int]	datediff([birth_dt], [image_report_dt], "d", "mdy")
Interval from Date of Birth to Date of Image Reference Scan Date	[image_ref_scan_int]	datediff([birth_dt], [image_ref_scan_dt], "d", "mdy")

PRISMM Radiology/Imaging Data Violations

	Incorrect scan date [+/- 1 month]
--	-----------------------------------

Major	Missed or incorrectly identified invasive cancer
	Missed or incorrectly identified progression
	Missed or incorrectly identified response
	Missed eligible imaging report(s)
Minor	Incorrect missing anatomic sites of cancer
	Incorrect report date [+/- 1 month]
	Incorrect scan type/scan sites
	Incorrect cancer status (between stable. mixed. or not stated/indeterminate)
Other	Incorrect/missing reference scan date
	Incorrect/missing procedure/report institution
	Other minor mistake/omission

General PRISSMM Medical Oncology Assessment

Provenance

- Internal (native) and External Medical Oncology outpatient provider notes
 - During the time of COVID:19, televisits with medical oncology providers can be used – starting March 2020
- Providers include physicians and mid-level providers like nurse practitioners and physician assistants

Selecting Med Onc Notes to Curate

- Find the date of diagnosis of the cancer of interest.
- Some patients have more than one cancer diagnosis. **Do** review and curate only notes for the cancer of interest.
- **Do** review and curate one clinical assessment per month, beginning at time of diagnosis.
 - **Do** choose first Medical Oncology note of the month that is authored by an MD actively following the patient for the cancer of interest.
 - If there is no note by a medical oncologist (MD) that month, **do** use the first note by a nurse practitioner or physician assistant (NP/PA) from a medical oncology practice.
 - Rarely** look at more than one note per month. If the patient has imaging scans after the first note in the month, curate the first note following the first scan.
 - **Do** give priority to internal visit notes. Only curate notes from an outside institution if there is no DFCI/Partners oncology note in that month. Prioritize as follows:
 1. Internal MD
 2. Internal NP/PA
 3. External MD
 4. External NP/PA
 - If a fellow has a note and an attending physician adds an addendum, review any information in the Summary/impression/Evaluation/Plan, including the addendum.
 - **Do not** use notes from Radiation Oncology, Surgery/Surgical Oncology, inpatient care, primary care, or other specialists not related to cancer (e.g. dermatology, cardiology).
 - Patients with early stage cancers are sometimes only followed by a surgical oncologist. This may mean that they are many months without notes that qualify for curation. That is ok!
 - **Rarely** curate notes by an oncologist from a different specialty than the cancer of interest. Occasionally a patient will transfer care to a different type of oncologist due to the particulars of their disease. For example, a patient with brain metastases from their primary cancer may be followed by Neuro Oncology. **Do** curate these notes if notes from the primary oncologist are not available. **Do not** curate notes from a different oncology specialist that are pertaining to a different cancer diagnosis.
 - When the cancer of interest is ovarian, fallopian tube, or primary peritoneal cancer: notes from Gynecologic Oncologists should be curated if the GynOnc administered antineoplastic drug therapy to the patient. Begin curating GynOnc notes in the month that antineoplastic therapy began, following the same guidelines as described for selection of Medical Oncology notes when more than one note is available in a given month.
- **Do** review only the Impression/Plan section at the bottom of the note as well as the reason for visit.
 - **Rarely** a medical oncology provider will summarize the cancer status directly above the Impression/Plan section, which can be reviewed for curation.
- **Do not** review any of the other sections, including the physical exam, interval history, lab results, etc.
 - The Impression/Plan section may have a different name in medical oncologist progress notes; other section headers could include Assessment, Summary, Conclusion, Problem List Items Addressed this Visit.
 - When there are no section headings in a provider's note, **do** review everything located beneath the physical exam.

PRISMM Medical Oncology Assessment Data Curation Directives

Outpatient Oncology Visit Date with Medical Oncology Provider

[md_onc_visit_dt] PN0609

Value

Date field. mmddyyyy

Directives

- Record the date of the outpatient oncology visit with a medical oncology provider. This is the date the visit occurred, not the date the note was signed or uploaded.

Data Standard

- PRISMM

Shared Field: No, a calculated interval in days from **Calculated Date of Birth** is generated

Institution Where Medical Oncology Visit Occurred

[md_inst] PN1074

Value

- Internal Institution
- External Institution

Directives

- Record whether the medical oncology visit took place at the internal (native) institution or at an external institution.
- For cancer centers, satellite offices count as internal.

Data Standard

- PRISMM

Shared Field: Yes

What type of cancer are you curating?

[md_type_ca_cur] PN0611

Values

1, Adrenocortical Carcinoma|3, Anal Cancer|5, Appendix Cancer|6, Bile Duct Cancer|7, Bladder Cancer|11, Brain Cancer|13, Breast Cancer|14, Breast Sarcoma|15, NET or Carcinoid|17, Cervical Cancer|97, Corpus Uteri Carcinoma and Carcinosarcoma|99, Corpus Uteri Sarcoma|19, Colon Cancer|20, Colon/Rectum Cancer|21, Esophagus Cancer|201, Ewing Sarcoma|23, Fallopian Tube Cancer|25, Gallbladder Cancer|205, Germ Cell Tumor|27, GIST|29, Head and Neck Cancer|30, Mesothelioma|35, Ill Defined/Cancer of Unknown Primary|39, Liver Cancer|41, Lung Cancer, NOS|45, Melanoma|47, Merkel Cell|211, Neuroblastoma|51, Non Small Cell Lung Cancer|215, Osteosarcoma|55, Ovarian Cancer|57, Pancreatic Cancer|58, Parathyroid Cancer|59, Penis Cancer|61, Peritoneum Cancer|63, Placenta Cancer|65, Prostate Cancer|67, Rectum and Rectosigmoid Cancer|69, Renal Kidney Cancer|71, Renal Pelvis Cancer|221, Retinoblastoma|225, Rhabdomyosarcoma|73, Scrotum Cancer|75, Small Cell Lung Cancer|77, Small Intestine Cancer|95, Soft tissue sarcoma|81, Stomach Cancer|83, Testis Cancer|85, Thymus Cancer|87, Thyroid Cancer|89,

Uterus Cancer NOS|91, Vagina Cancer|93, Vulva Cancer|231, Wilms Tumor|301, Leukemia|303, Non Hodgkins Lymphoma | 305, Hodgkins Lymphoma| 307, Multiple Myeloma | 309, Other hematopoietic or lymphoid neoplasm|1000, Other

Directives

- Record the cancer of interest representing the medical oncologist report that is being curated.
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Data Standard

- PRISMM

Shared Field: Yes

What is the patient's ECOG Performance Status as recorded by the Medical Oncologist in the Physical Exam section of the note?

[md_ecog] EXP01

Values

0, 0: Fully active, able to carry on all pre-disease performance without restriction | 1, 1: Restricted in physically strenuous activity but ambulatory and able to carry out work of a light or sedentary nature, e.g., light house work, office work | 2, 2: Ambulatory and capable of all selfcare but unable to carry out any work activities; up and about more than 50% of waking hours | 3, 3: Capable of only limited selfcare; confined to bed or chair more than 50% of waking hours | 4, 4: Completely disabled; cannot carry on any selfcare; totally confined to bed or chair | 9, Not documented in note

Directives

- Curate the ECOG performance status as recorded by the medical oncologist; if ECOG is not stated, select “9, Not documented in note” and curate the KPS score in the field below.
- Only explicitly stated scores should be recorded—do not try to score the patient based on information in the note.
- Make sure the performance status you record is from the current visit, and not from the past.

Provenance

- Physical Exam, Review of Systems and Impression/Plan sections of internal or external medical oncology notes.

Data Standard

- ECOG-ACRIN

Shared Field: Yes

What is the patient's Karnofsky Performance Status as recorded by the Medical Oncologist in the Physical Exam section of the note?

[md_karnof] EXP02

Values

100, 100: 100 Normal no complaints; no evidence of disease. | 90, 90: Able to carry on normal activity; minor signs or symptoms of disease. Able to carry on normal activity and to work; no special care needed. | 80, 80: Normal activity with effort; some signs or symptoms of disease. | 70, 70: Cares for self; unable to carry on normal activity or to do active work. | 60, 60: Requires occasional assistance, but is able to care for most of his personal needs. Unable to work; able to live at home and care for most personal needs; varying amount of assistance needed. | 50, 50: Requires considerable assistance

and frequent medical care. | 40, 40: Disabled; requires special care and assistance. | 30, 30: Severely disabled; hospital admission is indicated although death not imminent. | 20, 20: Very sick; hospital admission necessary; active; supportive treatment necessary. | 10, 10: Moribund; fatal processes progressing rapidly | 9, Not documented in the note

Directives

- Curate the Karnofsky Performance Status (KPS) as recorded by the medical oncologist. This field is only available for curation if the ECOG score is unavailable.
- Only explicitly stated scores should be recorded—do not try to score the patient based on information in the note.
- Make sure the performance status you record is from the current visit, and not from the past.

Provenance

- Physical Exam, Review of Systems and Impression/Plan sections of internal or external medical oncology notes.

Data Standard

- Karnofsky Performance Status

Shared Field: Yes

[In the Impression/Plan Section, how does the provider refer to the patient's hormone/ castrate/ androgen/ independent status at this visit?](#)

[\[md_pca_status\] PN0613](#)

Values

1, Hormone sensitive =castrate sensitive=androgen sensitive prostate cancer (HSPC=CSPC=ASPC) | 2, Hormone resistant=castrate resistant =androgen resistant/independent prostate cancer (HRPC=CRPC=ARPC) | 99, Not stated

Directives

- Record the patient's hormone status.
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Data Standard

- PRISMM

Shared Field: Yes

[In the Impression/Plan Section, what does the provider indicate about the change in the PSA level at this visit?](#)

[\[md_psa_status\] PN0614](#)

Values

1, PSA Nadir or undetectable | 2, PSA Stable | 3, PSA Increasing | 4, PSA Decreasing | 99, PSA change not mentioned

Directives

- Record the PSA status.
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Data Standard

- PRISMM

Shared Field: Yes

Is there any evidence of cancer in the Impression/Plan section?

[md_ca] PN0616

Values

- 1, Yes, the Impression/Plan states or implies there is evidence of cancer
- 2, No, the Impression/Plan states or implies there is no evidence of cancer
- 3, The Impression/Plan is uncertain. indeterminate. or equivocal
- 4, The Impression/Plan does not mention cancer

Directives

- Only read the “Impression and Plan” section of the visit note. This may also be called something like “Assessment and Recommendations.” If there are no section headers in the visit note, use everything after the description of the physical exam. Evidence of cancer may be explicit or implicit.
- Many “Impression and Plan” sections begin with a summary statement about the patient’s stage of disease and reason for visit.
 - If it says, “Patient with Stage IV cancer...” or “Patient with metastatic cancer...” this is implicit evidence of disease.

Only choose “no evidence of cancer” for a patient with metastatic disease if the provider explicitly states that the patient has had a complete response or that cancer is no longer present.

This may be written as: complete remission, NED/no evidence of disease, no active cancer.

Note: “no measurable disease” **does not** necessarily mean no evidence of cancer at all.

Example: “Ms. W is a woman with a 2.5+ year history of metastatic adenocarcinoma of the lung with bone involvement who has enjoyed a complete response to erlotinib monotherapy. She presents today with a recent CT of her chest and neck which confirms the ongoing CR.”
 - Stage I, II, or III disease does not necessarily imply evidence of cancer. There are also notes in which the provider does not mention the patient’s stage. In these cases. other clues may be present which imply evidence of cancer.

If a patient is on neoadjuvant chemo. consider this **yes, evidence of cancer.**

If the note mentions an upcoming planned resection/surgery for cancer, consider this **yes, evidence of cancer.**

Patients with ovarian, fallopian tube or primary peritoneal cancer: if a patient is on adjuvant chemotherapy following definitive resection or optimal debulking surgery (including patients with no gross residual disease or with residual tumor nodules 1 cm or less in size following debulking), consider this **no evidence of cancer** unless otherwise stated.

Interpreting ambiguous terms (i.e. words describing likelihood or suspicion of cancer):

Yes. the report states or implies there is evidence of cancer if any of the below terms are used to describe possibility of cancer:	The report is uncertain. indeterminate. or equivocal if any of the below terms are used to describe possibility of cancer:	No. the report states or implies there is no evidence of cancer if any of the below terms are used to describe possibility of cancer:
Concerning (for) <i>[cancer/malignancy/metastases/recurrence/relapse/progression]</i>	Bears watching (with mention of cancer)	Bears watching (with no mention of cancer)
Compatible with	Cannot be ruled out	Follow up recommended (with no mention of cancer)
Consistent with	Follow up recommended (with mention of cancer)	
Favors	Equivocal	
Likely/most likely	Possible	
Presumed	Potentially	
Probable	Questionable	
Suggests	Rule out	
Suspect(ed)	May/could (with mention of cancer)	
Suspicious (for)	Other words that indicate ≤ 50 % chance the finding is cancer	
Typical of		
Worrisome (for)		
Or other words that indicate >50% chance the finding is cancer		

- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Data Standard

- PRISSMM

Shared: Yes

Which best describes the medical oncologist's assessment of the change in the patient's cancer status recorded in the Impression/Plan/Assessment? [md_ca_status] PN0617

Values

- 1, Improving/Responding
- 2, Stable/No change
- 3, Mixed
- 4, Progressing/Worsening/Enlarging
- 5, Not stated/Indeterminate

Directives

- This field will only be available when “Yes, the Impression/Plan states or implies there is evidence of cancer” is chosen for **Is there any evidence of cancer in the Impression/Plan section?**
- Use the “Impression and Plan” section of the note to determine which option best corresponds to the oncologist’s impression of the patient’s overall cancer status.
- Report what the medical oncologist provider says the general or overall cancer status is.
- General guidelines:

- Use a general/summary statement or overall impression whenever possible.
 - Some notes include a recap of recent imaging. Only use this to evaluate status if the provider does not provide their own overall impression.
 - Some notes comment on tumor markers. Do not use this to evaluate status unless the provider states they are using it that way.

“Decrease in tumor markers indicates response to therapy” = Improving/Responding; “tumor markers have decreased” on its own should be ignored.
- Providers often copy part of the assessment from week to week. You may see a statement like “patient has completed X cycles of treatment and her scans show stable disease” repeated several times with only the number of cycles changing. It is likely the scans referenced are a few months in the past, but we do not always have a way to tell. Do not try to interpret what the provider is basing their assessment on or when scans happened etc. **If the provider gives a summary statement of disease status, even if it appears copied, use that as your response.**
- Choose the response that best reflects the patient’s status on that visit date. For instance, in the example below, we would say “improving/responding” since that is the status since the patient began treatment, and not “progressing” since that happened in the past.

Ms. G is a 65-year-old nonsmoking woman with metastatic lung adenocarcinoma here for re-evaluation after re-initiation of sunitinib therapy 2 weeks ago.

Will continue with therapy with plan to repeat CT scan at the end of this cycle. I was somewhat concerned about the degree of growth in the 5 weeks off of therapy (actual TIMC measurements were significant increase) but neck node has already dramatically decreased.

- Do not use descriptions of the patient’s general well-being or how they are tolerating treatment as indication of cancer status. For example, “doing well on chemo” may just mean absence of side effects, not cancer response. Similarly, “clinically improving/clinical improvement” should not be considered responding unless the provider directly states that it is a result of anti-neoplastic treatment or a sign of cancer response.
- Notes may include somewhat contradictory statements like, “progressing, but overall stable.” Choose the response that best corresponds to the favored or overall impression (in this example: stable).

Specific response guidelines:

- **Responding:** Cancer is responding to treatment, decreasing in size, improving, resolved.
 - Choose this if provider states the patient has decreased tumor burden or is responding to therapy generally.
 - In notes where no general statement is given, but specific sites of cancer are mentioned, choose responding if:
 - ☐ All sites are decreasing/resolved. If the medical oncologist states the patient has had a complete response to treatment, then evidence of cancer should be curated as “no”.
 - ☐ Some sites are decreasing/resolved and some are stable/unchanged.
 - In general, choose responding when the provider mentions response, even if it is not clear that it represents improvement at that visit. Common phrases include:
 - ☐ Ongoing response
 - ☐ Sustained response
 - ☐ Continuing response
 - ☐ Continued response
 - ☐ Lasting response
- **Stable/No Change:** Choose this option if the provider states the patient has stable disease or notes that tumor size/cancer burden is unchanged.
 - “Clinically stable” indicates stable disease – this includes statements like, “clinically stable to continue treatment/receive chemo”
 - “Stable response” indicates stable/no change
 - “Scans show good disease control”

- **Mixed:** Choose this if the provider states the patient has a mixed response or notes that some sites of cancer are decreasing or resolved, and some are new or increasing.
- **Progressing:** Cancer is growing. increasing in size. larger tumor/disease burden. progressing.
 - Choose this if the provider states the patient has recurred. relapsed. had progression/progressive disease. or increased tumor/cancer burden.
 - “Not responding” also indicates progressing/worsening.
 - In notes where no general statement is given. but specific sites of cancer are mentioned. choose progressing if:
 - ☐ All sites are increasing.
 - ☐ Some sites are increasing. and some are stable/unchanged.
 - ☐ A new site of cancer is noted.
- **Not stated/indeterminate:** Providers often do not describe a patient’s cancer status at each visit. especially if no scans or other tests evaluating extent of cancer have been performed. Choose this option if no general statement of cancer status is given. no imaging is described. and/or the provider states that the patient’s cancer status cannot be determined at that visit. Choose this if the patient was recently diagnosed and has not begun treatment or had definitive local therapy (i.e. for baseline visits).
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Data Standard

- PRISSMM

Shared Field: Yes

PRISMM Medical Oncology Assessment Calculated Fields

In order to share data in a de-identified manner, intervals in days are calculated in the REDCap project from the anchor date: Date of Birth as follows:

Interval Data Field	Interval	Calculation
Interval from Date of Birth to Medical Oncology Visit Date	[md_onc_visit_int]	datediff([birth_dt], [md_onc_visit_dt], "d", "mdy")

Medical Oncology Assessment Data QA

For Curator: Are the QA issues resolved?

[md_qacurator]

Values

0, No
1, Yes

Description

- Flag indicating that the curator has fixed the data quality issues found by the QA manager

Directives

- This field will only be available when “Yes, the Impression/Plan states or implies there is evidence of cancer” is chosen for **Is there any evidence of cancer in the Impression/Plan section?**
- Once the curator fixes data errors, “Yes” should be recorded.

Med Onc Assessment QA performed?

[md_qa]

Values

0, No
1, Yes

Description

- Flag indicating that the QA Manager reviewed the form looking at data quality issues.

Directives

- If the form is reviewed for data quality, the QA Manager should record “Yes.”

Were any data quality issues identified?

[md_qaissues]

Values

0, No
1, Yes

Description

- Flag indicating that the QA Manager found data quality issues after reviewing the form.

Directives

- If data quality issues are found, the QA Manager should record “Yes.”

Types of data quality issues identified

[md_qatype]

Values

Multi-select

- 1, Major violation(s)
- 2, Minor violation(s)
- 3, Other

Description

- Flags indicating the type of data quality issues found

Directives

- If data quality issues are found, the QA Manager should record “Yes.”

Medical Oncology Assessment Data Violations

Major	Incorrect visit date [+/- 1 month]
	Missed or incorrectly identified invasive cancer
	Missed or incorrectly identified progression
	Missed or incorrectly identified response
	Missed eligible note(s)
	Note associated with wrong cancer type
Minor	Incorrect cancer status (between stable, mixed, or not stated/indeterminate)
	Ineligible note included
Other	Did not choose best note in the month
	Other minor mistake/omission

General PRISSMM Tumor Markers Curation Directives

Provenance

Internal (native) and External Laboratory Test Results

- If the Tumor Marker section is in specified. all instances of these Tumor Markers from the internal/native EHR laboratory test section should be piped into REDCap. The data piping rules for tumor markers will vary by project. Some projects will include no tumor markers while others will include all or a subset of the tumor markers. See Appendix XX for suggested markers.
- The decision to curate external tumor markers will be set by the project.
- Tumor Markers include:
 - AFP
 - BhCG
 - CA125
 - CA15-3
 - CA19-9
 - CA2729
 - Calcitonin
 - CEA
 - Chromogranin A
 - LDH
 - PSA
 - NSE
 - Testosterone
 - Thyroglobulin
- Most of the data fields on this form are either piped in from either the EHR or data warehouse or are calculated.
- Some of the populated fields should not be modified. and this is indicated in the Field Note in the REDCap project.
- Certain date fields may be edited as follows if data is missing or incorrect; this is specified in the Field Note.

Tumor Marker Data Curation Directives

Tumor Marker Specimen Collection Date

[tm_spec_collect_dt] PN0629

Values

Date field, mmddyyyy

Directives

- The data element can either be piped into the REDCap project or manually curated.
- Record the date of the blood draw resulting in the specimen that was tested for this tumor marker.

Shared Field: No, contains PHI. The data element is flagged as an identifier in REDCap. Instead, a calculated interval in days [tm_spec_collect_int] from **Calculated Date of Birth** is generated.

Tumor Marker Lab Test

[tm_type_desc_2] PN0632

Values

1, AFP|2, BhCG|3, CA125|4, CA15-3|5, CA19-9|6, CA2729|7, Calcitonin|8, CEA|12, Chromogranin A|13, LDH|9, PSA|10, NSE|14, Testosterone|11, Thyroglobulin

Directives

- The data element can either be piped into the REDCap project or manually curated.
- Record the appropriate tumor marker.
- For the BPC the following cancers should have the following tumor markers collected.
 - Breast Cancer:
 - Colon Rectum Cancer: CEA
 - Pancreatic Cancer:
 - Prostate Cancer: PSA and Testosterone
 - Ovarian Cancer: CA-125
- Please note that the codes (in gray) for the choices are not visible in REDCap; however, the codes will be available at data export.

Data Standard

- NCI

Shared Field: Yes

Tumor Marker Numeric Result Value

[tm_result] PN0633

Values

Numeric

Directives

- The data element can either be piped into the REDCap project or manually curated.
- Record the numeric result for the tumor marker.
 - If the tumor marker does not have a numeric result, but does have a text result indicating less than the lower range of normal like "<0.5" record 0.

Shared Field: Yes

Tumor Marker Result Unit

[tm_result_uom] PN0634

Values

Text

Directives

- The data element can either be piped into the REDCap project or manually curated.
- Record the result units for the tumor marker.
 - Example: ng/ml

Shared Field: Yes

PRISMM Tumor Markers Piped Data

The tumor marker data elements are in large part piped into REDCap from either the institutional data warehouse or electronic medical record (EHR); the data fields are listed below. Do not edit these following fields.

Data Field	Can be curated?	HIPAA Identifier	To Be Shared	Source
Test Order ID	No	Yes	No	EHR
Specimen Collection Date	Yes	Yes	No	EHR
Tumor Marker Lab Test	Yes	No	Yes	EHR
Tumor Marker Numeric Result Value	Yes	No	Yes	EHR
Tumor Marker Result Unit	Yes	No	Yes	EHR
Tumor Marker Range of Normal Values	Yes	No	Yes	EHR
Tumor Marker Lower Value for Normal Range	Yes	No	Yes	EHR
Tumor Marker Upper Value of Normal Range	Yes	No	Yes	EHR
Tumor Marker Result Out of Range	Yes	No	Yes	EHR
Tumor Marker Source Institution	Yes	No	Yes	EHR

PRISMM Tumor Marker Assessment Calculated Fields

- In order to share data in a de-identified manner, several intervals are calculated in the REDCap project from the anchor date: Date of Birth as follows:

Interval Data Field	Interval	Calculation
Interval from Date of Birth to Specimen Collection Date	[tm_spec_collect_int]	datediff([birth_dt], [tm_spec_collect_dt], "d", "mdy")

General Cancer Panel Test Data Curation Directives

- Data related to each cancer panel test submitted to GENIE for each patient will be piped into the REDCap project; this is referred to as AACR Project GENIE Tier 1 data.
- Most of the data fields on this form are either piped in from either the EHR or data warehouse or are calculated.
- The populated fields should not be modified; curated fields should be used to correct any data issues.

Cancer Panel Test Data Curation Directives

Type of Test

[cpt_test_type] PN1088

Values

1, NGS

5, Germline

9, Other Genetic Test

Directives

- Indicate whether the test performed was NGS, germline, or other genetic test.

Data Standard

- PRISSMM

Shared Field: Yes

History: 07022024 Added option for “Other Genetic Test”

Germline Cancer Panel Test Fields

These questions are visible when **Type of Test** is “5, Germline”.

Is a germline sequencing report available for curation?

[cpt_germline_report] PN1399

Values

0, No

1, Yes

Directives

- If a germline report is available for curation, choose “yes”.
- If a germline report is not available for curation, choose “no”. You will have the option to record the MD report of germline testing results, if available. The MD report of germline results should only be curated if a germline sequencing report is unavailable for curation.

Data Standards

PRISSMM

Shared Field: Yes

History: 07022024 field added to dictionary

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Were any pathogenic alterations or variant of uncertain significance in any of these genes noted?

If both a pathogenic alteration and a VUS were noted for a given gene, choose the pathogenic alteration.

[cpt_atm], [cpt_bard1], [cpt_brca1], [cpt_brca2], [cpt_brip1], [cpt_chek2], [cpt_epcam], [cpt_fanca], [cpt_hoxb13], [cpt_mlh1], [cpt_msh2], [cpt_msh6], [cpt_nbn], [cpt_palb2], [cpt_pms2], [cpt_rad51c], [cpt_rad51d] PN1089, PN1375, PN1090, PN1091, PN1092, PN1093, PN1376, PN1094, PN1095, PN1096, PN1097, PN1098, PN1099, PN1100, PN1101, PN1377, PN1378

Matrix Values

- 1, Pathogenic alteration
- 2, Variant uncertain significance
- 3, Neither pathogenic alteration nor VUS
- 4, Not tested

Directives

- Linked to field **Is a germline sequencing report available for curation?** If “yes” is selected, this matrix field will be available for curation.
- Indicate whether any pathogenic alterations or variants of uncertain significance were reported for any of the following genes: ATM, BARD1, BRCA1, BRCA2, BRIP1, CHEK2, EPCAM, FANCA, HOXB13, MLH1, MSH2, MSH6, NBN, PALB2, PMS2, RAD51C, RAD51D

Data Standard

- PRISSMM

Shared Field: No

History: 07022024 Added BARD1, EPCAM, RAD51C, and RAD51D to matrix.

If another mutation that is not listed above was noted to be a pathogenic alteration, please record it here.

[cpt_pathogenic_oth] PN1102

Values

Free Text

Directives

- Record any other pathogenic alterations not captured in the field above.

Data Standard

- PRISSMM

Shared Field: No

If a germline sequencing report is unavailable: are germline mutation testing results mentioned in the medical oncology notes?

[cpt_germ_medonc] PN1400

Values

- 0, No
- 1, Yes

Directives

- Linked to field **Is a germline sequencing report available for curation?** If “no” is selected, this field will be available for curation.
- If the medical oncologist reports the results of germline testing in an outpatient medical oncology note, choose “yes”. If no germline testing is mentioned, choose “no”.

Data Standards

PRISSMM

Shared Field: Yes

History: 07022024 field added to dictionary

What were the reported results of germline testing?

[cpt_germ_medonc_result] PN1401

Values

- 0, MD notes absence of germline mutations
- 1, MD notes presence of germline mutations
- 2, MD notes germline testing was performed but results of germline testing are not reported
- 9, MD notes germline testing results are indeterminate or unknown

Directives

- Linked to field **If a germline sequencing report is unavailable: are germline mutation testing results mentioned in the medical oncology notes?** If “yes” is selected, this field will be available for curation.
- Choose the option that best represents the MD’s assessment of germline testing results.

Data Standards

PRISSMM

Shared Field: Yes

History: 07022024 field added to dictionary

Indicate all germline mutations in the following genes as reported by the medical oncologist

[cpt_germ_medonc_mutations] PN1402

Values

1, ATM | 2, BARD1 | 3, BRCA1 | 4, BRCA2 | 5, BRIP1 | 6, CHEK2 | 7, EPCAM | 8, FANCA | 9, HOXB13 | 10, MLH1 | 11, MSH2 | 12, MSH6 | 13, NBN | 14, PALB2 | 15, PMS2 | 16, RAD51C | 17, RAD51D | 18, Other gene | 99, Not stated or unknown

Directives

- Linked to field **What were the reported results of germline testing?** If “MD notes presence of germline mutations” is selected, this field will be available for curation.
- Select all clinically significant, pathogenic, or deleterious mutations or variants documented by the MD. Assume any germline mutations or variants that are documented by the MD are clinically significant unless otherwise stated.

Data Standards

PRISSMM

Shared Field: Yes

History: 07022024 field added to dictionary

Other Genetic Test Cancer Panel Test Fields

These questions are visible when **Type of Test** is “9, Other Genetic Test”.

Was either Homologous Recombination Deficiency (HRD) Status, Genomic Instability Score (GIS) or Loss of Heterozygosity (LOH) Score reported?

[cpt_hrd_status_yn] PN1392

Values

0, No
1, Yes

Directives

- Choose “yes” if a genetic test report is available that reports either HRD status, GIS score, or LOH score.

Data Standards

PRISSMM

Shared Field: Yes

History: 07022024 field added to dictionary

Homologous Recombination Deficiency Status

[cpt_hrd_status] PN1393

Values

0, Negative
1, Positive
2, Not Reported
9, Indeterminate or Unknown

Directives

- Linked to field **Was either Homologous Recombination Deficiency (HRD) Status, Genomic Instability Score (GIS) or Loss of Heterozygosity (LOH) Score reported?** If “yes” is selected, this field will be available for curation.
- Choose the option that best reflects the reported HRD status.

Data Standards

PRISSMM

Shared Field: Yes

History: 07022024 field added to dictionary

Genomic Instability Score (GIS)

[cpt_gis_status] PN1394

Values

- 0, Negative
- 1, Positive
- 2, Not Reported
- 9, Indeterminate or Unknown

Directives

- Linked to field **Was either Homologous Recombination Deficiency (HRD) Status, Genomic Instability Score (GIS) or Loss of Heterozygosity (LOH) Score reported?** If “yes” is selected, this field will be available for curation.
- Choose the option that best reflects the reported GIS score.

Data Standards

PRISMM

Shared Field: Yes

History: 07022024 field added to dictionary

Loss of Heterozygosity (LOH) Score (%)

[cpt_loh_score] PN1395

Values

Numeric 1-100

Directives

- Linked to field **Was either Homologous Recombination Deficiency (HRD) Status, Genomic Instability Score (GIS) or Loss of Heterozygosity (LOH) Score reported?** If “yes” is selected, this field will be available for curation.
- Enter the LOH score as a percentage.

Data Standards

PRISMM

Shared Field: Yes

History: 07022024 field added to dictionary

Genomic Scar Score (GSS)

[cpt_gss_num] PN1396

Values

Numeric 1-100

Directives

- Linked to field **Was either Homologous Recombination Deficiency (HRD) Status, Genomic Instability Score (GIS) or Loss of Heterozygosity (LOH) Score reported?** If “yes” is selected, this field will be available for curation.
- Enter the reported numeric result of the genomic scar score.

Data Standards

PRISMM

Shared Field: Yes

History: 07022024 field added to dictionary

Genomic Scar Score (GSS) Summary

[cpt_gss_summary] PN1397

Values

- 0, Low
- 1, High
- 2, Not Reported
- 9, Inconclusive

Directives

- Linked to field **Was either Homologous Recombination Deficiency (HRD) Status, Genomic Instability Score (GIS) or Loss of Heterozygosity (LOH) Score reported?** If “yes” is selected, this field will be available for curation.
- Enter the reported summary result of the genomic scar score.

Data Standards

PRISSMM

Shared Field: Yes

History: 07022024 field added to dictionary

NGS Cancer Panel Test Fields

These questions are visible when **Type of Test** is “1, NGS0”.

Cancer sequence number associated with the NGS test

[cpt_ca]

Values

Multiselect checkboxes; whole number 1-10 representing cancer diagnosis and “Unknown”

1, 1st | 2, 2nd | 3, 3rd | 4, 4th | 5, 5th | 6, 6th | 7, 7th | 8, 8th | 9, 9th | 10, 10th | 11, Unknown

Directives

- Record the appropriate **Curated Cancer Sequence Number**, which was recorded the Cancer Diagnosis form relating to the cancer specimen used for the Cancer Panel Test.
- Summary information about the cancer diagnoses is provided in REDCap to assist in selection, including the **Curated Date of Diagnosis, Cancer Type, Cancer Site**.
- If it is not possible to determine which cancer diagnosis, more than one diagnosis can be selected. Although more than one cancer diagnosis can be indicated, the actual diagnosis should be identified.
- REDCap lists details about each of the cancers to assist the curators.

Data Standard

- PRISSMM

Path Report Instance 1 | Pathology Report Instance with Specimen Tested

[cpt_path_report_instance] PN0717

Values

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1, Path Report Instance 1 | 2, Path Report Instance 2 | 3, Path Report Instance 3 | 4, Path Report Instance 4 |
5, Path Report Instance 5 | 6, Path Report Instance 6 | 7, Path Report Instance 7 | 8, Path Report Instance 8 |
9, Path Report Instance 9 | 10, Path Report Instance 10 | 11, Path Report Instance 11 | 12, Path Report Instance 12 |
13, Path Report Instance 13 | 14, Path Report Instance 14 | 15, Path Report Instance 15 | 16, Path Report Instance 16 |
17, Path Report Instance 17 | 18, Path Report Instance 18 | 19, Path Report Instance 19 | 20, Path Report Instance 20 |
21, Path Report Instance 21 | 22, Path Report Instance 22 | 23, Path Report Instance 23 | 24, Path Report Instance 24 |
25, Path Report Instance 25 | 26, Path Report Instance 26 | 27, Path Report Instance 27 | 28, Path Report Instance 28 |
29, Path Report Instance 29 | 30, Path Report Instance 30 | 31, Path Report Instance 31 | 32, Path Report Instance 32 |
33, Path Report Instance 33 | 34, Path Report Instance 34 | 35, Path Report Instance 35 | 36, Path Report Instance 36 |
37, Path Report Instance 37 | 38, Path Report Instance 38 | 39, Path Report Instance 39 | 40, Path Report Instance 40 |
41, Path Report Instance 41 | 42, Path Report Instance 42 | 43, Path Report Instance 43 | 44, Path Report Instance 44 |
45, Path Report Instance 45 | 99, Cannot track specimen to a pathology report | 100, Curated pathology report not
available

Directives

- Record the instance number of the pathology report that details the specimen that was used for the cancer panel test.
-

Data Standard

- PRISMM

Cancer Panel Test Calculated Data

In order to share data in a de-identified manner, several intervals are calculated in the REDCap project from the anchor date: Date of Birth as follows:

Interval Data Field	Interval	Calculation
Interval in Days from DOB to Cancer Panel Test Order Date	[cpt_order_int]	datediff([birth_dt], [[cpt_order_dt], "d", "mdy"])
Interval in Days from DOB to Cancer Panel Test Report Date	[cpt_report_int]	datediff([birth_dt], [[cpt_report_dt], "d", "mdy"])

Cancer Panel Test Data Violations

Major	Incorrect pathology report
	CPT associated with wrong cancer diagnosis
	Missed eligible germline test result or other genetic test
Minor	Results of germline or other genetic test incorrectly curated
Other	Other minor mistake/omission

Cancer Panel Test Piped Data

The tumor marker data elements are in large part piped into REDCap from either the institutional data warehouse or electronic medical record (EHR); the data fields are listed below. Do not edit these following fields.

Data Field	HIPAA Identifier	To Be Shared	Data Standard	Source	GENIE Clinical Data Field*
GENIE Sample ID	No	Yes		EHR	Yes
Sample Accession Number - Institutional Data	Yes	No		EHR	
Block Accession Number - Institutional data	No	Yes		EHR	
Requisition ID - Institutional Data	Yes	No		EHR	
GENIE Specimen OncoTree Diagnosis	No	Yes			Yes
Specimen Primary Diagnosis - Institutional Data					
Specimen Surgery Site - Institutional Data					
GENIE Specimen Biopsy Type	No	Yes			Yes
GENIE Sequencing Assay ID	No	Yes			Yes
NGS Report Date	Yes	No			
GENIE Sequence Date (quarter year)	No	Yes			Yes

Interval in Days from Date of Birth to NGS Report Date					
Summary of the Panel Test Findings - Institutional Data	No	No			

*If applicable.

Appendix 1: Acronyms and Abbreviations

Many acronyms and abbreviations are used in oncology research. Below is a listing of acronyms used in PRISSMM data collection effort.

Acronym/ Abbreviation	Term Reference Description
AACR	American Association for Cancer Research https://www.aacr.org
AJCC	American Joint Committee on Cancer https://www.facs.org/quality-programs/cancer/ajcc The American Joint Committee on Cancer (AJCC) provides worldwide leadership in the development, promotion and maintenance of evidence-based systems for the classification and management of cancer in collaboration with multidisciplinary organizations dedicated to cancer surveillance and to improving care.
ACS COC	American College of Surgeons Commission on Cancer https://www.facs.org/quality-programs/cancer/coc The Commission on Cancer (CoC), a program of the American College of Surgeons (ACoS), recognizes cancer care programs for their commitment to providing comprehensive, high-quality, and multidisciplinary patient centered care.
CDC	Centers for Disease Control and Prevention https://www.cdc.gov/ CDC works 24/7 to protect America from health, safety and security threats, both foreign and in the U.S. Whether diseases start at home or abroad, are chronic or acute, curable or preventable, human error or deliberate attack, CDC fights disease and supports communities and citizens to do the same.
CEA	Carcinoembryonic antigen
CS	Collaborative Stage
CT	computed tomography. computerized tomography https://www.cancer.gov/publications/dictionaries/cancer-terms/
EHR	electronic medical record. EHR. electronic health record; synonym EMR
EMR	electronic medical record. EHR. Electronic health record; synonym EHR
GENIE	Genomics Evidence Neoplasia Information Exchange https://www.aacr.org/Research/Research/PAGES/AACR-PROJECT-GENIE-FAQ.ASPX
IACR	International Association of Cancer Registries http://www.iacr.com.fr/index.php?option=com_content&view=category&layout=blog&id=100&Itemid=577
IARC	International Agency for Research on Cancer https://www.iarc.fr/ The International Agency for Research on Cancer (IARC) is the specialized cancer agency of the World Health Organization. The objective of the IARC is to promote international collaboration in cancer research.
ICD-O-3	International Classification of Diseases – Oncology, 3 rd version Important to note that the IARC provides codes, but these are not identical to US lists. https://seer.cancer.gov/icd-o-3/
MRI	magnetic resonance imaging. NMRI. and nuclear magnetic resonance imaging https://www.cancer.gov/publications/dictionaries/cancer-terms/
NAACCR	North American Association of Central Cancer Registrars https://www.naacr.org/
NCI	National Cancer Institute https://www.cancer.gov/ The National Cancer Institute (NCI) is the federal government's principal agency for cancer research and training.

Acronym/ Abbreviation	Term Reference Description
	With a team of approximately 3,500 is part of the National Institutes of Health (NIH), one of 11 agencies that make up the Department of Health and Human Services (HHS).
NDI	National Death Index https://www.cdc.gov/nchs/ndi/index.htm
NPCR	National Cancer Program Registries https://www.cdc.gov/cancer/npcr/about.htm CDC's National Program of Cancer Registries (NPCR) works to measure progress in preventing and treating cancer, a leading cause of death in the United States.
NM	nuclear medicine
PET	positron emission tomography (PET) https://www.cancer.gov/publications/dictionaries/cancer-terms/
PET/CT	positron emission tomography (PET) scan and a computed tomography (CT) scan https://www.cancer.gov/publications/dictionaries/cancer-terms/
PRISMM	P: Pathology R: Radiology for Loco-regional recurrence I: Imaging for distant metastasis S: Signs S: Symptoms M(k): Tumor Markers M: Medical Oncology Assessment
SEER	Surveillance, Epidemiology, and End Results (SEER) Program https://seer.cancer.gov/ The Surveillance, Epidemiology, and End Results (SEER) Program provides information on cancer statistics in an effort to reduce the cancer burden among the U.S. population. SEER is supported by the Surveillance Research Program (SRP) in NCI's Division of Cancer Control and Population Sciences (DCCPS).
TCGA	The Cancer Genome Atlas https://www.cancer.gov/about-nci/organization/ccg/research/structural-genomics/tcga The Cancer Genome Atlas (TCGA), a landmark cancer genomics program, molecularly characterized over 20,000 primary cancer and matched normal samples spanning 33 cancer types. This joint effort between the National Cancer Institute and the National Human Genome Research Institute began in 2006, bringing together researchers from diverse disciplines and multiple institutions.

Appendix 2: Data Standards

North American Association of Central Cancer Registrars (NAACCR)/SEER

- Ethnicity
- Race
- Sex
- ICD-O-3 Site/Topography
- ICD-O-3 Histology/Morphology
- ICD-O-3 Behavior

AJCC

- AJCC v7 and v8 Staging Manual
- Collaborative Staging
- Cancer Types

National Cancer Institute Thesaurus (NCIt)

- Cancer Directed Drug Therapy
- Cancer Regimens
- Cancer Types

Appendix 3: Synonymous Terms

Many terms can be use as synonyms in oncology. Below is a listing of synonymous used in PRISMM data collection effort.

Word	Synonym	Context
Adenocarcinoma	Carcinoma	Lung Cancer
Carcinoma	Adenocarcinoma	Lung Cancer
Site	Topography	ICD-O
And	With	Synonyms when describing multiple histologies within a single tumor
Simultaneous	Synchronous; Prior to first treatment; at the same time	At diagnosis of tumor/tumors

Appendix 4: Date Estimation Rules

Use the best information available; however, if no information exists, use the following rules:

Day is Unknown: Report the day of the month as the 15th. If the 15th does not make logical sense in relation to the other date fields reported on the form, then use either the 1st or 30th. Report month and year as documented in the medical record.

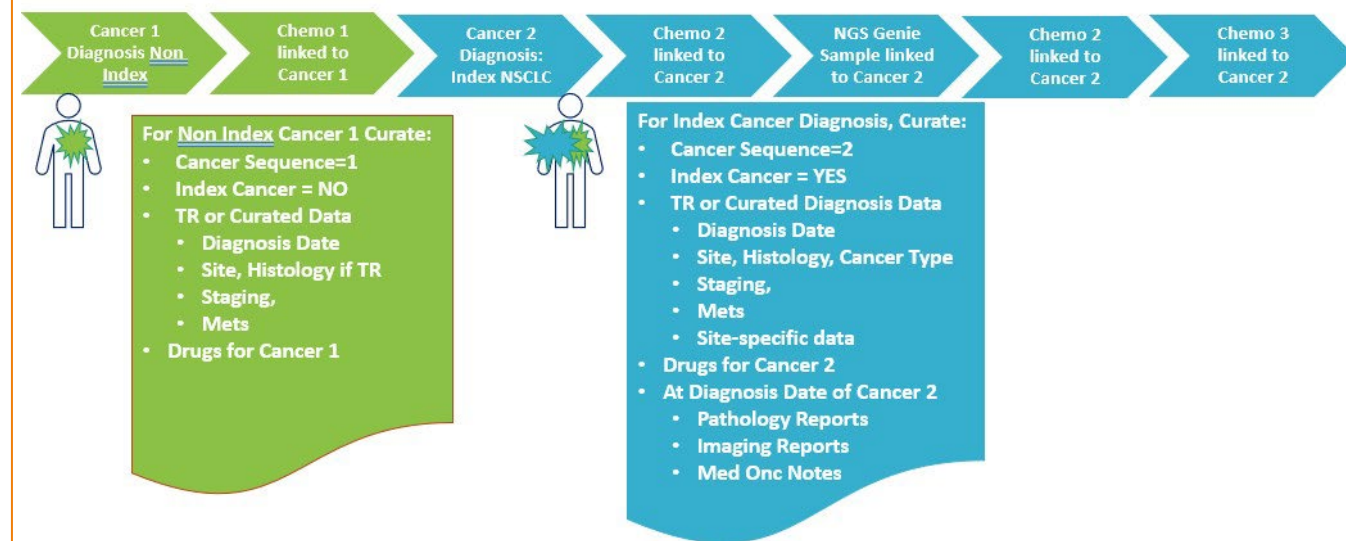
Month and Day Are Unknown: Report the month as June and the day as the 15th. If the 15th does not make logical sense in relation to the other date fields reported on the form, then report the day as either the 1st or the 30th. Report the year as documented in the medical record.

If dates of administration are Unknown. Cycles are Known: Estimate based on standard cycle length for the drug(s) given.

Appendix 5: Multiple Tumors and the Index Cancer

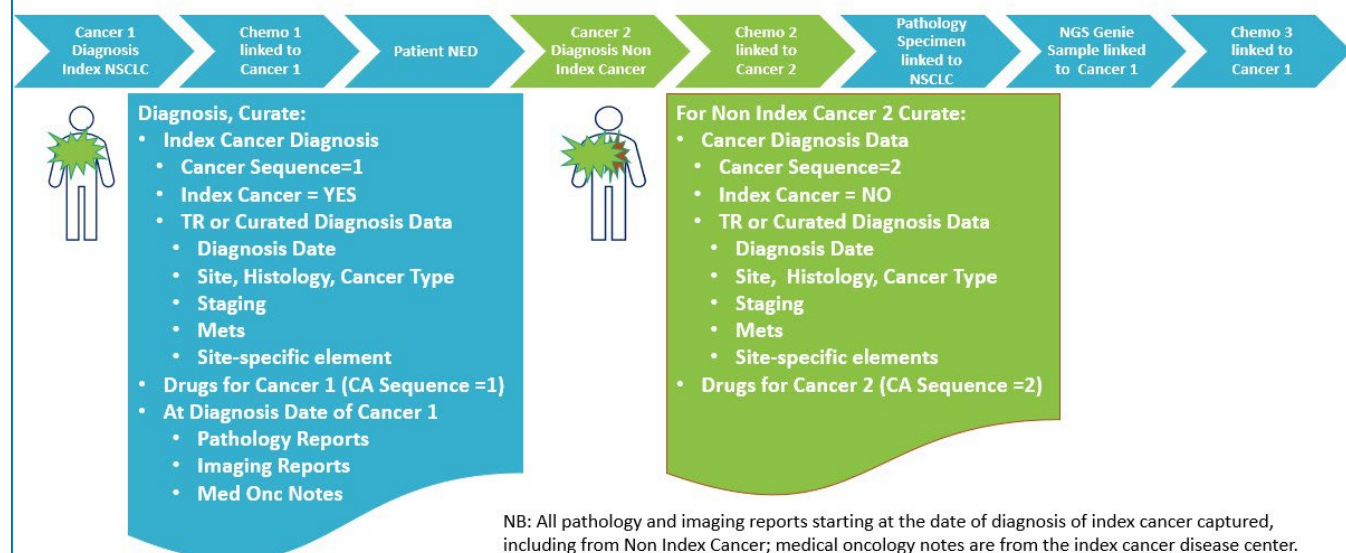
Use Case 1: Index Cancer is the Second Cancer with the first cancer not the Same Type as the Index

Background: Multiple Primaries With Non-Index NSCLC as First and Index NSCLC as Second Diagnosis

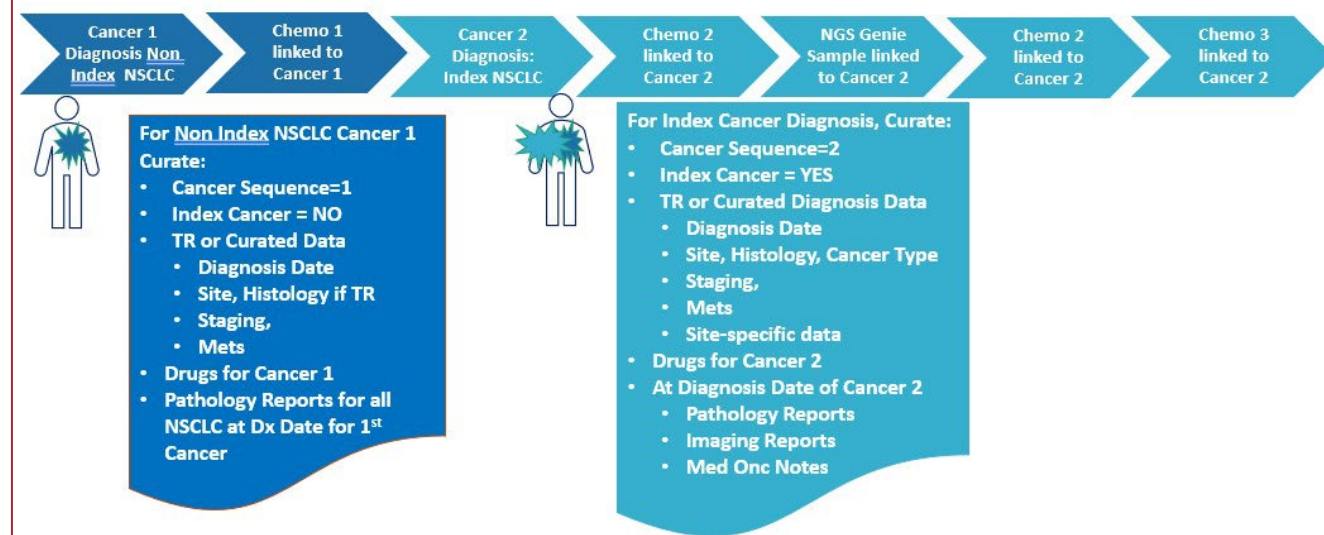


Extension of Use Case 1

Background: Multiple Primaries With Index NSCLC as Second Diagnosis



Background: Multiple Primaries With Non-Index NSCLC as First and Index NSCLC as Second Diagnosis



Appendix 6: Coding Tumor Grade

From SEER: <https://seer.cancer.gov/tools/grade/>

1. Use the Two-, Three- or Four-grade system information

a. Two-grade system

Term	Description	Grade Code	Exception for Breast and Prostate Grade Code
1/2. I/II	Low grade	2	1
2/2. II/II	High grade	4	3

b.

In transitional cell carcinoma for bladder, the terminology high grade TCC and low grade TCC are coded in the two-grade system.

c. Three-grade system

Term	Description	Grade Code	Exception for Breast and Prostate Grade Code
1/3	Low grade	2	1
2/3	Intermediate grade	3	2
3/3	High grade	4	3

d. Four-grade system: Any four-grade system including Edmondson and Steiner grade for liver.

Term	Description	Grade Code
1/4	Grade I; Well differentiated	1
2/4	Grade II; Moderately differentiated	2
3/4	Grade III; Poorly differentiated	3
4/4	Grade IV; Undifferentiated	4

2. Terminology: use the 'Description' column or the 'Grade' column to code grade. Breast & Prostate use the same grade code with a few noted exceptions.

Description	Grade	Assigned Grade Code	Exception for Breast and Prostate Grade Code
Differentiated. NOS	I	1	
Well differentiated	I	1	
Only stated as 'Grade I'	I	1	
Fairly well differentiated	II	2	
Intermediate differentiation	II	2	
Low grade	I-II	2	1
Mid differentiated	II	2	
Moderately differentiated	II	2	
Moderately well differentiated	II	2	
Partially differentiated	II	2	
Partially well differentiated	I-II	2	1
Relatively or generally well differentiated	II	2	
Only stated as 'Grade II'	II	2	
Medium grade. intermediate grade	II-III	3	2
Moderately poorly differentiated	III	3	
Moderately undifferentiated	III	3	
Poorly differentiated	III	3	
Relatively poorly differentiated	III	3	
Relatively undifferentiated	III	3	
Slightly differentiated	III	3	
Dedifferentiated	III	3	
Only stated as 'Grade III'	III	3	
High grade	III-IV	4	3

Description	Grade	Assigned Grade Code	Exception for Breast and Prostate Grade Code
Undifferentiated. anaplastic. not differentiated	IV	4	
Only stated as 'Grade IV'	IV	4	
Non-high grade		9	

Appendix 7a: ICD-O-3 Site Codes

The ICD-O-3 Site (Topography) Code list is the data standard for sites of cancer in PRISSMM, including the primary site of cancer, sites of distant metastases at diagnosis, anatomic sites of specimens in pathology, sites of cancer found on imaging.

The International agency for Research on Cancer provides ICD-O codes electronically on their website; however, the IARC version are not identical to the version used in the US. More details can be found here: <https://seer.cancer.gov/icd-o-3/>

PRISSMM uses the codes specified in the ICD-O-3 Site/Histology Validation List found at this site: <https://seer.cancer.gov/icd-o-3/>.

In addition, version 8 of the AJCC Staging Manual specify a subset of codes that should be associated with cancer types. These site codes are used in PRISSMM.

(Urinary) Bladder Cancer: Urothelial Carcinoma and Squamous Cell

- C67.0 Trigone of bladder
- C67.1 Dome of bladder
- C67.2 Lateral wall of bladder
- C67.3 Anterior wall of bladder
- C67.4 Posterior wall of bladder
- C67.5 Bladder neck
- C67.6 Ureteric orifice
- C67.7 Urachus
- C67.8 Overlapping lesion of bladder
- C67.9 Bladder, NOS

Breast Cancer Site Codes from AJCC Staging

- C50.0 Nipple
- C50.1 Central portion of breast
- C50.2 Upper-inner quadrant of breast (UIQ)
- C50.3 Lower-inner quadrant of breast (LIQ)
- C50.4 Upper-outer quadrant of breast (UOQ)
- C50.5 Lower-outer quadrant of breast (LOQ)
- C50.6 Axillary tail of breast
- C50.8 Overlapping lesion of breast
- C50.9 Breast, NOS

Colon Rectum Cancer Site Codes from AJCC Staging

- C18.0 Cecum
- C18.1 Appendix
- C18.2 Ascending colon
- C18.3 Hepatic flexure of colon
- C18.4 Transverse colon
- C18.5 Splenic flexure of colon
- C18.6 Descending colon
- C18.7 Sigmoid colon

C18.8 Overlapping lesion of colon
C18.9 Colon NOS
C19.9 Rectosigmoid junction
C20.9 Rectum NOS

Lung Cancer Site Codes from AJCC Staging

C34.0 Main bronchus
C34.1 Upper lobe lung
C34.2 Middle lobe lung
C34.3 Lower lobe lung
C34.8 Overlapping lesion of lung
C34.9 Lung NOS

Pancreas Cancer Site Codes from AJCC Staging (exocrine)

C25.0 Head of pancreas
C25.1 Body of pancreas
C25.2 Tail of pancreas
C25.3 Pancreatic duct
C25.7 Other specified parts of pancreas
C25.8 Overlapping lesion of pancreas
C25.9 Pancreas, NOS

Prostate Cancer Site Codes from AJCC Staging

C61.9 Prostate Gland

Renal Kidney Cancer Site Codes from AJCC Staging

C64.9 Kidney, NOS

Ovarian, Fallopian Tube, and Primary Peritoneal Cancer Site Codes from AJCC Staging

C56.9 Ovary
C57.0 Fallopian tube
C48.1 Specified parts of peritoneum
C48.2 Peritoneum
C48.8 Overlapping lesion of retroperitoneum and peritoneum

Appendix 7b: ICD-O-3 Site Coding Resolutions

How to code sites that are not listed explicitly as sites in ICD-O-3

Not every anatomic site encountered will be listed in the ICD-O-3 dropdown list; however, the ICD-O manual (https://apps.who.int/iris/bitstream/handle/10665/96612/9789241548496_eng.pdf;jsessionid=10A365E03C19501AD7D19CEB57A04475?sequence=1) should be searched to find the most relevant code. Below some common examples are listed.

Bone

C41 BONES, JOINTS AND ARTICULAR CARTILAGE OF OTHER AND UNSPECIFIED SITES

C41.2 Vertebral column

- Atlas
- Axis
- Bone of back
- Intervertebral disc
- Nucleus pulposus
- Spinal column
- Spine
- Vertebra

Chest Wall

Soft Tissue: C49.3 Connective, subcutaneous and other soft tissues of thorax (excludes thymus C37.9, heart and mediastinum C38._)

(includes adipose tissue, aponeuroses, artery, blood vessel, bursa, connective tissue, fascia, fatty tissue, fibrous tissue, ligament, lymphatic, muscle, skeletal muscle, subcutaneous tissue, synovia, tendon, tendon sheath, vein, vessel)

Connective, subcutaneous and other soft tissues of: (see list under C49)

- | | |
|--------------|--------------------------|
| • axilla | • thoracic wall |
| • chest | • infraclavicular region |
| • chest wall | • scapular region |
| • thorax | |

Includes:

Aorta, NOS
Axillary artery
Diaphragm
Intercostal muscle
Internal mammary artery
Latissimus dorsi muscle

Pectoralis major muscle
Subclavian artery
Superior vena cava
Thoracic duct
Trapezius muscle

Lymph Nodes

C77 LYMPH NODES

C77.2 Intra-abdominal lymph nodes

Abdominal lymph node Aortic lymph node Celiac lymph node Colic lymph node Common duct lymph node Gastric lymph node Hepatic lymph node Ileocolic lymph node Inferior mesenteric lymph node Intestinal lymph node Lumbar lymph node Mesenteric lymph node, NOS Midcolic lymph node Pancreatic lymph node, NOS Para-aortic lymph node Periaortic lymph node Peripancreatic lymph node Porta hepatis lymph node Portal lymph node Pyloric lymph node Retroperitoneal lymph node Splenic hilar lymph node

Splenic lymph node, NOS

Superior mesenteric lymph node

Appendix 8: ICD-O-3 Histology Codes

Invasive Cancer

The International agency for Research on Cancer provides ICD-O codes electronically on their website; however, the IARC version are not identical to the version used in the US. More details can be found here: <https://seer.cancer.gov/icd-o-3/>

PRISMM uses the codes specified in the ICD-O-3 Site/Histology Validation List found at this site: <https://seer.cancer.gov/icd-o-3/>.

In addition, version 8 of the AJCC Staging Manual specify a subset of codes that should be associated with cancer types. These site codes are used in PRISMM.

Breast Cancer Histology Definitions From SEER With ICD-O-3 Histology Codes for Noninvasive Cancer

Code	Description
8010	Carcinoma in situ
8050	Papillary carcinoma in situ
8070	Squamous cell carcinoma in situ
8201	Cribriform carcinoma in situ
8500	Ductal carcinoma in situ
8501	Comedocarcinoma, noninfiltrating
8503	Intraductal papilloma with ductal carcinoma in situ
8504	Encapsulated papillary carcinoma
8507	Intraductal micropapillary carcinoma
8520	Lobular carcinoma in situ
8522	Intraductal and lobular in situ carcinoma
8523	Intraductal mixed with other types carcinoma in situ
8543	Paget disease and intraductal carcinoma

Lung Cancer Histology Definitions From SEER With ICD-O-3 Histology Codes for Invasive Cancer: Lung Terms and Definitions

Adenocarcinoma with mixed subtypes (8255): A mixture of two or more of the subtypes of adenocarcinoma such as acinar, papillary, bronchoalveolar, or solid with mucin formation.

Adenosquamous carcinoma (8560): A single histology in a single tumor composed of both squamous cell carcinoma and adenocarcinoma.

Combined small cell carcinoma (8045): A small cell carcinoma that is combined with a non-small cell carcinoma. The combinations are small cell and adenocarcinoma, or squamous cell carcinoma, or large cell carcinoma.

Large cell carcinoma (8012): Large cell is a diagnosis that is used when the tumor is a non-small cell carcinoma that is undifferentiated. Because the tumor is undifferentiated, the pathologist cannot find glandular (adeno), or squamous differentiation.

Large cell neuroendocrine carcinoma (8013): A non-small cell carcinoma with neuroendocrine differentiation proven by immunohistochemical stain, currently classified as large cell carcinoma. These tumors require further study before being included as a separate category in a histologic classification.

Neuroendocrine carcinoma (8246): Neuroendocrine carcinoma is a group of carcinomas that include typical carcinoid tumor and small cell carcinoma. Code the specific histology when given. Code neuroendocrine carcinoma, NOS (8246) when no specific histology is documented.

Non-small cell carcinoma (8046): The term non-small cell is used two ways, as a group term describing all carcinomas that are not small cell; and as a default diagnosis when there isn't enough tissue to classify the tumor beyond the exclusion of small cell.

Pleomorphic carcinoma (8022): A poorly differentiated non-small cell carcinoma (squamous cell carcinoma, adenocarcinoma, or large cell carcinoma) containing spindle cells and/or giant cells or, a carcinoma containing only spindle cells and giant cells. These fall under the general category of sarcomatoid carcinoma.

Sarcomatoid carcinoma: A group of tumors that are non-small cell in type and contain spindle cells and/or giant cells. Depending on the histologic features the tumor may be designated: pleomorphic carcinoma (8022); spindle cell carcinoma (8032); giant cell carcinoma (8031), carcinosarcoma (8980); or pulmonary blastoma (8972)

Small cell carcinoma: Malignant epithelial tumor consisting of small cells. There are many types of lung cancer, but most can be categorized into one of two basic types, "small cell carcinoma" or "non-small cell carcinoma"

Undifferentiated carcinoma (8020): A high grade malignancy lacking glandular structures or other specific features that can be used to better classify the tumor. Undifferentiated carcinoma is used by pathologists when they believe the tumor is a carcinoma (not lymphoma, melanoma, or sarcoma) but they are not sure if the tumor is small cell or non-small cell.

Appendix 9: Common Antineoplastic Drugs and Regimens

Drugs approved for Cancer Care

Source: <https://www.cancer.gov/about-cancer/treatment/drugs>

The following sections are meant to orient curators to drugs and regimens commonly used in the treatment of specific cancers. These drugs have FDA approval for use in the cancers listed. This is not meant to be a comprehensive list of drugs that have ever been used to treat a given cancer; however, it reflects the drugs currently used to treat cancers.

Bladder Cancer Approved Drugs and Regimens

Source: <https://www.cancer.gov/about-cancer/treatment/drugs/bladder>

Bladder Cancer Approved Single Drugs

Generic Drug Name	Commercial Name/Synonym	Oral or IV Administration
Abemaciclib	Verzenio	Oral
Avelumab	Bavencio	IV
Alpelisib	Piqray	Oral
Anastrozole	Arimidex	Oral
Atezolizumab	Tecentriq	IV
Cisplatin	Platinol	IV
Doxorubicin Hydrochloride	Adriamycin	IV
Durvalumab	Imfinzi	IV
Gemcitabine*	Gemzar	IV
Erdafitinib	Balversa	Oral
Methotrexate*	Trexall	IV
Nivolumab	Opdivo	IV
Pembrolizumab	Keytruda	IV
Thiotepa	TESPA, Thiophosphoamide, TSPA	IV
Valrubicin	Valstar	IV
Vinblastine Sulfate*	Velban, Alkaban-AQ, VLB	IV

*included in regimen list, but not specified in the NCI single drug list

Bladder Cancer Approved Drug Combinations

Regimen Name	Included Drugs
GEMCITABINE-CISPLATIN	Gemcitabine Cisplatin
MVAC	M = Methotrexate V = Vinblastine Sulfate A = Doxorubicin Hydrochloride (Adriamycin) C = Cisplatin

Breast Cancer Approved Drugs and Regimens

Source: <https://www.cancer.gov/about-cancer/treatment/drugs/breast>

Breast Cancer Approved Single Drugs

Generic Drug Name	Commercial Name/Synonym	Oral or IV Administration
Abemaciclib	Verzenio	Oral
Ado-Trastuzumab Emtansine	Kadcyla	IV
Alpelisib	Piqray	Oral
Anastrozole	Arimidex	Oral
Atezolizumab	Tecentriq	IV
Capecitabine	Xeloda	Oral
Cyclophosphamide	Cytosan, Neosar	IV
Docetaxel	Taxotere	IV
Doxorubicin Hydrochloride	Adriamycin	IV
Epirubicin Hydrochloride	Ellence	IV
Eribulin Mesylate	Halaven	IV
Everolimus	Afinitor, Afinitor Disperz	Oral
Exemestane	Aromasin	Oral
Fam-Trastuzumab Deruxtecan-nxki	Enhertu	IV
Fluorouracil Injection	5-FU	IV
Fulvestrant	Faslodex	IM
Gemcitabine Hydrochloride	Gemzar	IV
Goserelin Acetate*	Zoladex	IV/SC
Ixabepilone	Ixempra	IV
Lapatinib Ditosylate	Tykerb	Oral
Letrozole	Femara	Oral
Leuprolide*	Lupron	IV
Megestrol Acetate	Megace	Oral
Methotrexate	Trexall	IV
Neratinib Maleate	Nerlynx	Oral
Olaparib	Lynparza	Oral
Paclitaxel	Taxol	IV
Paclitaxel Albumin-stabilized Nanoparticle Formulation	Abraxane	IV
Palbociclib	Ibrance	Oral
Pertuzumab	Perjeta	IV
Raloxifene Hydrochloride	Evista	Oral
Ribociclib	Kisqali	Oral
Talazoparib Tosylate	Talzenna	Oral
Tamoxifen Citrate	Nolvadex	Oral
Thiotepa	TESPA, Thiophosphoamide, TSPA	IV
Toremifene	Fareston	Oral
Trastuzumab	Herceptin	IV
Trastuzumab and Hyaluronidase-oysk	Herceptin Hylecta	IV
Vinblastine Sulfate	Velban, Alkaban-AQ, VLB	IV

- LHRH agonists

Breast Cancer Approved Drug Combinations

Regimen Name	Included Drugs
AC	A = Doxorubicin Hydrochloride (Adriamycin) C = Cyclophosphamide (Cytoxan)
AC-T	A = Doxorubicin Hydrochloride (Adriamycin) C = Cyclophosphamide (Cytoxan) T = Paclitaxel (Taxol)
CAF	C = Cyclophosphamide (Cytoxan) A = Doxorubicin (Adriamycin) F = Fluorouracil
CMF	C = Cyclophosphamide M = Methotrexate F = Fluorouracil
FEC	F = Fluorouracil E = Epirubicin Hydrochloride C = Cyclophosphamide
TAC	T = Docetaxel (Taxotere) A = Doxorubicin Hydrochloride (Adriamycin) C = Cyclophosphamide (Cytoxan)

Breast Cancer Other Drugs That Are Not Currently Being Collected as Antineoplastic Agents

Denosomab [Xgeva] IV

Pamidronate Disodium [Aredia] IV

Colon and Rectal Cancer Approved Drugs and Regimens

Source: <https://www.cancer.gov/about-cancer/treatment/drugs/colorectal#1>

Colon and Rectal Cancer Approved Single Drugs

Generic Drug Name	Commercial Name/Synonym	Oral or IV Administration
Bevacizumab	Avastin, Mvasi	IV
Capecitabine	Xeloda	Oral
Cetuximab	Erbitux	IV
Fluorouracil Injection	5-FU	IV
Ipilimumab	Yervoy	IV
Irinotecan Hydrochloride	Camptosar	IV
Leucovorin Calcium	Folinic Acid	IV
Nivolumab	Opdivo	IV
Oxaliplatin	Eloxatin	IV
Panitumumab	Vectibix	IV
Pembrolizumab	Keytruda	IV
Ramucirumab	Cyramza	IV
Regorafenib	Stivarga	IV
Trifluridine and Tipiracil Hydrochloride	Lonsurf	IV
Ziv-Aflibercept	Zaltrap	IV

Colon and Rectal Cancer Approved Drug Combinations

Regimen Name	Included Drugs
Regimens with Infused/IV Drugs	
FOLFIRI	FOL = Leucovorin Calcium (Folinic Acid) F = Fluorouracil IRI = Irinotecan Hydrochloride
FOLFIRI-BEVACIZUMAB	FOL = Leucovorin Calcium (Folinic Acid) F = Fluorouracil IRI = Irinotecan Hydrochloride BEVACIZUMAB
FOLFIRI-CETUXIMAB	FOL = Leucovorin Calcium (Folinic Acid) F = Fluorouracil IRI = Irinotecan Hydrochloride CETUXIMAB
FOLFOX	FOL = Leucovorin Calcium (Folinic Acid) F = Fluorouracil OX = Oxaliplatin
FOLFOX-BEVACIZUMAB	FOL = Leucovorin Calcium (Folinic Acid) F = Fluorouracil OX = Oxaliplatin BEVACIZUMAB
FOLFOX-CETUXIMAB	FOL = Leucovorin Calcium (Folinic Acid) F = Fluorouracil OX = Oxaliplatin CETUXIMAB
FU-LV	FU = Fluorouracil LV = Leucovorin Calcium
Regimens with Oral Drugs	
CAPOX	CAP = Capecitabine OX = Oxaliplatin
XELIRI	XEL = Capecitabine (Xeloda) IRI = Irinotecan Hydrochloride
XELOX	XEL = Capecitabine (Xeloda) OX = Oxaliplatin

NSCLC Cancer-Directed Drug Therapies

Source: <https://www.cancer.gov/about-cancer/treatment/drugs/lung>

Lung Cancer Single Drugs Curative (Adjuvant) Approved Chemotherapy Treatments

Generic Drug Name	Commercial Name/Synonym	Oral or IV Administration
Cisplatin	Platinol	IV
Vinorelbine	Navelbine	IV
Etoposide	Etopophos, Toposar	IV
Vinblastine	Alkaban-AQ, Velban	IV
Gemcitabine	Gemzar	IV
Docetaxel	Taxotere, Docecad	IV
Pemetrexed	Alimta	IV

Lung Cancer Drugs Palliative (Systemic) Approved Single Drugs

Generic Drug Name	Commercial Name/Synonym	Oral or IV Administration
Cisplatin	Platinol	IV
Carboplatin	Paraplatin	IV
Paclitaxel	Taxol, Onxal	IV
Docetaxel	Taxotere	IV
Vinorelbine	Navelbine	IV
Gemcitabine	Gemzar	IV
Etoposide	Toposar, VePesid, Etopophos	IV
Irinotecan	Camptosar and Onivyde	IV
Vinblastine	Alkaban-AQ, Velban	IV
Mitomycin	Mutamycin, Mitomycin-C	IV
Ifosfamide	Ifex	IV
Pemetrexed	Alimta	IV
Erlotinib	Tarceva	Oral
Bevacizumab	Avastin	IV
Albumin-bound paclitaxel	Abraxane	IV
Crizotinib	Xalkori	Oral
Afatinib	Gilotrif, Tomtovok, Tovok	Oral
Ceritinib	Zykadia	Oral
Ramucirumab	Cyramza	IV
Nivolumab	Opdivo	IV

Pancreatic Cancer Approved Drugs and Regimens

Source: <https://www.cancer.gov/about-cancer/treatment/drugs/pancreatic>

Pancreatic Cancer Approved Single Drugs

Drug	Trade Name/Synonyms	Route
Cisplatin*	Platinol	IV
Erlotinib	Tarceva	Oral
Everolimus	Afinitor, Afinitor Disperz	Oral
Fluorouracil Injection	5-FU	IV
Gemcitabine Hydrochloride	Gemzar	IV
Irinotecan Hydrochloride Liposome	Onivyde	IV
Leucovorin Calcium*	Folinic Acid	IV
Mitomycin C	Mutamycin, MTC	IV
Olaparib	Lynparza	Oral
Oxaliplatin*	Eloxatin	IV
Paclitaxel Albumin-stabilized Nanoparticle Formulation	Abraxane	IV
Sunitinib Malate	Sutent	Oral

*included in regimen list, but not specified in the NCI single drug list

Pancreatic Cancer Approved Drug Combinations

Regimen Name	Included Drugs
FOLFIRINOX	FOL = Leucovorin Calcium (Folinic Acid) F = Fluorouracil IRI = Irinotecan Hydrochloride OX = Oxaliplatin
GEMCITABINE-CISPLATIN	Gemcitabine Cisplatin
GEMCITABINE-OXALIPLATIN	Gemcitabine Oxaliplatin
OFF	O = Oxaliplatin F = Leucovorin Calcium (Folinic Acid) F = Fluorouracil

Prostate Cancer Approved Drugs and Regimens

Source: <https://www.cancer.gov/about-cancer/treatment/drugs/prostate>

Prostate Cancer Approved Single Drugs

Drug	Trade Name/Synonyms	Route
Abiraterone Acetate	Zytiga, Yonsa	Oral
Apalutamide	Erleada	Oral
Bicalutamide	Casodex	Oral
Cabazitaxel	Jevtana	IV
Cisplatin*	Platinol	IV
Darolutamide	Nubeqa	Oral
Degarelix	Firmagon	IV/SC
Docetaxel	Taxotere	IV
Enzalutamide	Xtandi	Oral
Flutamide	Eulexin	Oral
Goserelin Acetate	Zoladex	IV/SC
Leuprolide Acetate	Eligard, Lupron, Lupron Depot	IV
Mitoxantrone Hydrochloride	Novantrone	IV
Nilutamide	Nilandron	Oral
Radium 223 Dichloride	Xofigo	IV
Sipuleucel-T	Provenge	IV**

**leukapheresis

No regimens listed

Appendix 10: Tip Sheets

The following section contains 1-page tip sheets that summarize the critical information in the above sections. All curators should have access to the complete Manual Curation Directives document and should use that in training to thoroughly learn and understand the rules.

The tip sheets are designed to address the most critical data fields and the most common errors; they are a good resource for curators who are already somewhat familiar with the guidelines. The data violation tables have been copied below to help curators know which data elements are most crucial.

PRISSMM Curation Tip Sheet: Pathology

Curate **all** pathology reports from procedures performed on or after date of diagnosis of cancer of interest. **except** reports for cytogenetics, pap smears, non-diagnostic, or basal cell or squamous cell skin biopsies.

Always check for scanned pathology reports.

Report Date: Date report is filed under in EHR

Procedure Date: Date of the biopsy or surgery when the sample was taken.

Number of Specimens: Count each different specimen evaluated in the report. Group like with like.

Example: "3 of 5 lymph nodes evaluated contain cancer consistent with the patient's known colon adenocarcinoma" = 2 specimens (1. lymph node. no cancer 2. Lymph node. colon adenocarcinoma)

Anatomic Site: Choose most specific option.

In situ cancer present? Only yes if in situ is explicitly stated. otherwise assume no. Path reports often do not mention this.

Invasive cancer present? Most often will include the terms "carcinoma" or "malignant." See "Pathology Ambiguous Terms List" linked in REDCap for additional detail.

Invasive Cancer Type: Otherwise known as primary cancer; usually the organ or body part the cancer originated in. May not be explicitly stated on path report. Make reasonable assumptions as below.

Examples: "consistent with lung primary" = lung

Patient's only cancer diagnosis is breast. Breast biopsy report says "invasive ductal carcinoma" but does not name primary = breast

Patient has lung cancer and colon cancer. Biopsy of a different site just says "adenocarcinoma." This could be either! Read more detail of path report or check a clinic note to clarify.

Histology: Usually explicitly reported for surgical pathology. As above. "consistent with known primary" means you can assume the histology is the same as the initial diagnosis.

PRISSMM Pathology Data Violations

Major	Incorrect/missing procedure date [+/- 1 month]
	Incorrect/Missing cancer type
	Missed/incorrectly identified invasive cancer
	Missed eligible pathology report(s)
Minor	Incorrect/missing histology
	Incorrect/missing specimens
	Incorrect report date [+/- 1 month]
	Ineligible report included
Other	Incorrect/missing procedure institution
	Missing/incorrect “during diagnosis” status

PRISMM Curation Tip Sheet: Radiology/Imaging

Curate all imaging reports for cancer-specific scans performed on or after the date of diagnosis of cancer of interest. Do not include mammograms (unless cancer of interest is breast), ultrasounds, x-rays, images from procedures. Only read "Indication" and "Impression" sections of the report. Always check for scanned reports.

Scan Date: Date the scan was performed.

Report Date: Date report filed under in EHR.

Scan Type: Choose the imaging modality

Scan Site: Check all regions of the body that the scan covered.

Evidence of Cancer: Answer "Yes" if there is explicit OR implicit evidence of cancer.

Explicit evidence of cancer:

- Tumor or neoplasm
- An abnormality that is described as malignant, metastatic, or cancer.
- A nodule that is $\geq$ 1 cm unless otherwise noted to be benign.
- Lymph node $\geq$ 1 cm that the radiologist is following for change in size.
- Any finding that is described as recurrence, disease progression, change in disease/tumor burden.
- On PET, FDG-avid/uptake/increased activity = cancer

Implicit evidence of cancer:

- Context from "Indication" section of report, such as stating that the patient has a known active cancer.
- Context from previous scans. Assume sites previously described as cancer still are until otherwise stated.
- Treatment effect or response to therapy.

If new abnormality is mentioned:

- Yes, evidence of cancer if:
 - cancer is favored/likely (use ambiguous terms list)
- Indeterminate if:
 - Cancer is possible, non-cancerous condition favored/likely.
 - Cancer and other condition are possible, neither is favored.
- No evidence of cancer if:
 - Presence of abnormality mentioned with no further detail
 - Non-cancerous condition is favored, cancer is unlikely or not mentioned.

Status of Cancer: Use overall assessment or summary statement if given. Only use words describing cancer status (progressing, responding, relapse, recurrence) and changes in **volume** to evaluate. Ignore other descriptive words like "clarity," "prominent," and "conspicuous."

Do not use volume changes of fluid-based disease (e.g. pleural effusion) to evaluate status.

Improving = all sites smaller OR some sites smaller, some unchanged

Stable = all sites unchanged in size

Mixed = some sites increasing or new and, some decreasing or resolved

Progressing = all sites larger OR some larger, some unchanged OR any new sites of cancer

Where is cancer and Sites involved: Choose up to 15 sites of the body where the cancer is located. Do not record sites that would be "indeterminate" per the ambiguous terms list.

PRISSMM Radiology/Imaging Data Violations

Major	Incorrect scan date [+/- 1 month]
	Missed or incorrectly identified invasive cancer
	Missed or incorrectly identified progression
	Missed or incorrectly identified response
	Missed eligible imaging report(s)
Minor	Incorrect missing anatomic sites of cancer
	Incorrect report date [+/- 1 month]
	Incorrect scan type/scan sites
	Incorrect cancer status (between stable. mixed. or not stated/indeterminate)
Other	Incorrect/missing reference scan date
	Incorrect/missing procedure/report institution
	Other minor mistake/omission

PRISSMM Curation Tip Sheet: Medical Oncology Assessment

Beginning on or after date of diagnosis of cancer of interest. curate one medical oncology visit note per month. Choose earliest note in the month from (1) Internal MD. If none, then (2) Internal NP/PA. If none, then (3) External MD. If none, then (4) external NP/PA. **Do not** use notes from other provider types (RNs, radiation oncologists, surgeons, etc) even if this means one or months with no notes curated. There is one exception to this rule – when a patient is re-staged during the month, try to curate this visit regardless of order to record the newest cancer status.

Only read the “Impression and Plan” section.

Always check for scanned visit notes when there is not a visit with the internal medical oncologist.

Visit Date: Date the provider saw the patient.

Evidence of Cancer: Answer Yes if there is explicit OR implicit evidence of cancer in the Impression and Plan. Use what you know from prior notes to inform your answer.

Assume yes for:

- Stage IV/metastatic patients unless explicitly stated otherwise.
- Patients on neoadjuvant chemo or upcoming surgery for cancer is planned.
- Provider describes treatment response or change in disease burden.

Assume no for:

- Patients on surveillance/follow-up
- Patients on adjuvant therapy
- We know the only site of cancer was surgically removed

Status of Cancer: Use overall assessment or summary statement if given. Choose response that best reflects status on that visit date. Do not use tumor markers or descriptions of general wellness as cancer status.

Improving/Responding: responding to treatment. decreased in size. improving. resolved.

- If no general statement. all sites decreasing OR some decreasing and some unchanged.
- Include phrases like ongoing/sustained/continuing/lasting response

Stable: stable disease. tumor burden unchanged

- May say “clinically stable” or “Stable response” or “disease is controlled.”

Mixed: mixed response. some sites increasing. some decreasing

Progressing: Cancer is growing. increasing in size. larger tumor/disease burden. progressing

- May say patient recurred. relapsed. has progressive disease (PD). or cancer is not responding
- Patient has new site of cancer (NOT new cancer diagnosis of a different primary)
- If no general statement given. all sites larger OR some larger. some unchanged OR any new sites of cancer

Not Stated/Indeterminate: most common choice! Provider often does not describe imaging. comment on cancer status. or describe sites of disease. Also choose this for baseline visit (patient is recently diagnosed. has not had definitive local therapy or begun systemic treatment).

PRISMM Medical Oncology Assessment Data Violations

Major	Incorrect visit date [+/- 1 month]
	Missed or incorrectly identified invasive cancer
	Missed or incorrectly identified progression
	Missed or incorrectly identified response
	Missed eligible note(s)
	Note associated with wrong cancer type
Minor	Incorrect cancer status (between stable, mixed, or not stated/indeterminate)
	Ineligible note included
Other	Did not choose best note in the month
	Other minor mistake/omission

Appendix 11: Tumor Markers

Suggested Tumor Markers by Cancer Type

Cancer	Histology	Tumor Marker	Sample for Testing	Source	Type
Bile Duct Cancer		CA 19-9: Cancer Antigen 19-9	blood	NCI SEER	Cancer antigen proteins & glycoproteins
Bladder Cancer C67.0-C67.9	8000-9136,9141-9582,9700-9701 (no mention in cs)	CEA: Carcinoembryonic Antigen	blood		Oncofetal antigen
Breast Cancer C50.0-C50.6, C50.8-C50.9	8000-9136,9141-9582,9700-9701	CA 15-3: Cancer Antigen 15-3	blood	NCI SEER	Cancer antigen proteins & glycoproteins
Breast Cancer C50.0-C50.6, C50.8-C50.9	8000-9136,9141-9582,9700-9701	CA 27.29	blood	NCI	Cancer antigen proteins & glycoproteins
Breast Cancer C50.0-C50.6, C50.8-C50.9	8000-9136,9141-9582,9700-9701	CEA: Carcinoembryonic Antigen	blood	SEER UNM	Oncofetal antigen
Cervical Cancer C53.0, C53.1, C53.8, C53.9	8000-8576 8940-8950, 8980, 8981	CA 15-3: Cancer Antigen 15-3	blood	SEER UNM	Cancer antigen proteins & glycoproteins
Colon/Rectal Cancer C18.0, C18.2 -C18.9. C19.9, C20.9	8000-8152, 8154-8231, 8243-8245, 8247, 8248, 8250-8576, 8940-8950, 8980, 8981	CEA: Carcinoembryonic Antigen	blood	NCI SEER	Oncofetal antigen
Gallbladder Cancer C23.9	8000-9136, 9141-9582, 9700-9701	CA 19-9: Cancer Antigen 19-9	blood	NCI SEER	Cancer antigen proteins & glycoproteins
Gastric Cancer C160-C166,C168-C169	8000-8152,8154-8231,8243-8245,8247,8248,8250-8934,8940-9136,9141-9582,9700-9701	CA 19-9: Cancer Antigen 19-9	blood	NCI SEER	Cancer antigen proteins & glycoproteins
Gastric Cancer C160-C166,C168-C169	8000-8152,8154-8231,8243-8245,8247,8248,8250-8934,8940-9136,9141-9582,9700-9701	CEA: Carcinoembryonic Antigen	blood	SEER	Oncofetal antigen
Head and Neck Cancer		CEA: Carcinoembryonic Antigen	blood	SEER	Oncofetal antigen

Cancer	Histology	Tumor Marker	Sample for Testing	Source	Type
Liver Cancer C22.0	8000-8157,8162-8175,8190-9136,9141-9582,9700-9701	AFP: Alpha-fetoprotein	blood	AACC NCI SEER	Oncofetal antigen
Melanoma	8720-8790	LD: Lactate dehydrogenase	blood	NCI	
Neuroblastoma		LD: Lactate dehydrogenase	blood	NCI	
Neuroblastoma		Neuron-specific enolase (NSE)	blood	NCI	
Neuro-endocrine tumors C24.1, C18.0, C18.2-C18.9, C19.9, C20.9, C17.0-C17.3, C17.8, C17.9, C16.0-C16.6, C16.8, C16.9	8153, 8240-8242, 8246, 8249	CgA: Chromogranin A	blood	NCI	
Non-Small Cell Lung Cancer C34.0-C34.3, C34.8-C34.9		CEA: Carcinoembryonic Antigen	blood	SEER	Oncofetal antigen
Small Cell Lung Cancer C34.0-C34.3, C34.8-C34.9		Neuron-specific enolase (NSE)	blood	NCI	
Ovarian Cancer (epithelial) C56.9	8000-8576, 8950-8671, 8930-9110	CA 125: Cancer Antigen 125	blood	NCI SEER	Cancer antigen proteins & glycoproteins
Ovarian Cancer (epithelial) C56.9	8000-8576, 8950-8671, 8930-9110	CEA: Carcinoembryonic Antigen	blood		Oncofetal antigen
Ovarian Cancer (germ cell) C56.9		AFP: Alpha-fetoprotein	blood	SEER	Oncofetal antigen
Ovarian Cancer (germ cell) C56.10		B-hCG: Beta Human chorionic gonadotropin	blood urine	NCI	Hormones
Ovarian Cancer (germ cell) C56.11		LD: Lactate dehydrogenase	blood	NCI	
Pancreatic Cancer C25.3-C25.4, C25.7-C25.9		CA 19-9: Cancer Antigen 19-9	blood	NCI	Cancer antigen proteins & glycoproteins
Pancreatic Cancer C25.3-C25.4, C25.7-C25.9		CEA: Carcinoembryonic Antigen	blood	SEER	Oncofetal antigen
Prostate Cancer C61.9	8000-9136,9141-9582,9700-9701	PSA: Prostate Specific Antigen	blood	NCI SEER	Enzyme
Testes Cancer C62.0-C62.1, C62.9	8000-9136,9141-9582,9700-9701	AFP: Alpha-fetoprotein	blood	SEER	Oncofetal antigen

Cancer	Histology	Tumor Marker	Sample for Testing	Source	Type
Testes Cancer C62.0-C62.1, C62.9	8000-9136,9141-9582,9700-9701	B-hCG: Beta Human chorionic gonadotropin	blood urine	NCI	Hormones
Testes Cancer C62.0-C62.1, C62.9	8000-9136,9141-9582,9700-9701	LD: Lactate dehydrogenase	blood	NCI	
Thyroid Cancer C73.9	8000-9136,9141-9582,9700-9701	Calcitonin (for medullary)	blood	NCI SEER	Hormone
Thyroid Cancer C73.9	8000-9136,9141-9582,9700-9701	CEA: Carcinoembryonic Antigen	blood	SEER	Oncofetal antigen
Thyroid Cancer C73.9	8000-9136,9141-9582,9700-9701	Thyroglobulin	blood	NCI	Protein

Appendix 12: REDCap Set-up and Data Piping

The PRISSMM Redcap data dictionary is provided to sites for upload into the site's REDCap instance; optimal use of the project requires repeating instances.

- 1) Project Set-up
 - a. Enable optional modules and customizations
 - b. Set a custom record label
 - c. Add training documentation to PRISSMM Pathology, Imaging, and Medical Oncologist assessment
 - d. Implement Data Quality module, if using
- 2) Customize data fields to accommodate site staff
 - a. Curation Initiation Eligibility – add names of site curators, PM, QA Manager to data field **PRISSMM Curator** [cur_curator]
 - b. Quality Assurance – Add names of site PMs and/or QA Managers as appropriate to data fields **QA Reviewer** [qa_full_reviewer] and **QA Reviewer for SDV of Dual Curation Case** [qa_full_reviewer_dual]
- 3) Data Quality Rules: add project logic checks

The screenshot shows the 'Project Setup' page for a project named 'BPC PRISSMM Comprehensive Project'. The page has a navigation bar with 'Project Home', 'Project Setup' (active), 'Other Functionality', and 'Project Revision History'. Below the navigation bar, the project status is 'Development' and 'Completed steps 0 of 7'. The main content area is divided into three sections, each with a 'Not started' status icon and an 'I'm done!' button.

- Main project settings:** Includes 'Enable' buttons for 'Use surveys in this project?' and 'Use longitudinal data collection with defined events?'. There is a link to a video: 'VIDEO: How to create and manage a survey'. A text field 'Modify project title, purpose, etc.' is also present.
- Design your data collection instruments:** Includes a 'Go to' section with 'Online Designer' and 'Data Dictionary' buttons, and a link to 'Explore the REDCap Shared Library'. Below this, there is a question about checking the 'Check For Identifiers' page and links to 'Smart Variables', 'Piping', and 'Action Tags'.
- Enable optional modules and customizations:** Includes a list of modules with 'Enable' buttons: 'Repeatable instruments', 'Auto-numbering for records', 'Scheduling module (longitudinal only)', 'Randomization module', and 'Designate an email field for sending survey invitations'. There is also an 'Additional customizations' button.

Project Set-up: Enabling Optional Modules and Customizations

- 1) Activate Multiple Instances with Custom Labels in the Repeatable Instances.

Eight of the forms should have the repeatable instruments activated; it is recommended that these forms have custom labels applied to assist curators. See below.

- 2) To assist the curators, it is recommended that certain data elements are added to the instances. In the example below, Cancer Diagnosis date [ca_dx_dt] is added so that the curator will see the date associated with the form.

Repeatable instruments

An excellent way to collect repeating data in REDCap is to use repeatable instruments and/or repeatable events. This is sometimes called one-to-many data collection. Some examples may include but are not limited to the following: data from multiple visits or observations, concomitant medications, adverse events, or repetitive surveys (daily, weekly, etc.).

Below you can specify a data collection instrument to be infinitely repeatable, which means that an instrument can be repeated over and over again (a different number of times for each record) even without enabling REDCap's longitudinal module. Once an instrument is set to repeat, you will see options on the Record Home Page to add another instance of the instrument for the currently selected record. All instances of a repeating instrument will then be displayed as a table near the bottom of the Record Home Page, thus allowing viewing of the instances and easy navigation within them.

Repeat this instrument?	Instrument name	Custom label for repeating instruments (optional)
		Example: [visit_date], [weight] kg
☐	Curation Initiation Eligibility	
☐	Patient Characteristics	
☒	Cancer Diagnosis	[ca_dx_dt]
☒	Ca Directed Drugs	[drugs_startdt_drug_1]
☒	Ca Directed Radtx	[rt_start_dt]
☒	Prissmm Pathology	[path_proc_dt]
☒	Prissmm Imaging	[image_scan_dt]
☒	Prissmm Med Onc Assessment	[md onc_visit_dt]
☒	Prissmm Tumor Marker	[tm_spec_collect_dt]
☒	Cancer Panel Test	[cpt_report_dt],[cpt_genie_sample_ic]
☐	Curation Completion	
☐	Quality Assurance	

Save

Cancel

Prissmm Imaging		
1		11-05-2013,CT
2		12-14-2013,MRI
3		02-06-2014,PET or PET-CT
+ Add new		

Project Set-up: Custom record labels

To assist curators reviewing and curating the patient's data from the electronic medical record, add the last name and the internal mrn as a custom label. (See below)

Once activated, the name and the medical record of the patient will appear in the record status dashboard alongside the Record ID, allowing the curator to readily identify the patient in the EHR.

Make additional customizations to your project

You may use the options below to make customizations to the project. When done, click Save to save your changes.

☒ Set a custom record label

You may append other data and/or static text to any record name (e.g., Study ID) as the record is displayed on your data collection instruments, such as inside the drop-down lists when choosing a record and at the top of the page after being selected. Simply provide the text you wish to display below, and place any variable names inside square brackets [], after which the data collected for those variables for that record will replace the variable in the text.

Custom record label:

Example: if ([last_name], [first_name]) were entered, then for record '102' it would display '102 (Doe, Jon)'.

Project Set-up: Training Material

To assist curators reviewing and curating the patient's data from the electronic medical record, each of the PRISMM forms has a place for to insert the relevant training material, the slides and the ambiguous terms, which can be accessed during curation.

Pathology

 Variable: prissmmp_doc

PRISMM Pathology Directives and Training Materials

For directives and training on this section click on the document

 Variable: prissmmpterms_doc

PRISMM Pathology Ambiguous Terms

For guidance on interpreting ambiguous terms, click on the document

Imaging

 Variable: prissmmri_doc

PRISMM Imaging Directives and Training Materials

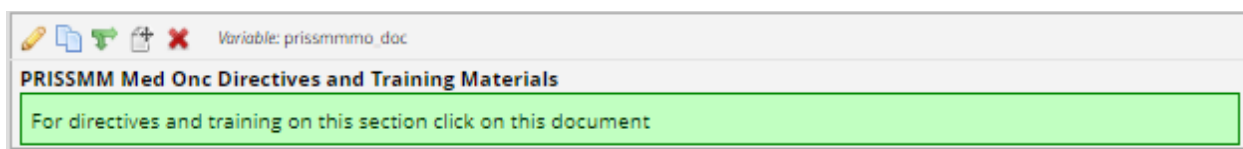
For directives and training on this section click on the document

 Variable: prissmmriterms_doc

PRISMM Imaging Ambiguous Terms

For guidance on interpreting ambiguous terms, click on the document

Medical Oncology



Data Quality Rules and Data Resolution Workflow – optional

☒ **Enable the Field Comment Log or Data Resolution Workflow (Data Queries)?**

For this project, you may enable either the Field Comment Log or Data Resolution Workflow (also known as the Data Queries module). The Field Comment Log (enabled by default) allows users to leave comments for any given field on a data entry form by clicking the balloon icon next to the field. All comments can also be viewed, searched, and downloaded on the Field Comment Log page. Alternatively, if the Data Resolution Workflow is enabled, users will be allowed to open a workflow for documenting the process of resolving issues with data in the project (i.e. opening, responding to, and closing data queries). [View more details](#)

Enable:

☒ Allow users to edit or delete Field Comments (excludes Data Resolution Workflow comments)?

Customizing Data Fields to Include Site Staff

The customization could occur in either the PRISMM REDCap Data Dictionary prior to uploading or during set-up prior to moving the project into production.

Data Dictionary

If the customization is done in the .csv data dictionary prior to installation, look for the Field Label of the three data fields requiring customization and edit them.

Example: Before

Variable / Field Name	Form Name	Section Header	Field Type	Field Label	Field Note
qa_full_reviewer_dual	quality_assurance	dropdown	QA Reviewer for SDV of Dual Curation Case	1, QA Manager - customize 2, PM	Record the name of the person who SDV QA'd the dual curation record.

Example: After customization with 3 QA managers specified (Jackie, Monica, and John)

Variable / Field Name	Form Name	Section Header	Field Type	Field Label	Field Note
qa_full_reviewer_dual	quality_assurance	dropdown	QA Reviewer for SDV of Dual Curation Case	1, Jackie 2, Monica 3, John	Record the name of the person who SDV QA'd the dual curation record.

Data Sources for PRISSMM Project

The PRISSMM Redcap project has data from different sources

- 1) Piped: Preferred
- 2) Piped / Curated
- 3) Curated
- 4) Calculated

The **Field Note** column in the PRISSMM REDCap Data Dictionary notes whether the data could/should be piped from the institutional data sources. The following list of patient data collection instruments indicates which could have piped data.

Patient Characteristics

Piped: Preferred

Cancer Diagnosis

Piped or Curated

Cancer Directed Drug Therapy

Piped or Curated

Cancer Directed Radiation Therapy

Piped or Curated

PRISSMM Pathology

Piped / Curated

PRISSMM Imaging

Piped / Curated

PRISSMM Medical Oncology Assessment

Piped / Curated

PRISSMM Tumor Markers, *if included*

Piped / Curated

Cancer Panel Test

Piped / Curated

Laboratory Tests, *if included*

Piped / Curated

Appendix 13: Retired Data Elements

CRC: Tumor Regression Grade (CS SSF 5)

[ca_crc_trg]

Form Name: cancer_diagnosis

Version Retired: 3.4

CRC: KRAS (CS SSF 9)

[ca_crc_kras]

Form Name: cancer_diagnosis

Version Retired: 3.4

LUNG: Pleural/Elastic Layer Invasion (PL) by H and E or Elastic Stain (CS SSF 2)

[ca_lung_pl_el_inv] PN0135

Form Name: cancer_diagnosis

Version Retired: 3.6

Mapping:

[ca_lung_pl_el_inv] value : [ca_lung_pl_el_inv_2] value

0	: 0
10	: 4
20	: 4
30	: 5
40	: 6
988	: 9
999	: 9

CRC: Perineural Invasion (CS SSF 8)

[ca_crc_per_inv] PN0131

Form Name: Cancer Diagnosis

Version Retired: 3.7

Mapping:

[ca_crc_per_inv] value : [ca_crc_per_inv_2] value

0	: 0
10	: 1
988	: 8
998	: 9
999	: 9

CRC: Circumferential Resection Margin (CS SSF 6)

[ca_crc_crm] PN0130

Form Name: Cancer Diagnosis

Version Retired: 3.7

Mapping:

[ca_crc_crm] value : [ca_crc_crm_2] value

XX1	: XX0
XX2	: XX2
XX3	: XX3
XX4	: XX4
XX5	: XX5
XX6	: XX6
XX7	: XX7

XX8 : XX8
XX9 : XX9
XX11 : XX1